Blacktown
City Council







WSUD developer handbook

MUSIC modelling and design guide

2026

Copyright

Blacktown City Council 2026, Water sensitive urban design (WSUD) developer handbook – MUSIC modelling and design guide.

Copyright © 2026 Blacktown City Council.

Blacktown City Council grants a non-exclusive, revocable licence to any person to reproduce and use this handbook solely in connection with property developments with the Blacktown City region. This licence does not permit: a) commercial exploitation of this handbook or any of its content; b) distribution or communication of this handbook or any of its content for a fee or other commercial advantage; or c) incorporation of this handbook or any of its content into any commercial product or service, without the prior written consent of Blacktown City Council. All other rights are reserved.

Acknowledgements

We acknowledge the Dharug People as the Traditional Custodians of the land on which Blacktown City is built. We pay our respects to Elders past and present, and acknowledge Aboriginal people for their custodianship of this land. We will work together for a united city that respects this land and values the contribution to our community and culture of all people of Aboriginal and Torres Strait Islander heritage.

The first version of the WSUD developer handbook was prepared by AECOM in November 2013. Blacktown City Council acknowledges the work of Tony Merrilees in reviewing the first version and providing technical expertise in developing this updated document. Technical support and advice were provided by Mark Liebman. The third version has been reviewed by Mark Liebman and Alex Kwok.

MUSIC icons were provided courtesy of eWater.

Disclaimer

This handbook has been developed in good faith, after careful review and consultation.

Use of this handbook or any of its content in connection with a development application does not mean the development application will be approved or that a particular outcome will be achieved.

While every care has been taken in compiling this handbook, Blacktown City Council accepts no liability whatsoever for any loss (including without limitation direct or indirect loss and any loss of profit, data or economic loss) occasioned to any person nor for any damage, cost, claim or expense arising from reliance on this handbook or any of its content.

The handbook will be reviewed and updated periodically and the updated version will be available on Blacktown City Council's website. Please ensure you are using the current version.

Please provide any comments to Blacktown City Council, email wsud@blacktown.nsw.gov.au

Date	Version	Change description	Approved by
2020	2	Section 12- Stormwater quality improvement devices	Keysha Milenkovic
2026	3	Updates from MUSIC into MUSICX Section 12 - Stormwater quality improvement devices Section 15 - Flood modelling of local catchments Section 16 - WSUD checklist Section 17 - Glossary and acronyms	Keysha Milenkovic

Table of Contents

1 Introduction 1

2 Compliance pathways for WSUD (excluding flooding) 3

2.1 Precinct scale water quality offset scheme for infill development 3

2.2 S3QM deemed-to-comply tool 3

2.3 Model for urban stormwater improvement conceptualisation (MUSIC) 4

2.3.1 MUSIC-*link* 4

3 Stormwater management report 6

3.1 Water quality modelling report 6

3.2 Voluntary planning agreement or S7.11 development 7

3.3 Water conservation report 8

3.3.1 Determination of actual non-potable reuse and percentage reuse 9

3.4 Stream erosion index (SEI) 9

3.5 On-site stormwater detention 10

3.5.1 Special modelling requirements for the growth areas or other DCPs that allow pre to post modelling 11

3.6 GPT calculations 12

4 Small scale stormwater quality model (S3QM) 13

5 Groundwater assessment report 14

5.1 Applicable development 14

5.2 No adverse impact to surrounding or adjacent properties 14

5.3 No adverse impact to infrastructure 15

5.4 No adverse impact to the environment 15

5.5 No adverse impact to groundwater dependent ecosystems 15

5.6 Design and maintenance of groundwater management systems 16

6 MUSIC model setup 17

7 Rainfall and evaporation inputs 18

7.1 Rainfall data for water quality modelling 18

7.2 Rainfall data for hydrologicmodelling 19

7.3 Potential evapotranspiration (PET)data 20

7.4 Digital (electronic) modelling 20

8 Source nodes and pollutant generation 21

9 Rainfall runoff parameters 22

10 Link routing 26

11 Stormwater management measures 27

11.1 Specific requirements for MUSIC modelling in Blacktown 27
11.2 Standard requirements for MUSIC modelling nodes in Blacktown 28
11.3 Wetlands 28
11.3.1 Wetland design considerations 29
11.3.2 Wetland MUSIC modelling requirements 34
11.4 Ponds 37
11.4.1 Types of ponds 37
11.4.2 Water quality ponds 37
11.4.3 Storage ponds 37
11.4.4 MUSIC modelling requirements for ponds 37
11.5 Sedimentation basins 39
11.6 Detention basins 41
11.6.1 Conventional detention basins 41
11.6.2 Detention basins used in conjunction with water quality devices 41
11.6.3 Additional design information 42
11.6.4 MUSIC modelling requirements for detention basins 45
11.7 Infiltration systems 46
11.8 Bioretention systems 47
11.8.1 General description 47
11.8.2 Temporary bioretention systems 47
11.8.3 Hydraulic loading rates for bioretention basins 47
11.8.4 Solar access for bioretention systems and thermal impacts 48
11.8.5 Additional design information for bioretention systems 50
11.8.6 Protecting bioretention basins from sediment loading 51
11.8.7 MUSIC modelling requirements for bioretention basins 51
11.9 Media filtration 53
11.9.1 Media filtration using conventional underground sand filters 53
11.9.2 Permeable pavement filtration 53
11.10 Green roofs 55
11.11 Design of gross pollutant traps (GPTs) 56
11.11.1 Design standards 57
11.12 Buffers 57
11.13 Swales 58
11.13.1 Design standards 58
11.13.2 MUSIC modelling requirements for swales 59

11.14 Rainwater and stormwater reuse 60
11.14.1 General description and considerations 60
11.14.2 Design of rainwater and stormwater tanks 61
11.14.3 Non-potable reuse rates for modelling reuse tanks in MUSIC 62
11.14.4 Tank size allowances 63
11.14.5 Achieving 80% non-potable demand for business and industrial development 64
11.14.6 Stormwater tank modelling constraints 64
11.14.7 Improving the cost effectiveness of rainwater tanks 66
11.14.8 Special requirements for private hospitals, aged care centres and nursing homes 66
11.14.9 Modelling rainwater and stormwater tanks in MUSIC 67
11.14.10 Additional stormwater tank MUSIC modelling requirements 68
11.15 Generic node 68
11.16 Hydrocarbons and oils 68
11.16.1 General removal requirements for lower risk development 69
11.16.2 Removal requirements for higher risk developments 73
12 Stormwater quality improvement devices 75
12.1 Approved stormwater quality improvement devices for use in Blacktown City 75
12.2 AtlanBasin (Atlan Stormwater) 79
12.2.1 General description 79
12.2.2 Inspections and testing 80
12.3 AtlanFilter (Atlan Stormwater) 80
12.3.1 General description 80
12.3.2 Characteristics 81
12.3.3 MUSIC modelling requirements for AtlanFilters 82
12.3.4 Specific requirements for AtlanFilters with on-site stormwater detention 83
12.3.5 On-site stormwater detention problems with AtlanFilters and solutions 84
12.4 Flowceptor (Atlan Stormwater) 84
12.5 FlowGuard (Atlan Stormwater) 86
12.5.1 General description 86
12.5.2 Conditions of approval and MUSIC modelling 86
12.5.3 Maintenance, inspection and testing 87
12.6 Spillceptor (Atlan Stormwater) 87
12.7 StormSack (Atlan Stormwater) 89
12.8 Vortceptor (Atlan Stormwater) 91

12.9 HumeCeptor (Humes) 92
12.10 HumeFilter UPT (Humes) 94
12.10.1 General description 94
12.11 HumeGard (Humes) 95
12.12 Filterra (Ocean Protect) 96
12.12.1 General description 96
12.12.2 Inspections and testing 98
12.13 Jellyfish (Ocean Protect) 98
12.13.1 General description 98
12.13.2 MUSIC modelling requirements for Jellyfish 98
12.13.3 Characteristics 99
12.13.4 Achieving the 90% average annual load reduction in total hydrocarbons 102
12.14 OceanGuard (Ocean Protect) 102
12.15 OceanSave (Ocean Protect) 104
12.16 StormFilter (Ocean Protect) 106
12.16.1 General description 106
12.16.2 Characteristics 106
12.16.3 MUSIC modelling requirements for StormFilters 107
12.16.4 Hydraulic loading rates for StormFilters 108
12.16.5 Specific requirements for StormFilters with on-site stormwater detention 109
12.16.6 On-site stormwater detention problems with StormFilters and solutions 110
12.17 CDS GPT (Civil Mart – Rocla) 110
12.18 Enviropod 200 (Civil Mart – Rocla) 111
12.19 Specific requirements for maintenance of filter cartridges 112
12.19.1 Failure mechanisms for filter cartridges 112
12.19.2 Minimum performance standard 113
12.19.3 Flow test method – individual cartridge constant head 113
12.19.4 Flow test method – individual cartridge constant flow (flow meter) 114
12.19.5 Maintenance contract 115
12.20 Requirements for approval of new stormwater quality improvement devices 115
13 Calculation of the Stream Erosion Index (SEI) 118
13.1 When to use the SEI 118
13.2 Deemed-to-comply solutions for the North West Growth Area 118
13.3 How to estimate the SEI 118
13.4 Estimating the critical flow for the receiving waterway 119

13.5 Developing SEI MUSIC models for pre and post-development 120
13.6 Estimate the mean annual flow for pre and post-development and calculating the SEI 120
13.6.1 Method 1. Generic node with high flow by-pass 120
13.6.2 Method 2. Modification of the generic node flow transfer function 121
14 WSUD signage 122
15 Flood modelling of local catchments 123
15.1 Flood impact and risk assessment 123
15.2 Categories of flood affectation 124
15.3 Design standards 124
15.3.1 Design flood/flood planning level 124
15.3.2 Minimum floor levels (residential-habitable, business/industrial) 125
15.3.3 Minimum carport/carpark level 125
15.3.4 Minimum driveway level 125
15.3.5 Minimum garage floor level 125
15.3.6 Minimum basement carpark/below-ground garage crest level 125
15.3.7 Farm sheds (farm equipment and material storage only) 125
15.3.8 Pools 126
15.3.9 Additions to existing buildings in flood areas 126
15.3.10 Building on land located 2.5 m or more below the 1% AEP flood level 126
15.3.11 Subdivision of land within riverine flood risk precinct 127
15.3.12 Subdivision of land within overland flow flood risk precinct 127
15.3.13 Building on C4 land 127
15.4 Information available from Blacktown City Council 127
15.4.1 Free maps online 127
15.4.2 Flood related information that you can purchase 128
15.4.3 Flood advice letter 128
15.4.4 Flood Control Lot Certificate 128
15.4.5 Aerial laser survey 128
15.4.6 Blacktown City Council flood model 129
15.4.7 Blacktown City Council flood model- results only 129
15.4.8 Blacktown City Council flow information 129
15.4.9 Contour plans and general pipe layout 129
15.5 General design and flood modelling guidelines 129
15.5.1 Site survey 129
15.5.2 Manning's n roughness value 130

15.5.3 Buildings, structures and extensions as obstructions 130
15.5.4 Fences as obstructions 130
15.5.5 Pit blockages 130
15.5.6 Pipe blockages 131
15.5.7 Floodplain storage and filling in the floodplain 131
15.5.8 Fencing in the floodplain – urban zones 132
15.5.9 Fencing in the floodplain – E2, E3 and E4 zones 133
15.6 Flood modelling techniques 134
15.6.1 Hydrology 134
15.6.2 Pre to post modelling and tolerances 134
15.6.3 HEC-RAS 1D 134
15.6.4 TUFLOW and HEC-RAS 2D 135
15.7 Flood emergency evacuation and management plan 135
15.7.1 General principles 135
15.7.2 Protection of people from injury 136
15.7.3 Protection of building structures from damage 137
15.7.4 Protection of building contents 137
15.7.5 Flood risk management plan 137
15.7.6 Sample flood emergency plan 138
15.7.7 Sample flood emergency plan – State Emergency Service 141
16 WSUD checklists 142
17 Glossary and acronyms 144
18 References 151

List of figures

Figure 1. MUSIC model setup for Blacktown City Council 17
Figure 2. 6-minute rainfall station comparison 18
Figure 3. Estimating fraction impervious for land-use 23
Figure 4. Treatment options available in MUSIC 28
Figure 5. Wetland zones 32
Figure 6. Example of a wetland node in MUSIC 36
Figure 7. Example of a pond node in MUSIC 38
Figure 8. Example of properties of a sedimentation basin in MUSIC 40
Figure 9. Examples of properties of a detention basin node in MUSIC 46
Figure 10. Example of hydraulic loading rate in MUSIC 47
Figure 11. Typical layout of batters and walls to allow sunlight to plants and reduce heat effects . 49
Figure 12. Example of applying the shadow requirements in bioretention basins 49
Figure 13. Example of properties of a bioretention node in MUSIC 52
Figure 14. Example of properties of permeable pavers in MUSIC 55
Figure 15. Example of TSS removal in a Vortex style CDS node in MUSIC 57
Figure 16. Swale node for MUSIC 59
Figure 17. Rainwater percentage reuse 64
Figure 18. Example of different types of purple pipes 66
Figure 19. Example of properties of a rainwater tank in MUSIC 67
Figure 20. Oil baffle configuration (no weir) 71
Figure 21. Oil baffle configuration (with weir) 71
Figure 22. Section through a full-height AtlanFilter chamber 81
Figure 23. Example of MUSIC setup for AtlanFilter 82
Figure 24. Example of MUSIC setup for Filterra 97
Figure 25. Standard Jellyfish arrangement for Blacktown City Council 100
Figure 26. StormFilter chamber arrangement 107
Figure 27. Example of MUSIC setup for StormFilters 107
Figure 28. Example of MUSIC with flow bypass 120
Figure 29. Example of MUSIC SEI setup with modified generic node 121
Figure 30. Single vane louvers and double vane louvers 133
Figure 31. FloodSafe Hawkesbury-Nepean brochure 141

List of tables

Table 1. Required percentage reductions in post development annual average load of pollutants .. 6
Table 2. Recommended 6-minute rainfall station 19
Table 3. Selected daily rainfall gauges in Blacktown City Council 19
Table 4. Recommended daily rainfall station 20
Table 5. Monthly evapotranspiration for the Sydney region 20
Table 6.Stormwater quality parameters for MUSIC Source Nodes 21
Table 7.Rainfall-runoff parameters 22
Table 8.Planting zones 34
Table 9. Filter median particle size and saturated hydraulic conductivity 55
Table 10. Green roof parameters 56
Table 11. List of currently approved stormwater quality improvement devices for use in Blacktown City - Current as of June 2026 78
Table 12. AtlanBasin Model numbers and treatable flow rates 79
Table 13. Basin pollutant removal rates 80
Table 14. AtlanBasin Generic Node input and output values 80
Table 15. AtlanFilter characteristics and controls 81
Table 16. AtlanFilter pollutant removal rates 82
Table 17. AtlanFilter Generic Node input and output values 83
Table 18 Flowceptor treatable flow rates 85
Table 19. Flowceptor pollutant removal rates 86
Table 20. Flowceptor GPT Node input and output values 86
Table 21. Approved pollutant removal rates (from SQIDEP) 87
Table 22. FlowGuard MUSIC Generic Node input and output values 87
Table 23. Spillceptor pollutant removal rates 88
Table 24. Spillceptor GPT Node input and output values 88
Table 25. Spillceptor treatment flow rates and construction specifications 89
Table 26. StormSack constraints > 300 mm 90
Table 27. StormSack constraints > 170 mm, but < 300 mm 90
Table 28. StormSack pollutant removal rates 90
Table 29. StormSack GPT Node input and output values 91
Table 30. Vortceptor pollutant removal rates 91
Table 31. Vortceptor GPT Node input and output values 92
Table 32. Vortceptor treatable flow rates 92
Table 33. HumeCeptor pollutant removal rates 93

Table 34. HumeCeptor GPT Node input and output values 93
Table 35. HumeFilter Model numbers and key design information 94
Table 36. HumeFilter pollutant removal rates 94
Table 37. HumeFilter Generic Node input and output values 95
Table 38. HumeGard pollutant removal rates 95
Table 39. HumeGard GPT Node input and output values 95
Table 40. HumeGard treatable flow rates 96
Table 41. Filterra pollutant removal rates 98
Table 42. Filterra Generic Node input and output values 98
Table 43. Jellyfish Generic Node input and output values 99
Table 44. Jellyfish Generic Node input and output values 99
Table 45. Jellyfish treatable flow rates 101
Table 46. OceanGuard constraints (type M2, L2, L3, XL2 or XL3) 103
Table 47. OceanGuard constraints (type M1, L1 or XL1) 103
Table 48. OceanGuard pollutant removal rates 104
Table 49. OceanGuard GPT Node input and output values 104
Table 50. OceanSave pollutant removal rates 105
Table 51. OceanSave GPT Node input and output values 105
Table 52. OceanSave treatable flow rates 105
Table 53. StormFilter characteristics and controls 106
Table 54. StormFilter pollutant removal rates 108
Table 55. StormFilter Node input and output values 108
Table 56. CDS pollutant removal rates 111
Table 57. CDS GPT Node input and output values 111
Table 58. Approved Enviropod models 111
Table 59. Enviropod pollutant removal rates 112
Table 60.Enviropod GPT Node input and output values 112
Table 61. Items/factors needed for Blacktown City Council assessment of SQIDs 115

1 Introduction

This is a technical handbook and is intended for the use of civil engineers experienced in the design and construction of water sensitive urban design (WSUD) systems, infrastructure, floodplain assessment and modelling.

Assessments and modelling, as noted in this handbook, requires an engineer to have both Chartered Professional Engineer (CPEng) and National Engineering Register (NER) qualifications and be practicing in their area of expertise. In some specific areas such as water conservation, specialists in irrigation and ultraviolet (UV) treatment may also be required.

Controls

Blacktown City Council's *Development Control Plan (DCP) Part J Water Sensitive Urban Design and Integrated Water Cycle Management 2015* describes Blacktown City Council's development controls relating to the management of water. Throughout this handbook, it will be referred to as Part J.

Based on Part J, the principles and methods highlighted within this handbook are applicable throughout the Blacktown local government area. This includes the Blacktown growth centres and Western Sydney Employment Area (WSEA).

Part J includes the following development controls:

- water conservation
- water quality
- on-site stormwater detention (OSD)
- groundwater assessment
- stream erosion index (SEI).

To demonstrate how a proposed development complies with Blacktown City Council's development controls, it is necessary to identify which controls are applicable to your development site. This depends on:

- the location of the proposed development
- the land use zone
- the type of development proposed.

Using the handbook

This handbook provides a detailed explanation of the methods and assumptions that we require applicants to adopt when preparing their engineering designs and when carrying out Model for Stormwater Improvement Conceptualisation (MUSIC) or other water quality modelling.

Blacktown City Council encourages applicants to use the Small-Scale Stormwater Quality Model (S3QM) tool to determine which development controls apply to their site. It is important to note that S3QM shall be used to model the following types of development only:

- smaller developments less than 4,000 square metres
- developments with simple treatment trains such as roof to rainwater tank and impervious areas to a bioretention system with or without a gross pollutant trap (GPT) upstream
- developments with 1 or 2 catchments and where there is no flow from one catchment to another.

S3QM is not suitable for modelling any sites which involve proprietary treatment devices such as filter cartridges.

To access S3QM: https://www.s3qm.com.au

When applicable development controls have been determined for a proposed development, the applicant is required to demonstrate, using sound mathematical modelling, that their proposed design satisfies Blacktown City Council's development controls. There are several different ways that applicants can demonstrate compliance with Blacktown City Council's development controls.

The handbook also references flood assessment and floodplain management. For this, Blacktown City Council's *Development Control Plan (DCP) Part A Introduction and General Guidelines* should be referred to.

Urban development will increase water usage, alter flow regimes in our local waterways and increase pollutant loads in stormwater runoff. Blacktown City Council has adopted a WSUD approach to the management of water across our city. As such, sound policies are required to ensure a sustainable environment for present and future generations.

2 Compliance pathways for WSUD (excluding flooding)

Blacktown City Council has simplified the development application (DA) submission pathway and created:

- a Precinct scale water quality offset scheme where applicants can contribute to the cost of regional water quality treatment schemes in lieu of undertaking works on-site
- an online S3QM tool that provides information on a lot, applicable controls under Part J and helps to design a complying solution to meet your requirements
- MUSIC source nodes and treatment nodes, including proprietary device nodes specific to our city so you can use MUSIC to design your stormwater treatment train.

2.1 Precinct scale water quality offset scheme for infill development

The water quality offset scheme enables developers to pay a contribution to Blacktown City Council in lieu of undertaking water quality treatment works on-site. Water quality treatment will be provided in a regional facility managed by Blacktown City Council.

You can determine if your property is within the scheme boundary by referring to the map at Appendix 1 in Part J of Council's Development Control Plan (DCP), or by using the S3QM tool. The scheme includes a Section 7.11 Contribution Plan No.19 area and a voluntary area. In the voluntary area,d you can choose to opt into the Precinct scale water quality offset scheme by entering into a Voluntary planning agreement (VPA) with Blacktown City Council.

The scheme applies to all development proposals/applications that requires water quality controls under Part J, except for large lots (greater than 40,000 square metres) of land zoned business and industrial, which must meet their water quality control requirements on-site. Smaller industrial and business lots (between 2,000 square metres to 40,000 square metres) that wish to opt into the voluntary planning scheme, or those that are required to contribute to a plan, must also: install basic treatment measures in the form of an approved gross pollutant and hydrocarbon trap, in addition to making a contribution.

2.2 S3QM deemed-to-comply tool

S3QM is an online deemed-to-comply assessment tool which has all of Blacktown City Council's mathematical modelling requirements hardwired. S3QM is based on the MUSIC model and produces practically identical results. This tool can be used to assess on-site stormwater detention, water conservation and water quality requirements, based on location. It then determines the required size of the treatment. It includes a selection of devices, for example gross pollutant traps (GPTs), filter baskets, rainwater tanks and bioretention systems. Other devices may be added to S3QM in the future.

S3QM can be used to model the following types of development:

- smaller developments less than 4,000 square metres
- developments with simple treatment trains such as roof to rainwater tank and impervious areas to a bioretention system, with or without a GPT upstream
- developments with 1 or 2 catchments and where there is no flow from one catchment to another.

S3QM is not suitable for modelling any sites which involve proprietary treatment devices such as filter cartridges.

The applicant does not need to do any further modelling to demonstrate compliance with the development controls. All that is required is an S3QM compliance certificate and a matching set of drawings which document the compliant design.

Note: The S3QM compliance certificate must be submitted to Council via the 'submit to Council' button within S3QM. Council will not accept certificates of compliance that have been 'cut and paste' from S3QM into a stormwater management report in the submission of the DA.

To access the S3QM tool go to www.s3qm.com.au

2.3 Model for urban stormwater improvement conceptualisation (MUSIC)

MUSIC was developed by the Cooperative Research Centre for Catchment Hydrology (CRCCH, 2001). MUSICX is an upgraded version of MUSIC. Throughout this handbook it will be referred to as MUSIC.

MUSIC predicts the performance of water sensitive urban design (WSUD) systems and assists in the planning and design of stormwater strategies. Blacktown City Council requires that a MUSIC model is developed for sites where water quality, water conservation and on-site stormwater detention controls identified in Part J apply.

Blacktown City Council uses MUSIC to assess the validity of proposed water quality treatment and harvesting strategies to meet the targets in Part J. Please ensure a submission is prepared in an up-to-date MUSIC version.

This handbook ensures a consistent and uniform approach to water quality and harvesting modelling within our city. It provides specific guidance on rainfall and evaporation inputs, source node selection, rainfall runoff parameters, pollutant generation parameters and treatment nodes.

2.3.1 MUSIC-*link*

Blacktown City Council has worked with eWater to establish MUSIC-*link*. To use MUSIC-*link*:

- open MUSICX version 1.40.0 or later (MUSIC 6 is no longer accepted)
- on the top menu, choose 'Configure' and click on the Blacktown City Council MUSIC-link. If you are unfamiliar with Blacktown's adopted nodes and data, click on the 'Blacktown City Council Default Nodes' to download a model with data specific to our city. This will include a climate file, source nodes and the various treatment nodes.

These treatment nodes include proprietary devices assessed by Blacktown City Council as suitable for use within our city. Note: The pollutant removal rates may vary from the treatment rates on the manufacturer's website or elsewhere.

Blacktown City Council has reviewed the use of the standard MUSIC nodes and where applicable, adopted lower pollutant removal rates (k factors) for a number of treatment nodes. This includes detention basins, sedimentation basins, media filtration and swales.

Alternatively, click on 'Blacktown Development.mlb' to go to a blank screen, populated with Blacktown climate, source nodes and the amended standard MUSIC treatment nodes.

Some treatment systems in MUSIC can be constructed but will not be counted as meeting your performance targets. These include:

- infiltration systems (not suitable for heavy clay soils found locally)

- media filtration (for example, sand filters in subsurface tanks)
- buffers.

These systems are not justified as a treatment system in our city for various reasons including:

- excessive pollutant removal in MUSIC for a primary treatment system
- the results independent of slope or vegetation cover
- heavy clay soils with very low hydraulic conductivity
- the difficulty to maintain resulting in ineffective performance.

MUSIC-*link* will produce a report outlining the effectiveness of the model (for example, if it has reached the minimum target pollutant removal rates). It will also provide the total area, percentage site impervious and whether individual devices are within Blacktown City Council guidelines.

Where there are differences in the model to Blacktown City Council's standards there is an opportunity to explain within the MUSIC-*link* report. For example, the rainwater usage target is 80% directed at business/industrial use. However, it will give an error in residential developments as the reuse is typically around 40%. The report would justify the lesser rate, as residential development is subject to BASIX and the 80% does not apply.

There is a help guide found at Using MUSIC-*link* for Blacktown City Council:

https://toolkit.eWater.org.au/Dropbox/music/metadata/AuthorityData/Blacktown%20City%20Council/BlacktownCityCouncil-UsingMUSIC-link.pdf.

3 Stormwater management report

Part J outlines Blacktown City Council's requirements for water quality treatment, water conservation measures, erosion sediment and pollution controls, on-site stormwater detention (permanent or temporary), and the stream erosion index (SEI). A Stormwater management report comprehensively sets out the assumptions and modelling outcomes to ensure a prompt assessment of a DA.

The Stormwater management report should include:

- water quality modelling report
- water conservation report
- stream erosion index (where applicable)
- on-site stormwater detention
- flood report – see Chapter 15 of this handbook
- GPT calculations.

3.1 Water quality modelling report

A Water quality modelling report is required under Part J for all development that meets the water quality controls through an on-lot approach. For a development that uses a deemed-to-comply solution (S3QM), there is no need to submit a Water quality modelling report. Refer to Chapter 4 for S3QM submission requirements.

The water quality targets under Part J and principally the *Blacktown Growth Centres Development Control Plan* (DCP) are noted in Table 1.

Pollutant	Post development pollutant reduction targets (percentage of post development annual average load)
Gross pollutants (GP)	90
Total suspended solids (TSS)	85
Total phosphorous (TP)	65
Total nitrogen (TN)	45
Total hydrocarbons	90

Table 1. Required percentage reductions in post development annual average load of pollutants

The Water quality modelling report should include:

- a description of the stormwater treatment train that is proposed to meet the reduction targets in Table 1
- a MUSIC catchment plan showing land use types (for example roof and driveway) overlain by a plan showing what areas go to which treatment devices and any areas of bypass. Ensure the plan size is large enough to demonstrate all the elements and that the whole site is covered including any areas of bypass. A simple land use plan is not sufficient. For complicated arrangements, separate land use and water quality device plans should be provided
- details of what devices are proposed for treatment

- calculations to justify any sizing of any proprietary devices, including the determination of the high flow bypass flow
- specification of a minimum 4 average number of exceedance per year (4 EY) (3 month) flow for gross pollutant traps upstream of a bioretention basin, or a wetland
- specification of a minimum 2 EY (6 month) flow for gross pollutant traps not part of a local treatment train
- the use of specific programs such as Personal Computer Storm Water Management Model (PCSWMM) for HumeCeptor. Include the results in the body of the report and the PCSWMM output as an attachment
- details of the treatment train and what devices flow into which other devices
- details of how the hydrocarbon removal targets will be achieved. Refer to Section 11.16 for empirical solutions
- a screen shot of the MUSIC model layout in the report if MUSIC is used
- MUSIC-*link* printout as an attachment (if MUSIC-*link* is used)
- information on sites > 2,000 square metres subject to the water quality offset scheme requiring an approved gross pollutant and hydrocarbon trap treating the 2 EY (6 month) flow. Specify the size and type of the unit in accordance with Section 11.16
- information on rainwater tanks that are specified as part of the on-site water quality works. Detail all non-potable uses such as landscape watering areas and rates. Detail the number of toilets/urinals and rates used. For variable uses such as vehicle washing, general wash down and dust suppression, provide details of all rates and assumptions used
- a detailed non-potable water supply and irrigation plan for non-potable water uses on the site including all toilet/urinal flushing and landscape watering (and other non-potable uses as appropriate). This is to justify and ensure that the pollutant removal through the rainwater or stormwater tank can be achieved and is required for submission prior to construction certificate. See details below in Section 3.3
- any other information that will assist in the assessment of the application.

3.2 Voluntary planning agreement or S7.11 development

For sites subject to the Precinct scale water quality offset scheme (Section 7.11 Contribution Plan No. 19), the report is to simply state that this site is subject to this scheme or the intention to enter into a voluntary planning agreement (VPA). To avoid delays in executing the VPA, start negotiations with Blacktown City Council through council@blacktown.nsw.gov.au early in the DA process to enter into the VPA prior to determination of the DA.

For business and industrial development which meets its water quality obligations through a VPA or S7.11 contribution and which is larger than 2,000 square metres there is an obligation to have an on-site gross pollutant trap (GPT) capable of trapping 90% of hydrocarbons and gross pollutants. Refer to Section 11.16 for detailed requirements.

To identify how the development will comply with this control, the water quality section of the Stormwater management report is to be submitted. It should note the choice of devices and the GPT sizing calculations.

If the applicant does not enter a VPA prior to determination and should the proposed be approved, the requirement to enter a VPA will become a deferred commencement condition. Alternatively, it may result in the refusal of the proposal. All VPAs need to be placed on public exhibition for 30 days which may cause delays in the application process.

3.3 Water conservation report

Business or industrial sites (or the business component of mixed-use development) must supply 80% of their non- potable demand using non-potable water. This can be assessed using the Node Water Balance feature in MUSIC or the S3QM. This may principally include roof rainwater but may include other non-potable sources. See Section 11.14.3 for further details. The Water conservation report is to detail whether the target has been achieved and if not, provide justification and consider alternative sources.

For residential sites within the water quality offset scheme, the site must meet NSW BASIX (https://pp.planningportal.nsw.gov.au/basix-water-section) requirements. The Water conservation report can simply note that the site is subject to BASIX.

For residential sites which have an on-site water quality obligation and where the rainwater tanks are being used as part of the water quality treatment train, then the tank size has to match the tank size specified in BASIX (if it has been specified). The water demand must match the requirements of Section 11.14.4.

To ensure that the specified non-potable requirements are achieved, an experienced hydraulic engineer must prepare and certify a detailed Non-potable water supply and irrigation plan for non-potable water uses on the site. This must include all toilet/urinal flushing, landscape watering and other non-potable uses. This must be provided prior to applying for the construction certificate. As a minimum, the plan is to show the pipe and tank arrangement including schematic that includes:

- a first flush or pre-treatment system (typically 0.2 L/m^2 of roof area)
- plumbing arrangement compliant with latest Sydney Water plumbing regulations
- a control panel with warning light to indicate pump failure
- a Sydney Water approved mains water bypass for toilet flushing or other internal uses
- flow meters on the bypass line and the pump outflow line, to measure the volume of non-potable and potable water usage
- an irrigation controller for landscape watering, allowing for seasonal variations
- a plan showing the proposed irrigation system arrangement
- treatment measures to ensure the water that is supplied is fit for purpose - high risk applications including aged care and childcare facilities are not recommended
- warning signs fitted to all external taps using rainwater or stormwater - see sign details in Section 14.

Every development must ensure that every non-potable water source, including stormwater, is fit for purpose. This may require additional treatment such as UV disinfection or alternative treatment mechanisms. Where disinfection is required, the information in Table 6.4 of Managing urban stormwater: Harvesting and reuse 2006 by the NSW Government will apply.

Prior to occupation, certification will be needed stating that the requirements of the detailed non-potable water supply and irrigation plan for non-potable water uses have been satisfied.

Architectural plans are required for buildings, or parts of buildings, that are not affected by BASIX. This is to demonstrate compliance with the minimum standards defined by the Water Efficiency Labelling and Standards (WELS) Scheme for any water use fittings. Minimum WELS ratings are:

- 4-star dual-flush toilets
- 3-star showerheads
- 5-star taps (for all taps other than bath outlets and garden taps)

- 3-star urinals
- 3-star water efficient washing machines and dishwashers.

3.3.1 Determination of actual non-potable reuse and percentage reuse

For small scale residential development, Blacktown City Council's WSUD standard drawing 6012 provides a typical arrangement for the rainwater tank and ancillary equipment. There is no requirement to determine non-potable usage or percent reuse for individual allotments.

For business and industrial development and larger multi-storey residential development using landscape watering, there is a requirement to provide regular reports (at least annually) on non-potable water usage and non-potable water percentage reuse. These reports should be sent to Blacktown City Council's WSUD Compliance Officer at wsud@blacktown.nsw.gov.au. The method of determining the non-potable reuse, the reuse percentage for business and industrial development and larger multi-storey residential development using landscape watering, depends on the configuration of the reuse tank. This includes how potable water is added to the system and how this is distributed to the various non-potable end uses.

3.4 Stream erosion index (SEI)

The SEI is used to measure the impacts of urbanisation on the local creeks and streams that lead to both higher peak flows and higher volumes of runoff. These factors lead to an increase in stream erosion.

Under Part J there is no specific requirement to demonstrate compliance with the SEI target. This is because it is deemed-to-comply due to the OSD or regional detention requirements within the older areas of our city.

The requirement to calculate the SEI applies to areas of the growth centres and Western Sydney Employment Areas where on-lot water treatment is required. Typically, this is for areas zoned R3 or higher; business and industrial development and some SP2 lands.

In the above areas (excluding the Little Creek catchment), Blacktown City Council will accept the following approaches for a deemed-to-comply solution in lieu of providing the full SEI calculations and SEI MUSIC models:

- Residential development R3 or R4:
  - bioretention on-lot to meet the water quality targets
  - a rainwater tank that supplies a minimum of 80% non-potable demand for all landscape watering including the bioretention basin.
- Business and industrial development and some SP2 lands:
  - full bioretention on-lot to meet the water quality targets
  - a rainwater tank that supplies a minimum of 80% non-potable demand for all on-site demand including toilet reuse, other on-site demand (such as vehicle washing) and all landscape watering including the bioretention basin.

Where a deemed-to-comply solution is not available, Part J requires the post development duration of stream forming flows will be no greater than 3.5 times the pre-developed duration of stream forming flows, with a stretch target of 1 (Note: Little Creek SEI < = 1). The method of calculation is detailed in Section 13.

To achieve a SEI <= 1 within the Little Creek catchment is very difficult. A scheme has been developed under the contribution plan to direct the frequent flows to an alternate catchment thereby protecting Little Creek from the frequent stream forming flows. When this is in place, the SEI <= 3.5 will then apply.

3.5 On-site stormwater detention

On-site stormwater detention (OSD) controls are required dependent upon the locality of the development site and the types of development which is defined in Part J of the Blacktown City Council *Development Control Plan (DCP) 2015*.

In relation to the locality, newer release areas have S7.11 stormwater infrastructure (i.e. regional detention basins) and therefore a permanent OSD is typically not required in these areas. An OSD is usually required in older parts of our city which have no S7.11 stormwater infrastructure. Some exemptions may be applied in accordance with Part J.

Permanent and temporary stormwater infrastructure

In areas where OSD is not required, a contribution under Section 7.11 of the *Environmental Planning and Assessment Act 1979* may be required to assist Blacktown City Council with the provision of S7.11 stormwater infrastructure.

Permanent OSD is required for Sydney Water sites and NSW Government development (for example, schools, prisons, and hospitals). Alternatively, a specific written agreement can be negotiated with Blacktown City Council for payment of the appropriate Section 7.11 contribution. In addition, there are some sites within the growth centres and Western Sydney Employment Areas that require a permanent OSD.

Temporary stormwater infrastructure may be required for the growth centres or Western Sydney Employment Areas where:

- the proposed S7.11 stormwater infrastructure has not yet been built
- the downstream drainage system connecting a property to a S7.11 stormwater infrastructure has not been constructed.

The temporary stormwater infrastructure requirements are considered satisfied with the use of Blacktown City Council's S3QM and WSUD standard drawings.

Development type

Some types of development do not require an OSD while other types do.

Generally, a smaller development is not required to have an OSD, however there are exceptions to this in the Parramatta River catchment. There is also a deemed-to-comply solution for small business and industrial development which is less than 500 square metres in area.

When residential developments and business and industrial development have a development footprint of greater than 150 square metres, they are required to have an OSD. Further clarification can be found in Part J of the Blacktown City Council *Development Control Plan (DCP) 2015*.

OSD design and S3QM deemed-to-comply tool

The S3QM deemed-to-comply tool can be used to firstly determine if your proposed development needs to have an OSD. If it is required, the tool can calculate the size and characteristics of the OSD storage for sites of any size. The detailed design requirement of the OSD can be found in Council's *Engineering guide for development 2005* and WSUD standard drawings 6101-6103.

There are set discharge and storage rates for all of our city. This includes::

- an environmental control at 40 L/s/ha (50% AEP) which requires storage at a rate of 300 cubic metres/ha
- a flood control at 190 L/s/ha (1% AEP) which requires storage at a rate of an additional 155 cubic metres/ha for a total of 455 cubic metres/ha.

OSD can be contained in basins, above ground structures and below ground structures. However, Blacktown City Council does not accept any above ground basin that exceed 600 millimetres depth as part of childcare development. For the layout of conventional detention basins and specifications, see Blacktown City Council's WSUD standard drawings 6101-6103.

Applicants must provide an OSD catchment plan (including calculations) showing the areas draining to the detention basin and clearly show areas of bypass, including any upstream swale areas.

Exceptional cases

Under the *Local Government Act 1993*, Blacktown City Council is prohibited from approving a development that may have adverse impacts. Should Blacktown City Council become aware of an adverse drainage impact caused by the proposed development, we may require an OSD. This includes instances where the development is located in a catchment which does not normally require an OSD.

Examples of where this may apply, include:

- areas subject to local runoff and the unrestrained site discharge will contribute to overland flows
- sites with limited downstream capacity such as an existing inter-allotment drainage line
- discharging to a Blacktown City Council drainage system against the natural fall of the land or into a differing catchment.

3.5.1 Special modelling requirements for the growth areas or other DCPs that allow pre to post modelling

Where agreed to by Blacktown City Council, the following parameters can be considered for computer modelling. This is based on satisfying the pre-to-post modelling for all storms from 50% AEP (2 year ARI) to 1% AEP (100 year ARI), with particular reference to the 5% AEP (20 year ARI).

Based on the 1987 version of *Australian Rainfall and Runoff (ARR)* the parameters for the pre-development catchments are:

DRAINS

ILSAX or ILCL hydrology and kinematic wave can only be used for sites where the length of overland flow paths is less than 500 metres. Where the length of the overland flow path is longer than 500 metres, use Laurenson Hydrology within DRAINS using the RAFTS parameters identified in the RAFTS section below:

- pervious area depression storage 15 millimetres
- travel time use Kinematic Wave with n* retardance coefficient. Allow:
    - open space (natural bushland) n* = 0.3
    - rural development (no dams) n* = 0.4
    - rural development (with dams) n* = 0.5

- the post developed criteria are as per *Blacktown City Council's Engineering guide for development 2005*.

RAFTS

- Blacktown City Council only uses Australian Representative Basin Model (ARBM) parameters
- Depression Storage Capacity (DSC) (initial loss) 15 millimetres
- allow for a minimum Pern roughness of 0.06
- allow for an existing 5% site imperviousness
- the post developed criteria are as per *Blacktown City Council's Engineering guide for development 2005*.

Calculations/spreadsheets are to be provided to justify the stage discharge relationship used in the DRAINS/RAFTS model overflow based on the dual orifice combination for the detention basin.

3.6 GPT calculations

The Stormwater management report is to include relevant GPT calculations. These would typically include:

- defining the obligation for a GPT, if the development is greater than 2,000 square metres and off-setting its water quality obligations, it requires a GPT a 6 month return period storm design objective. If it is carrying out works on-site, it then has a 3 month return period design objective
- modelling results which define the relevant design flow
- the proposed GPT, including the model number and treatable flow rate of the unit
- the sizing of the weir or diversion chamber and noting if there are backwater impacts that need to be addressed. For example, if a bioretention basin is located downstream, demonstrate that the top level of the extended detention depth remains below the invert of the GPT. This ensure the bioretention basin does not drown the GPT or force it into bypass prematurely.

4 Small scale stormwater quality model (S3QM)

When submitting a DA which includes reliance on an S3QM certificate, it is also necessary to include with your submission (where applicable):

- an OSD catchment plan showing the areas (and calculations) draining to the OSD basin/tank and areas of bypass. These must match the values shown on the S3QM certificate
- a Water quality catchment plan showing areas draining to any water quality treatment device(s) and any areas of bypass. This includes identifying the differing treatment train catchments relied upon in S3QM
- any necessary calculations supporting the design. For example, GPT calculations showing that the GPT has been sized in accordance with the design standards specified in this handbook
- for water conservation, detail the number of toilets and justify where the effective number of toilets is reduced (for example, working 5 days a week). More details can be found in section 11.14 in this handbook
- submissions must consider the water demand of the toilets. High-use toilets shall be chosen in accordance with section 11.14 of this handbook. For example, a toilet in a hotel lobby would be a high-use toilet while a toilet within a private room would be considered a low-use toilet. S3QM enables modelling of both high and low water-use toilets.

5 Groundwater assessment report

Where triggered by excavation or filling, a desktop Groundwater assessment report is required in accordance with Section 4.6 of Part J. This must be provided from a geotechnical engineer registered on NER or a hydrogeologist. Where the desktop Groundwater assessment report determines there is potential for interaction with groundwater, a Groundwater management plan that meets Blacktown City Council's DA requirements must be prepared.

The applicant must submit the Groundwater assessment report (and Groundwater management plan if applicable) with the DA.

The applicant must demonstrate (as noted in Sections 5.2 - 5.5) that there are no adverse impacts on:

- surrounding or adjacent properties
- infrastructure
- the environment
- groundwater dependent ecosystems.

For further information refer to the *Groundwater Management Handbook* (Sydney Coastal Councils Group 2016) as it applies to areas away from the coast.

5.1 Applicable development

The groundwater controls in Part J apply to:

- all development that is equal to, or less than, 40 metres from the top of the bank of a watercourse that has a cut or fill proposed which is equal to or greater than 1 metre over the existing pre-developed surface
- all development that is greater than 40 metres from the top of the bank of a watercourse that has a cut or fill proposed which is equal to or greater than 1.5 metres over the pre-existing developed surface
- any development that impacts on groundwater quantity and works associated with groundwater which may require licence under Sections 89-91 of *Water Management Act 2000.*

5.2 No adverse impact to surrounding or adjacent properties

Where below ground structures are close to each other (typically less than 3 metres), provision must be made for the natural flow of groundwater to be included:

- in the design of the perimeter
- through drainage to offset any restriction to flow.

Such restriction may otherwise incur a localised or possibly wider impact.

If the nature of construction methods or the bulk of a below-ground structure creates an impediment to natural flow paths, artificial drains may be used.

Perimeter, through drainage or artificial drains may only be used where it is demonstrated that the natural groundwater flow regime is restored both up-gradient and down-gradient of the site. Further, there must be no adverse effects on surrounding property or infrastructure.

Where zonings would allow future basement construction on either side of the development, allowance will be made for such future works as part of the application.

5.3 No adverse impact to infrastructure

Blacktown City Council will not permit the discharge of any intercepted groundwater to its infrastructure, such as concrete pits and pipes. This discharge is prohibited if it will have an adverse impact on the maintenance or lifespan of the infrastructure.

Such discharge may include various contaminants including flow off acid-sulphate soils, or otherwise, very low pH or saline flows.

Where such flow would otherwise be intercepted, it is to be directed to a sewer through a trade waste agreement (subject to approval of Sydney Water). Alternatively, tank a basement or design a retaining wall to be self-supporting under saturated conditions.

Blacktown City Council will generally not allow the discharge of groundwater flows to the kerb. However, we will allow the discharge of infrequent clean and uncontaminated groundwater to Blacktown City Council pits.

5.4 No adverse impact to the environment

Blacktown City Council will not permit the discharge of any intercepted groundwater to its waterways that will have an adverse impact on the environment. This includes pollutants that will adversely impact the health of these systems including the discharge of highly saline groundwater.

Where such flow would otherwise be intercepted, it is to be directed to a sewer through a trade waste agreement (subject to approval of Sydney Water). Alternatively, tank a basement or design a retaining wall to be self-supporting under saturated conditions.

Salinity is usually a result of elevated concentrations of sodium and chloride, although other salts may also contribute. The salinity of groundwater is usually expressed in units of parts per million (ppm) or milligrams per litre (mg/L) of total dissolved solids (TDS). A similar measurement, electrical conductivity (EC), defines the ability of the fluid to transmit an electrical current. Values of EC are commonly reported in microSiemens per centimetre (µS/cm).

Any discharge > 1,000 ppm of salt must discharge to sewer.

5.5 No adverse impact to groundwater dependent ecosystems

Blacktown City Council will not permit any adverse impact to groundwater dependent ecosystems. The NSW Government has identified a number of known groundwater dependent ecosystems in our city. Refer to mapping conducted by the NSW Department of Planning and Environment:

- Sydney metro (including Georges and Parramatta River catchments): https://datasets.seed.nsw.gov.au/dataset/hevae-vegetation-groundwater-dependent-ecosystems-sydney-metro
- Hawkesbury – Nepean: https://datasets.seed.nsw.gov.au/dataset/hevae-vegetation-groundwater-dependent-ecosystems-value-hawkesbury-nepean

5.6 Design and maintenance of groundwater management systems

Any groundwater management systems proposed are to have a minimum design life of 50 years. Details of the method of construction must be provided where construction occurs on a hillside and involves the construction of permanent structures other than piles or footings below the water table.

All components of the structure, including subsoil drainage, must be located entirely within the property boundary. The groundwater regime is to be maintained as close as possible to the pre-development condition during the construction and operational phases of any development. Construction techniques, where possible, will eliminate the need for dewatering.

All groundwater management activities, including monitoring, must be conducted in line with the groundwater assessment and management plan and as agreed to by Blacktown City Council.

6 MUSIC model setup

MUSIC is developed by ewater™. MUSIC predicts the performance of water quality management systems and assists in the planning and design of stormwater strategies.

There are several steps in running a MUSIC model, as summarised in Figure 1.

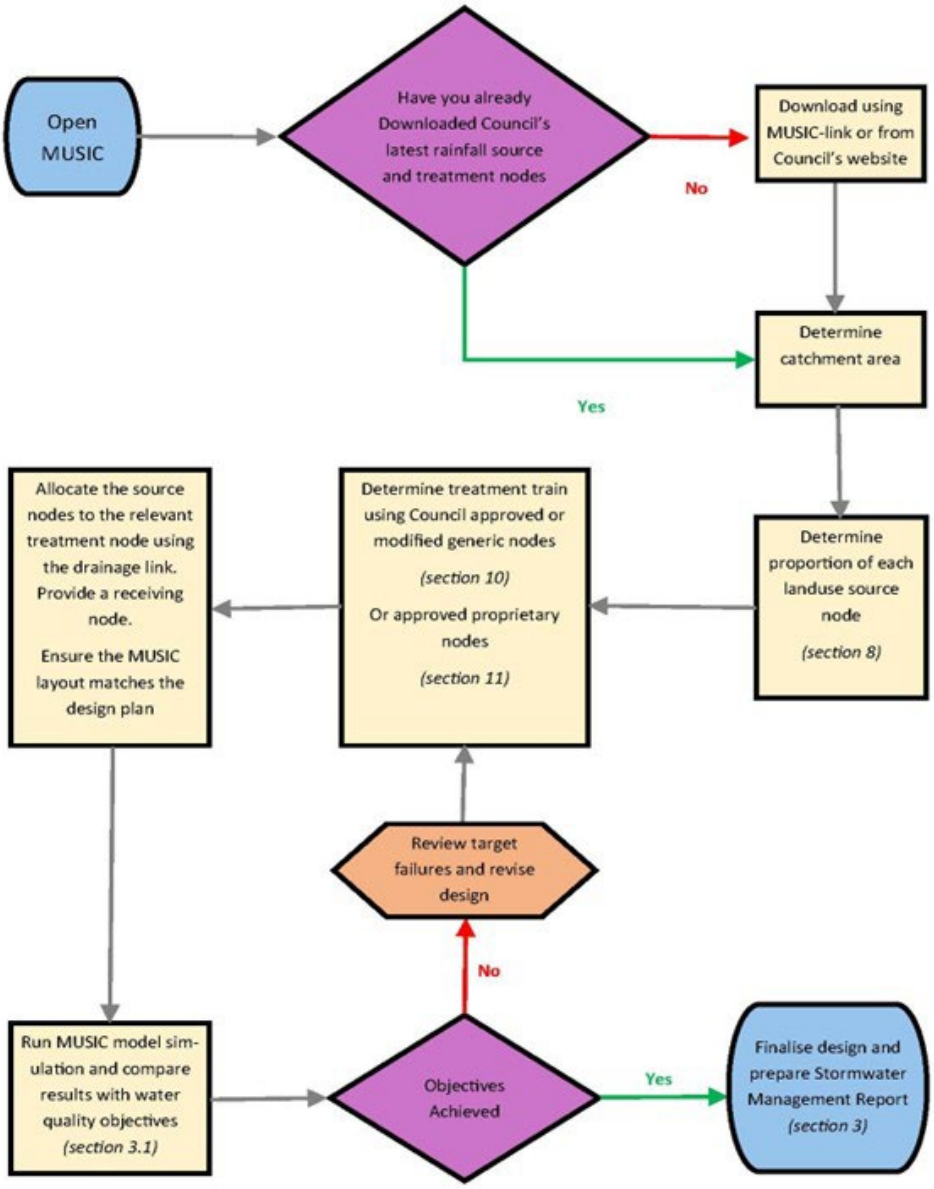


Figure 1. MUSIC model setup for Blacktown City Council

7 Rainfall and evaporation inputs

Blacktown rainfall is typically 700 to 900 millimetres per year, with maximum rainfall in summer and minimum in winter.

Stormwater runoff (represented as surface runoff and baseflow) is generated in MUSIC through the interaction of rainfall, evapotranspiration and the MUSIC Rainfall-Runoff Model - see eWater MUSIC User Guide for a full description of Rainfall-Runoff Model. The following sections outline Blacktown City Council's preferred rainfall and evapotranspiration datasets.

7.1 Rainfall data for water quality modelling

Blacktown City Council requires rainfall simulation be adopted for modelling, using:

- continuous simulation of a minimum of 10 years
- 6-minute time step to allow for the appropriate definition of storm hydrograph movement through small-scale treatment measures such as vegetated swales and bioretention systems.

To ensure a consistent approach to modelling, Blacktown City Council has identified an appropriate rainfall station for our city and periods of modelling to be utilised within the MUSIC model. Two 6-minute data stations were investigated for their suitability and they are located at:

- 067033 Richmond RAAF Base, located approximately 8 kilometres north-west of the city
- 067035 Liverpool (Whitlam Centre) located approximately 11 kilometres south of the city.

Rainfall data from each of these stations was compared to daily data available at Blacktown (gauge no. 067059), to see which bore a closer resemblance to rainfall conditions within our city. A common period was compared for all stations: 1964 to 1992. The results of this investigation are shown in Figure 2.

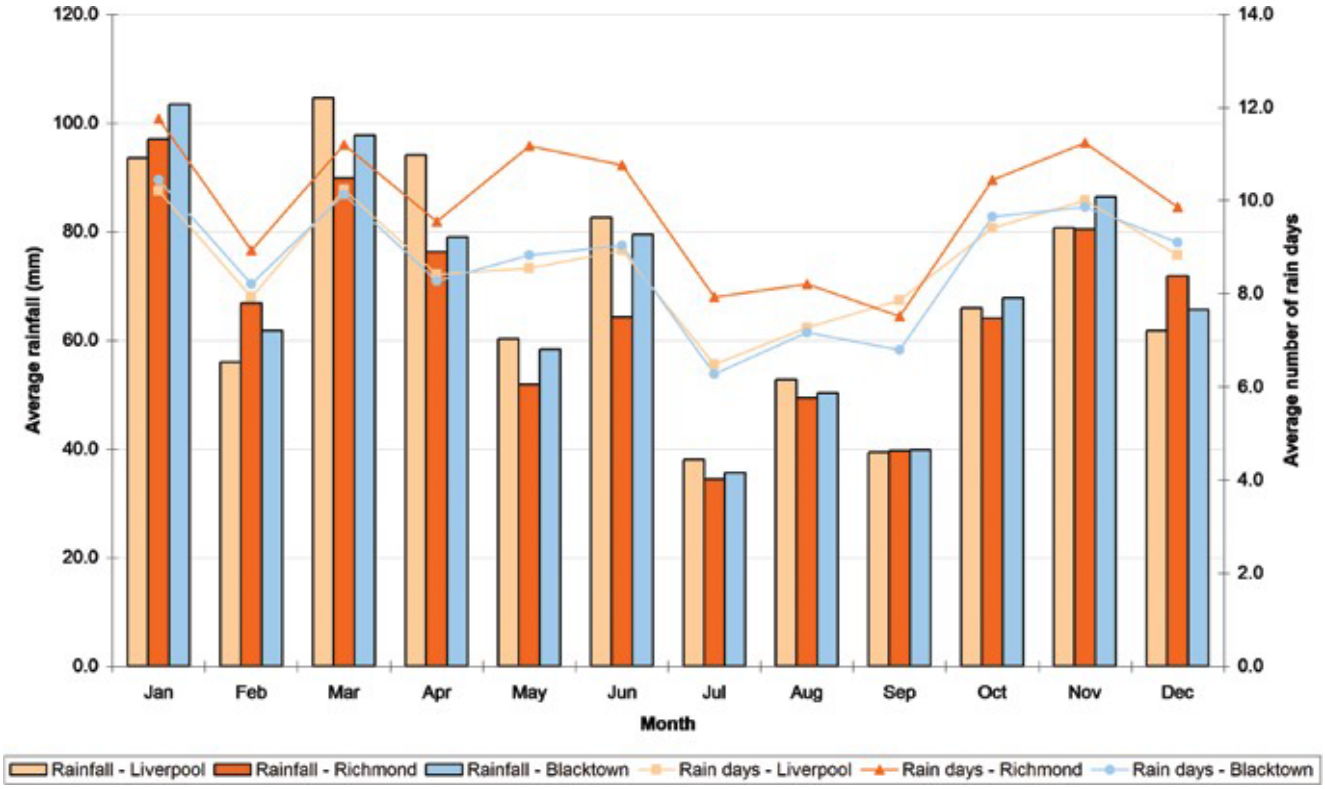


Figure 2. 6-minute rainfall station comparison

Both the Liverpool and Richmond stations provide a reasonable match to Blacktown in terms of average monthly rainfall. However, the Liverpool data matches Blacktown's rain day pattern better than the Richmond data.

A reasonable length of record is available from Liverpool, with 6-minute records starting in 1965 and continuing to 2001 (with 1 significant gap during 1978 to 1980). In 2001, the station was closed but was replaced with Station 067020 Liverpool (Michael Wenden Centre).

The recommended 6-minute rainfall station for use within our city is 067035 Liverpool (Whitlam Centre). Blacktown City Council requires all stormwater quality modelling in MUSIC to be undertaken using this data. A modelling period of 1967 to 1976 is recommended because the annual rainfall is representative of the long-term average.

Table 2 includes details of the recommended data.

Rainfall station	Modelling period	Annual rainfall (mm)
067035 Liverpool (Whitlam Centre)	1967 to 1976	857

Table 2. Recommended 6-minute rainfall station

7.2 Rainfall data for hydrologic modelling

Blacktown City Council requires rainfall simulation to be adopted for hydrologic assessment modelling, major stormwater harvesting and stormwater storage design sizing on a catchment basis, using:

- continuous simulation of 20 years
- daily time step for simulating rainwater/stormwater storage sizes and estimating supply reliability.

A number of daily rainfall stations were investigated for use, as shown in Table 3. The gauges investigated were those with longer available records.

Station	Approximate location in the city	Data availability	Mean annual rainfall (mm)	Mean number of days per year with equal to or greater than 1 mm rain
067059 Blacktown	Central	1963 to 1993	854	84
067076 Quakers Hill Treatment Works	Central	1948 to date	851	77
067016 Minchinbury	South west	1901 to 1970	778	59
067026 Seven Hills (Collins Street)	East	1950 to date	926	86

Table 3. Selected daily rainfall gauges in our city

To provide a consistent approach to modelling, Blacktown City Council has identified 2 appropriate daily rainfall stations for our city and periods of modelling to be utilised within the MUSIC model. The preferred station is 067059 Blacktown, due to its longer record of good quality data (1963 to 1993). However, 067076 Quakers Hill Treatment Works is also acceptable for the years specified due to its location within the catchment. The 1971 to 1992 period has been recommended to avoid significant gaps in the data.

The recommended daily rainfall stations are shown in Table 4. For sub-daily simulation, the Liverpool rainfall station must be used. However, Liverpool is not recommended for daily simulation as:

- there is a gap in the data from 1978 to 1981 and the 1981 to 2001 period (the longest unbroken period of record available)
- it has a relatively low average annual rainfall.

Minchinbury and Seven Hills gauges are not recommended as they exhibit average rainfall conditions somewhat different to those recorded at Blacktown.

Rainfall station	Modelling period	Mean annual rainfall (mm)
067059 Blacktown (preferred)	1963 to 1993	854
067076 Quakers Hill Treatment Works	1971 to 1992	832

Table 4. Recommended daily rainfall station

7.3 Potential evapotranspiration (PET) data

Blacktown City Council requires the following potential evapotranspiration (PET) data in MUSIC:

- where available, local PET information is preferred. In most cases, local data will not be available so average monthly data from Sydney (available within the MUSIC model) can be used
- average Sydney PET data is suitable for use in modelling water quality and hydrology. The monthly PET values for the Sydney region, including Blacktown, are shown in Table 5.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
PET mm	180	135	128	85	58	43	43	58	88	127	152	163

Table 5. Monthly evapotranspiration for the Sydney region

Evaporative loss should normally range from 75% of PET for completely open water, to 125% of PET for heavily vegetated water bodies.

7.4 Digital (electronic) modelling

The digital (electronic) MUSIC file that includes the source nodes and treatment nodes acceptable to Blacktown City Council is available for download through MUSIC-*link*.

8 Source nodes and pollutant generation

After the meteorological data has been put into the MUSIC model, the user must then define the source nodes to reflect the details (that is, area and land use) of the contributing catchments. MUSIC currently has 5 standard source nodes:

- Urban
- Agricultural
- Forest
- User defined
- Imported data.

Of these 5 source nodes, Urban is the most commonly used in our city. The Agricultural and Forest nodes are identical to Revegetatedland.

The urban node is broken down into 4 components, specifically developed by Blacktown City Council:

- Roof
- Sealedroad
- Unsealedroad (previously Other Impervious Areas)
- Revegetatedland (previously Pervious Areas).

As outlined in the eWater *MUSIC User Guide*, a comprehensive review of stormwater quality in urban catchments was undertaken by Duncan (1999). This review forms the basis for the default values of event mean concentrations in MUSIC for TSS, TP and TN. More recently, Fletcher et al. (2004) has updated the values provided in Duncan (1999) and specifically provides guidance on an appropriate land use breakdown.

Table 6 presents the recommended model defaults for various land-use categories. These values are consistent with those recommended by the Growth Centres Commission. For all simulations, the MUSIC model must be run with pollutant export estimation method set to 'stochastic generation'.

Land-use category		Log10 TSS (mg/L)		Log10 TP (mm/L)		Log10 TN (mg/L)	
		Storm flow	Base flow*	Storm flow	Base flow*	Storm flow	Base flow*
BCC Roof areas (Roof)	Mean	1.30	1.20	-0.89	-0.85	0.30	0.11
	Std Dev	0.32	0.17	0.25	0.19	0.19	0.12
BCC Road Areas (Sealedroad)	Mean	2.43	1.20	-0.30	-0.85	0.34	0.11
	Std Dev	0.32	0.17	0.25	0.19	0.19	0.12
BCC Other Impervious areas (Unsealedroad)	Mean	2.15	1.20	-0.60	-0.85	0.30	0.11
	Std Dev	0.32	0.17	0.25	0.19	0.19	0.12
BCC Pervious Areas (Revegetatedland)	Mean	2.15	1.20	-0.60	-0.85	0.30	0.11
	Std Dev	0.32	0.17	0.25	0.19	0.19	0.12

Table 6.Stormwater quality parameters for MUSIC Source Nodes

* Base flows are only generated from pervious areas. Therefore, these parameters are not relevant to impervious areas. However, MUSIC, requires a number to be entered even though it is not applied.

9 Rainfall runoff parameters

As outlined in Chapter 7, stormwater runoff (represented as storm flow and baseflow) is generated in MUSIC through the interaction of rainfall, evapotranspiration and the MUSIC Rainfall-Runoff Model. A full description of the MUSIC Rainfall-Runoff Model is provided in the eWater *MUSIC User Guide*.

If the reader of this handbook has no MUSIC modelling experience, they should review the eWater *MUSIC User Guide* before reading further.

MUSIC rainfall-runoff parameters have been derived for the Western Sydney region from model calibration studies. The parameters recommended in Table 7 are the same as those recommended by the Growth Centres Commission for use in growth areas. The Growth Centres Commission also suggest that a sanity check can be performed on total runoff volumes by comparing with the values presented in Figure 2.3 of the CRCCH's Technical Report 04/8 Stormwater Flow and Quality, and the *Effectiveness of Non-Proprietary Stormwater Treatment Measures – A Review and Gap Analysis*, Fletcher et. al., 2004.

Parameter	Recommended values
Rainfall threshold (mm)	1.4
Soil capacity (mm)	170
Initial storage (percentage)	30
Field capacity (mm)	70
Infiltration capacity coefficient a	210
Infiltration capacity coefficient b	4.7
Initial depth (mm)	10
Daily recharge rate (percentage)	50
Daily baseflow rate (percentage)	4
Deep seepage (percentage)	0

Table 7.Rainfall-runoff parameters

The steps for setting up the rainfall-runoff parameters are described below.

Step 1: Estimate fraction impervious

An initial estimate of the impervious fraction for the particular land-use should be made.

The impervious area should be based on building density controls developed by Blacktown City Council as well as the development's urban planners and architects.

The building density controls that are of relevance include minimum soft landscaping area, maximum building envelopes, floor space ratios and road design guidelines. These estimates should also be compared to aerial photos of similar recent developments in the vicinity of the proposed development.

Where differences between the estimates and the on-ground impervious area are significant, then estimates should be revised, or the differences justified.

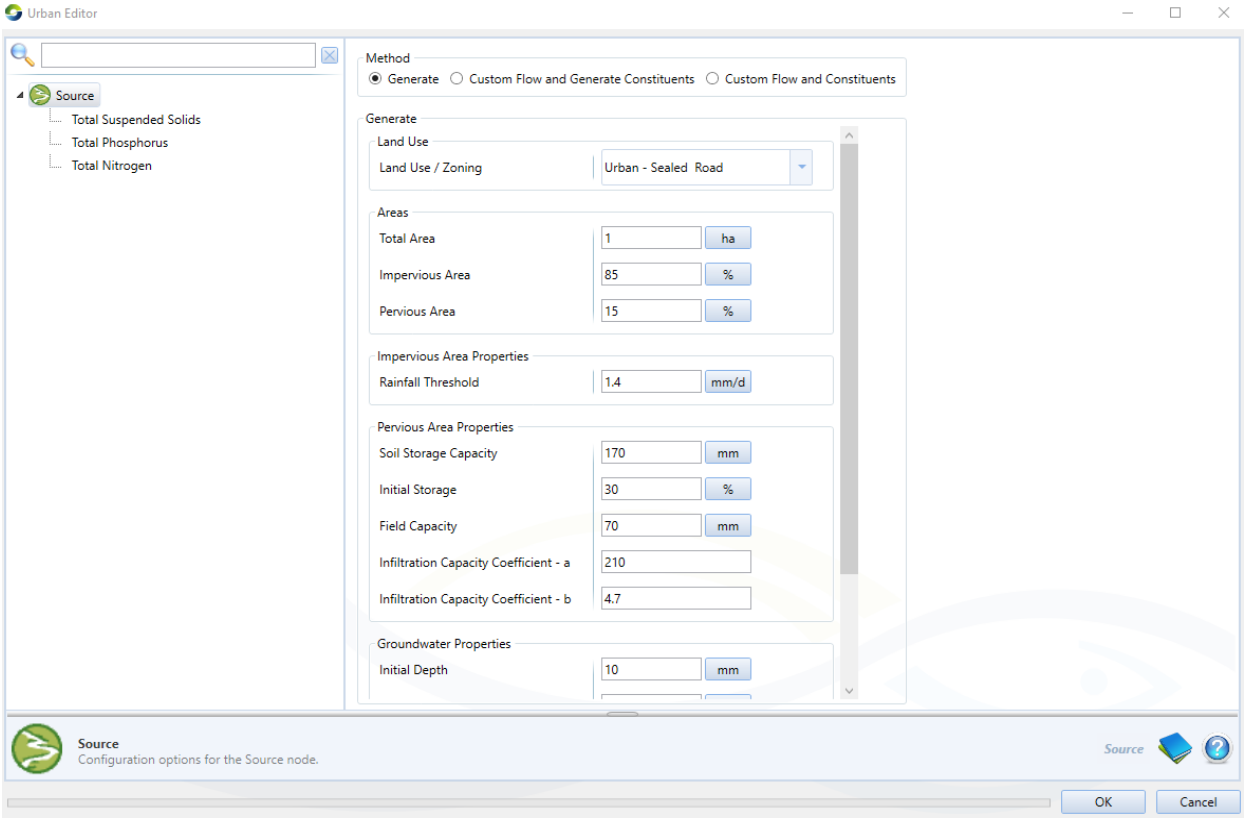


Figure 3. Estimating fraction impervious for land-use

Table 3.3 (page D-12) of the *Blacktown City Council's Engineering guide for development 2005* describes the minimum fraction impervious for different development types as follow:

- private areas under major electricity transmission easements – 15% (minimum)
- public recreation areas – 50%
- new residential lot only (outside growth centres) – 80%
- new residential lot only (inside growth centres) – 85%
- medium density development lot only (for example villas, home units) – 85%
- half width road reserve – 95%
- industrial areas/business areas – 90 to 100%
- any landscape area < 0.4 metres deep over a concrete podium is considered as impervious.

Step 2: Split MUSIC catchments into land use types

The catchment must be split into the various land types (that is, roads, roofs and other impervious and pervious surfaces). Each individual source Node, with the exception of the imported data node, requires the total area and impervious percentage of the site to be defined.

For a specific development, the site area is to be split into the 4 land-use source nodes from Section 8. For new subdivisions or staging, calculate the area of new roads and the area of new lots. The individual lots can be combined into the 4 source nodes upstream of the treatment devices, representing the entire area.

For low-density residential subdivisions, allow the following percentages for land-use for the new lots only considering 85% impervious:

- Roof – 60% (of which 50% goes to the rainwater tank)
- Sealedroad (driveways) – 10%
- Unsealedroad (courtyards, paths) – 15%
- Revegetatedland (pervious areas) – 15%.

When utilising this approach for residential subdivision of R2 land in the growth centres:

- roof areas are to be modelled as 100% impervious
- if there is a rainwater tank, then it should be modelled immediately downstream of the roof
- if only a portion of the roof drains to the rainwater tank, then the roof will need to be split into 2 separate nodes, 1 of which bypasses the rainwater tank. Generally, Blacktown City Council will only consider a maximum of 50% of the roof area of new low-density residential developments draining to the rainwater tank (including new subdivisions). This is unless there is specific information that provides a different figure when considering a particular development. In such cases, the roof areas must match with the BASIX certificate for residential development. Allow for a 2kL rainwater tank in MUSIC per lot
- reuse will be modelled in accordance with Section 11.14.3
- the Sealedroad Node includes roads, driveways, car parks and other trafficable areas. Any pervious areas (for example, verges) associated with impervious areas such as roads and car parks should be included in the Revegetatedland node. However, future Blacktown City Council roads may be considered with the Sealedroad node as 95% impervious and 5% pervious
- the Unsealedroad node should include areas such as footpaths, courtyards and decks
- all pervious areas should be included in the Revegetatedland node. Pervious areas should be connected to the treatment systems for treatment otherwise considered as bypass. The area of the treatment device itself (such as a bioretention basin, swale, or wetland) also needs to be included as a pervious source node. Such nodes must include any percentage imperviousness such as pathways or driveways, pits and scour protection
- the MUSIC model must account for all the areas being developed. Where areas cannot drain to a treatment device, these areas are considered as bypass and the specific land use(s) identified

For industrial/business subdivisions or staging where proposed:

For the lot, only consider the following proportions:

- Roof – 45% (no rainwater tank)
- Sealedroad (driveway or parking areas) – 45%
- Unsealedroad (paths) – 5%
- Revegetatedland (pervious areas) – 5%.

When utilising this approach for subdivision/staging of industrial or business land consider:

- the public roads as Sealedroad at 95% impervious
- the on-lot treatment required for the area of lot is represented by a generic node set to the required pollutant removal rates (Table 1 in Section 3.1). There is no rainwater tank required.

Step 3: Set soil properties

For 100% impervious source nodes, the only rainfall-runoff parameter that plays a part is the rainfall threshold, which should be set to 1.4 millimtres.

For all pervious source nodes, the soil characteristics shown in Table 6 should be adopted in MUSIC.

For all treatment nodes the exfiltration rate (millimetres/h) is to be set to 0.

10 Link routing

Drainage links are used in MUSIC to connect source nodes to treatment nodes and/or collection points. The drainage links account for the passage of stormwater and the time of travel between 2 nodes.

There are 3 options for the routing of stormwater available within the drainage link:

- no routing
- translation of the flood wave (only)
- Muskingum Cunge method of stream routing

For localised development assessment, Blacktown City Council does not permit any routing. Therefore, the only option is no routing.

For very large catchments and the design of regional facilities, routing may be considered where directed by Blacktown City Council's Manager Asset Design.

11 Stormwater management measures

11.1 Specific requirements for MUSIC modelling in Blacktown

Specific requirements for MUSIC modelling in our city are noted below:

- minimum percentage impervious for the site is detailed in Chapter 9, Step 1 - estimate fraction of directly connected impervious
- water quality removal targets are as per Section 3.1 Table 1
- provide a MUSIC catchment plan that shows both the land use and the areas contributing to each specific device. In many cases, it may be easier to split these into 2 separate plans, particularly with complex arrangements of land-uses and treatment systems
- high backwater levels downstream of the device, determined through hydraulic grade line analysis, will affect performance. Consequently, Blacktown City Council has adopted 1 EY as the design water level, immediately downstream of the device, as the required standard to set the device above. This is except where noted for specific systems
- detention basins are not to be modelled for pollutant removal but only modelled to improve buffering through downstream treatment systems, if required. Such buffering needs to be considered carefully as the use of such systems to reduce the flow rate may result in overloading of downstream water quality treatment devices. This could lead to higher maintenance costs and possible premature failure. The long-term maintenance risks need to be effectively managed. Blacktown City Council restricts the maximum permissible hydraulic loading rate for bioretention basins in Section 11.8.3
- for residential developments, Blacktown City Council does not permit any treatment device (other than rainwater tank) to be located in private courtyards or rear yards. They must be positioned in common areas or front yards to enable easy inspection by Blacktown City Council's WSUD Compliance Officer
- in the growth centres, on-lot water quality treatment is required to meet the required water quality targets for R3, R4, SP2 (special uses excluding drainage), business and industrial zoned areas
- within the growth centres, where small-lot Torrens title subdivision occurs in R3 zones, it is unacceptable for the water quality devices to be located within the small individual lots. To ensure ongoing compliance and provide for inspection by Blacktown City Council's WSUD Compliance Officers, the water quality treatment system is to be amalgamated into a single lot within a residential block. In certain areas, based on contours, 2 treatment devices may be permissible where the catchment grades in different directions. Inter-allotment drainage schemes will be required along the lot low points to direct flow off each lot to the treatment system. To ensure the ongoing maintenance costs are equitably distributed, the remaining lots will be encumbered by a positive covenant and community title
- for small-lot R3 subdivision in the growth centres, an alternative for off-site water quality treatment may exist in some areas where there may be space available for additional bioretention filter area in the S7.11 stormwater infrastructure. Further information is available under Blacktown City Council Policy 000520- NWGC- voluntary stormwater treatment contributions. An application may be made to Blacktown City Council's Asset Design section, with the appropriate fee to investigate the availability of such space and to provide a quote for the installation costs and one-off maintenance cost.

11.2 Standard requirements for MUSIC modelling nodes in Blacktown

Following the determination of a site water quality and hydrologic objectives, and where full on-lot treatment is required, the user is to develop an appropriate treatment train for the development. This will be dependent on any site constraints and opportunities.

Within the current version of MUSIC, the user has several treatment options available:

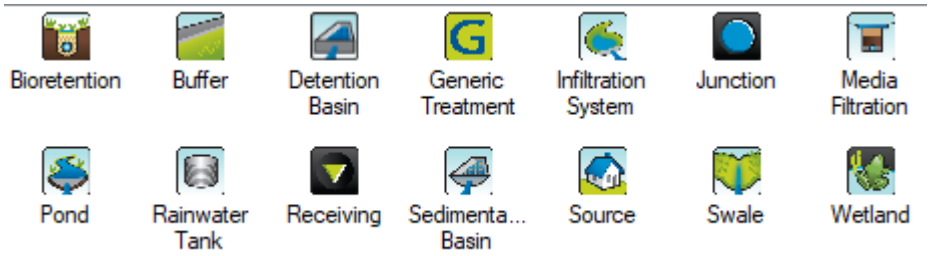


Figure 4. Treatment options available in MUSIC

The default parameters in MUSIC for the equilibrium value or background constant C*, used to define the treatment efficiency of each treatment measure, is not to be changed. However, for the decay function, the exponential rate constant k values have been modified significantly for a number of treatment nodes. These treatment nodes include Sedimentation Basin, Detention Basin, Swales and Media Filtration (media use is restricted) for particular use in our city, based on local experience.

MUSIC gives the option under the Advanced tab to access the Advanced Properties for each treatment node, in addition to k-C* values. This includes orifice discharge and weir coefficients, void ratio and number of continuously stirred tank reactor (CSTR) cells. Blacktown City Council does not permit these MUSIC default values to be changed except where specified in the paragraph above.

The following measures are not to be modelled in MUSIC as treatment systems: natural or artificial waterways, natural wetlands, naturalised channel systems, creeks, trunk drainage, environmental buffers, infiltration systems, media filtration (other than playing fields), buffers and ornamental lake/pond systems.

11.3 Wetlands

The priority of design criteria for constructed wetland systems are:

- stormwater treatment using enhanced sedimentation, fine filtration and pollutant uptake processes to remove pollutants from stormwater
- enhanced aesthetic, recreational and cultural values
- habitat provision.

Constructed wetland systems (without on-site stormwater detention) consist of:

- an upstream GPT
- a high flow bypass channel from the GPT to a location downstream of the wetland
- an inlet pond (considered as a separate sedimentation basin in MUSIC)
- a macrophyte zone (a heavily vegetated area to remove fine particulates of varying shallow depths and uptake of soluble pollutants)

- intermediate pools to transition between marsh zones
- a deep-water zone.

Constructed wetland systems (with on-site stormwater detention) consist of:

- an upstream GPT
- a high flow bypass channel from the GPT to the detention storage where provided at a lower level than the wetland, or to a deep-water zone
- an inlet pond (considered as a separate sedimentation basin in MUSIC)
- a macrophyte zone (a heavily vegetated area to remove fine particulates of varying shallow depths and uptake of soluble pollutants)
- intermediate pools to transition between marsh zones
- a deep-water zone.

11.3.1 Wetland design considerations

General design considerations

The constructed wetland must be designed in line with best management practices as defined below. Only when these are satisfied can the wetland be modelled in MUSIC.

Constructed wetlands are to be designed generally in line with the principles set out in the *Wetland Design Manual* by Melbourne Water (latest version) except where noted below.

Parameters and sizing

- for the purposes of an initial assessment, a typical wetland surface area covers a minimum of 2% of the catchment area draining to it. In some areas, this percentage may be significantly higher. Lower available surface area will require either substantial higher pre-treatment, or supplementary water quality treatment such as bioretention further downstream
- the wetland system is to be generally designed for the 4 EY (3 month) design storm
- if the treated outfall from the wetland is not being directed to a storage pond for reuse and is being directed to downstream receiving waters, the invert level from the outfall from the wetland is to be positioned above the 50% AEP AHD level in the adjacent watercourse. No backwater intrusion into the device is permitted from the 50% AEP level in the adjacent floodway
- floating wetlands can provide significant pollutant removal, particularly when retrofitted to an existing wetland or pond. To model these systems using MUSIC in our city, they are to be modelled as a pond, and not as a wetland. This is providing the floating wetland has a minimum coverage of 10% of the pond area.

Gross pollutant trap (GPT)

- a GPT with diversion weir is required upstream of the inlet pond as initial treatment and sized for the 4 EY (3 month) storm. To protect the wetland, the GPT should be capable of removing 50% TSS
- direct the treated flow from the GPT to the inlet pond (sediment basin in MUSIC) with the invert set to about a third up the pond basin, and as far from the overflow to the macrophyte zone as possible

- where the wetland sits to the side and forms part of a larger detention basin, direct the untreated piped high flow bypass from the wetland GPT (in excess of the 4 EY flow) to the detention basin and bypass the wetland. This will protect the wetland macrophytes from the higher velocity flows. Similarly, position any trapped low points in the road opposite the detention basin and away from the wetland to ensure flood flows, in excess of the piped drainage capacity, do not damage the wetland
- where the wetland forms part of a detention basin and the detention basin storage is positioned above the wetland extended detention depth (EDD), direct the untreated piped high flow bypass from the wetland GPT (in excess of the 4 EY flow) to the deep-water zone via a silt trap. This ensures the high flows 'backwater' across the wetland, protecting the macrophyte vegetation from scour by high velocity flows. Similarly, for any trapped low points in the road opposite the combined wetland/detention basin, design the wetland where possible to convey the high flood flows via a swale around the macrophyte zone to the deep-water zone to minimise damage to the wetland
- where the wetland is independent of a detention basin, direct the untreated piped high flow bypass > 4 EY downstream of the wetland controls
- ensure the GPT is not drowned by setting the pipe invert out of the GPT, to the top of the wetland EDD and ensuring the weir height is sufficient to drive the treatable flow rate through the GPT in line with the manufacturer's requirements
- in some very constrained systems, it may not be possible to raise the GPT invert to the EDD. Subject to Blacktown City Council approval, it may be possible to raise the diversion weir level to a higher level to account for the working tailwater level in the inlet pond by liaising with the GPT supplier
- where the invert of the outlet pipe from the GPT is below the tailwater level in the inlet pond, provide a sluice gate or butterfly valve between the GPT and the pond. So the GPT can be cleaned independently without draining the inlet pond water. Where the inlet pond water extends in the pipe beyond the GPT to a depth greater than 0.5 metres, provide a sluice gate on the upstream side as well. The sluice gate is to be constructed in stainless steel within its own lockable pit. Operate the gate by a screw thread spindle and key.

Inlet pond (sediment basin)

Design the inlet pond using either a MUSIC sediment basin node or a MUSIC wetland node using the 'Estimate Inlet Volume' function. If using the wetland node, it assumes a fixed inlet zone depth of 2 metres. To vary this, use the sediment basin node instead which has more flexibility to alter the depth.

Size the inlet volume with a 4EY design flow, assuming a sediment loading rate of 1m^3/ha/year to remove 95% (R = 0.95) of all suspended sediment down to a particle size of at least 125 micrometres (µm). Where land area is very restricted this could be reduced to 85% with Blacktown City Council approval (R = 0.85). An ability to retain collected sediment for a period of up to 5 years between maintenance is a minimum requirement.

- the sediment storage zone is to occupy the bottom ⅓ of the inlet pond total depth below the permanent water level (PWL). The upper ⅔ is considered as the permanent pool volume

- to aid with maintenance, the base of the inlet pond must be constructed in minimum 150 millimetres 40 MPa reinforced concrete. The sides must be constructed in either concrete, or reinforced concrete blockwork up to at least the sediment storage zone (bottom ⅓). Where split face blockwork is used, this would generally be positioned with the rough face external to the inlet pond. Where space permits the area above the sediment storage can be battered and more aesthetically pleasing treatments adopted to be reflective of adjacent landscaping
- the inlet pond must be completely isolated from the macrophyte zone to enable it to be drained whilst maintaining the macrophyte zone water level at the Permanent Water Level (PWL)
- provide weir overflow to the macrophyte zone across the full width of the flow path. Set the low weir level to a minimum of 0.2 metres above the PWL. Where a 0.4 metres EDD is used, raise the low weir to 0.25 to 0.3 metres above the EDD providing the 4 EY weir flow depth to the macrophyte zone is at, or below, the EDD
- provide pipe(s) from the inlet pond to the macrophyte zone across the flow path set to the PWL. Size the pipe(s) to achieve a notional inlet pond (sedimentation basin) detention time of up to 8 hours
- where the wetland forms part of a detention basin and surface flows in excess of the pipe flow are directed to the inlet pond, provide a high flow bypass set to the EDD direct to the deep-water zone. Isolate the remainder of the inlet pond to prevent these high flows discharging back into the macrophyte zone
- a minimum 3.5 metres wide heavy-duty ramp at 10% (5% cross fall or flatter) is to be provided to the base of the inlet pond. This will enable a 20 tonne excavation plant to enter the sediment pond for clean out operations. Ramps can be a minimum of 150 millimetres thick, plain, coloured or feature-finish concrete. Ramps can also be in rock for aesthetics (d50 > 400 millimetres) encapsulated using a sandstone log edge confinement course
- a hardstand area, with a minimum turning circle appropriate to the types of maintenance vehicles to be used, must be provided. This should be located adjacent to the inlet pond maintenance access ramp to enable maintenance vehicles to safely reverse and exit the sediment loading area to the street in a forward direction.

Wetland and macrophyte zone

- the macrophyte zone EDD must be ≤ 400 millimetres
- where combined with a detention basin, the maximum depth over the plants (including macrophyte EDD) is 1,000 millimetres
- the macrophyte zone must provide a minimum residence time between 24 hours up to 72 hours. If the wetland's outlet control is not an overflow pipe invert set to the PWL, such as a submerged orifice in an overflow pit with a protective screen, then the residence time calculation changes. In that case, it can be assessed using a program such as DRAINS and modelling the wetland as a detention basin
- the total permanent pool volume (cubic metres) divided by the surface area of the wetland at the permanent water level (square metres) must be less than 0.40 metres
- dieback of macrophytes is commonly caused by the permanent water being too deep. Ensure the deep marsh zone depth is a maximum of 300 millimetres before transitioning to the submerged marsh zone

- provide an internal system to create a sinuous flow path within the wetland to increase residence time and avoid short circuiting. The top is to be a minimum of 100 millimetres above the wetland EDD. Larger wetlands are to use internal clay/soil berms with sandstone blocks or similar above as required to achieve the minimum level. Timber baffle walls will not be permitted
- the length of the macrophyte zone must typically be ≥ 4 times the average width of the macrophyte zone
- velocities within the macrophyte zone must not exceed 0.05 m/s at the EDD during 4 EY inflow events. Consider the flow area as the EDD plus the shallowest depth multiplied by the narrowest width of the macrophyte zone at PWL
- the velocity through the macrophyte zone is ≤ 0.5 m/s during the peak 1% AEP flow assuming:
    - the macrophyte zone is at EDD if the wetland is not within a retarding basin or floodplain
    - the water level is at the peak 10% AEP water level if the wetland is within a retarding basin or floodplain.
- maintain a minimum 100 millimetres depth at the edge of the macrophyte zone via a concrete, rock or hard edge to reduce the risk of mosquito breeding during dry periods when the water levels drop
- dense vegetation bands and flat bathymetry orientated perpendicular to the flow path are required for even flow distribution and to reduce short-circuiting through the macrophyte zone
- a grade of between 1:200 and 1:400 must be provided between marsh zones (longitudinally through the macrophyte zone as shown below in Figure 5) to enable the wetland to freely drain. Intermediate pools will generally be needed to transition between marsh zones. Balance pipes must be placed between all open water zones (inlet, intermediate and outlet pools) to enable water levels to be drawn down for maintenance or water level management purposes

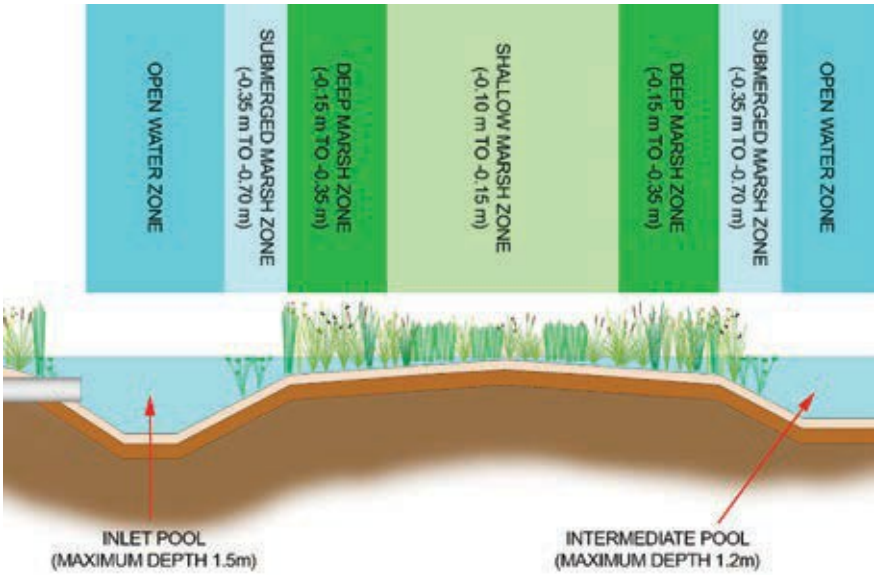


Figure 5. Wetland zones

(Source Melbourne Water, Wetland Design Manual 2017)

- provide a minimum 3.5 metres wide heavy-duty ramp at maximum 10% gradient (5% crossfall or flatter) to the macrophtye zone. This will enable heavy machinery to enter for clean out operations. Ramps can be in rock, plain or feature concrete. The ramp is to extend 2 metres into the macrophyte zone itself to enable boat access
- the wetland must be configured to enable maintenance vehicles to drive around at least 50% of the wetland perimeter. This may be achieved via subdivisional road networks. Vehicular access must be provided as close as possible to wetland structures that may catch debris
- provide an independent drainage pipe direct to the downstream discharge point from a level as low as possible in the deep-water zone, controlled by a gate valve. This will enable, as much as practical, the wetland to be drained for cleaning. No independent drainage pipe outlet is required from the inlet pond
- provide a recirculation system to maintain the wetland during prolonged dry periods. It will depend on where the water that enters is coming from, how much and how often, including any baseflows. For regional facilities, consult with Blacktown City Council's Green Space Service section. Reticulation will increase aeration, reduce mosquitoes and flush stagnant water
- provide at least 200 millimetres of topsoil over a clay liner and a minimum of 300 millimetres of topsoil over a high-density polyethylene (HDPE) or geosynthetic clay (GCL) liner in all areas of the macrophyte zone. On downward slopes towards the deep-water zone at the change of slope, provide a 200-300 millimetres lip in clay to retain the topsoil
- HDPE liners are to be 1.5 millimetres thick with A44 Bidim or approved equivalent above and below the liner on a 50 millimetres sand bed. Extend to a minimum of 100 millimetres above the EDD.
- clay liner to comprise of a material with a maximum permeability of 1×10^{-9} m/s with the sample remoulded to a standard maximum dry density of 95%. This should be in line with AS1289.5.5.5 at standard optimum moisture content in line with AS1289.6.7.3. Clay is to be non-dispersive Emerson class 5 or 6. Where this cannot be achieved, generally develop the clay liner by over-ripping the existing clay base to a nominal depth of 500 millimetres. Stockpile and dry the material before stabilisation through the addition of a suitable agent (normally gypsum or ag-lime) before re-installation as a low permeability clay liner. Extend to a minimum of 100 millimetres above the EDD
- all clay liners are subject to technical direction by a geotechnical or suitably experienced civil engineer registered on NER
- for the outlet control for the wetland, in addition to the design wetland control pipe (equivalent pipe diameter) provide a mechanism to temporarily adjust the water level within the wetland to assist in plant establishment. This could be through a series of staggered maximum 50 millimetres pipes with sealed screw caps on the inside of the control pit at various lower levels below the permanent water level. Provide maximesh rubbish screens to minimise blockages over the 50 millimetres pipe, plus over any orifice or an outlet pipe controls
- at all outlet pits, ensure all pipes/penetrations from the basin are epoxy sealed and watertight using puddle flange or similar
- no formal access to water will be permitted unless appropriate safety benching is provided. As a minimum safety edging will be provided where ponded water depths exceed 300 millimetres deep
- consider access safety controls in areas:
    - adjacent to zones of deep-water (greater than 300 millimetres at PWL)

- where safety benches do not meet the width criteria
- adjacent to potentially unsafe structures
- where high velocities may be encountered
- where batters are steeper than 1V:4H.

- engage an experienced and qualified aquatic ecologist to design at least 8 permanent habitat features per 1,000 square metres of wetland surface area. For example, place rocks or logs (which can be salvaged from the land development process) around and within a wetland to provide shelter, perches and basking areas for native wildlife such as turtles, frogs and birds. Logs must be suitably anchored to avoid buoyancy and movement in high flows
- a water level marker (or water depth gauge) must be installed, typically near the outlet, to show the wetland water level relative to normal water level. Ensure this does not penetrate and damage the liner.

Wetland vegetation

Planting zones are noted in Table 8.

ID	Planting zone	Depth range
TE	Terrestrial	> + EDD
EB	Ephemeral batter	PWL to + EDD
SM	Shallow marsh	PWL to -0.2 m
DM	Deep marsh	-0.2 m to -0.4 m
SU	Submerged marsh	-0.4 m to -0.7 m

Table 8.Planting zones

- approximately equal amounts of shallow marsh (100-200 millimetres deep) and deep marsh (200-400 millimetres deep) in the macrophyte zone are required for effective wetland function
- the macrophyte zone must contain a minimum of 80% cover of emergent macrophytes comprising of shallow and deep marsh zones. Open water areas (maximum 20% of the wetland area) must include submerged marsh vegetation
- suitable plant species and densities are available from Blacktown City Council's website: https://www.blacktown.nsw.gov.au/Plan-build/Stage-2-plans-and-guidelines/Stormwater-management-and-water-sensitive-urban-design-WSUD/Toolkit-for-water-sensitive-urban-design-WSUD-application-stage/Water-sensitive-urban-design-WSUD-planting-and-landscaping/Wetlands-planting
- typha is considered a virulent weed and must not be allowed to establish at any time. Prior to handover of any wetland to Blacktown City Council, the wetland is to be Typha free for a minimum of 6 months.

11.3.2 Wetland MUSIC modelling requirements

To model a wetland in Blacktown, 3 to 5 MUSIC nodes are required:

Step 1: provide a GPT node (see Section 11.11) with a high flow set to the 4 EY flow. Direct the low flows to the inlet pond (sediment basin) and the > 4 EY high flow bypass downstream of the Wetland node via a secondary link

Step 2: provide a separate Sedimentation Basin node (Section 11.5) to represent the inlet pond with characteristics based on the empirical modelling requirements and details outlined for the inlet pond in Section 11.3.1 above

Step 3: provide the Wetland node as described below in the Input values section

Step 4 (optional)**:** provide a Rainwater Tank node (Section 11.14) to represent any reuse storage independent from the wetland. Provide a link representing the treated pipe flow from the wetland to the tank with a secondary flow representing the high flow bypass to the next node down. Note: the reuse option in the Wetland node is unacceptable

Step 5 (optional)**:** provide a Detention node (Section 11.6) downstream of the wetland if detention is used and there is a water quality device downstream that would benefit from the reduced flows, otherwise ignore.

Input values into the wetland MUSIC catchment node, including the following:

- there is no Low Flow By-pass in the wetland. Leave default as 0.00
- the High Flow By-pass was considered as part of the GPT node. Leave default as 100.00
- in our city, the Inlet Pond Volume is set to 0.0
- enter the proposed surface area of wetland macrophyte zone under ‘Storage Properties’. It is not the actual surface area, but it is the area that when multiplied by the Extended Detention Depth will give the volume of storage. Where the sides of the basin are battered, the Surface Area is the area at half the Extended Detention Depth. That is, the average basin area
- set the Extended Detention Depth between 0.25 metres to a maximum 0.4 metres. Any on-site stormwater detention storage is measured above the wetland Extended Detention Depth and must not be included in the wetland Extended Detention Depth
- in MUSIC set the Permanent Pool Volume as 90% of the actual volume of water. This allows for sediment storage over time. This ensures that the hydraulic retention time during a storm event is not overestimated, as this would also overestimate the removal of contaminants that flow out of the wetland
- set the Initial Volume to match the Permanent Pool Volume.

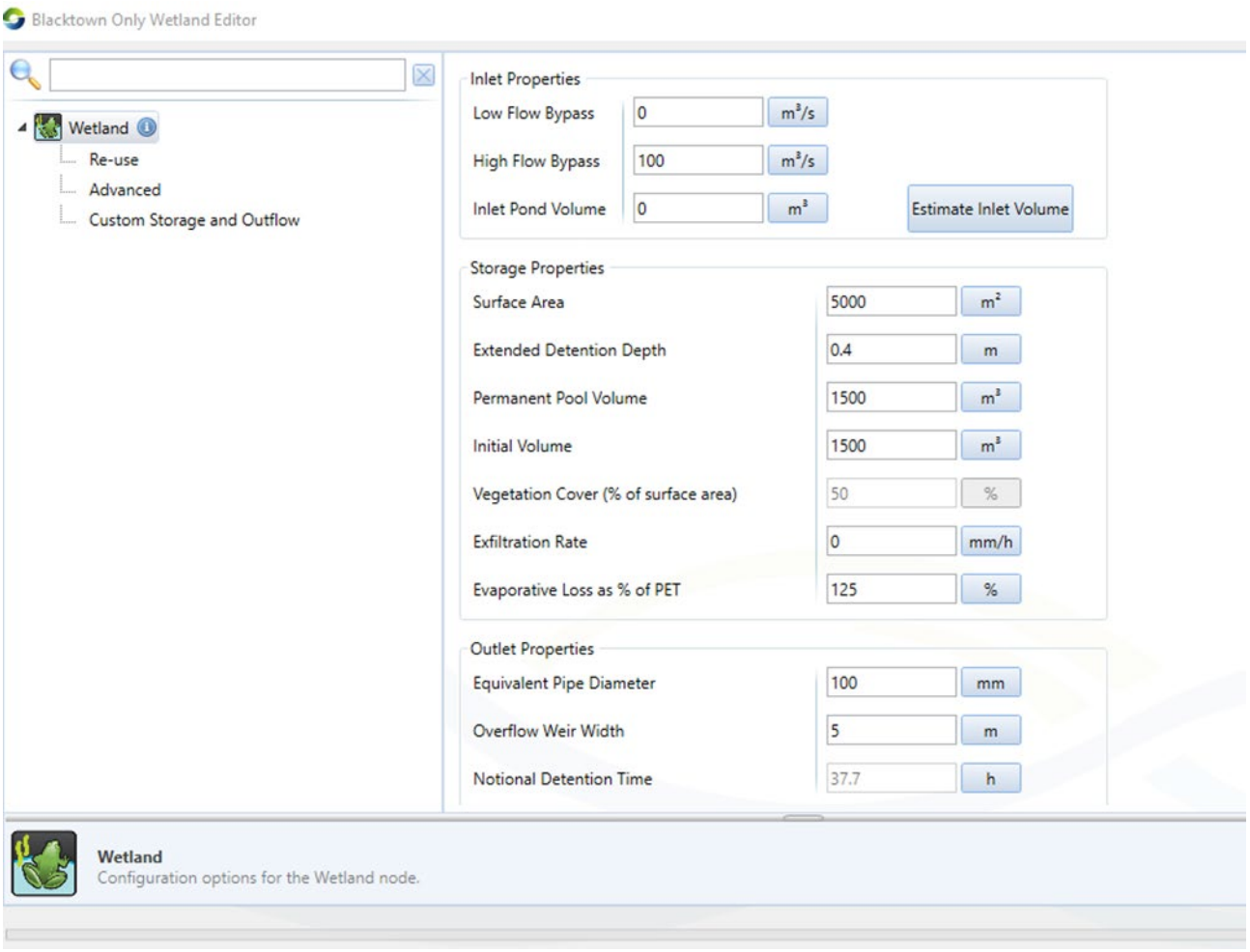


Figure 6. Example of a wetland node in MUSIC

- Exfiltration is the water lost from the treatment measure into the surrounding soil (Blacktown City Council requires 0 millimetres per hour for wetlands) because they are lined
- adjust the Equivalent Pipe Diameter to ensure the treatment measure has a Notional Detention Time of a minimum of 24 hours to preferably 72 hours or more. This assumes the overflow pipe is set to the permanent water level
- match the Overflow Weir Width to the actual weir length. Choosing too small a weir width will overestimate the performance of the wetland
- tick the Custom Storage and Outflow where an overflow pit is used with the orifice set at a lower level than the permanent water level. Then add the Stage/Discharge/Storage spreadsheet details
- using the Re-use option in the Wetland node is unacceptable. A separate Rainwater Tank node is to be used to represent a separate storage area clear of the wetland, such as a pond or tank
- the default MUSIC ‘k’ m/yr values of TSS 1,500, TP 1,000 and TN 150 are acceptable.

11.4 Ponds

11.4.1 Types of ponds

Ponds can be sized for 3 different purposes, which include:

- pollutant removal
- stormwater storage for reuse
- ornamental.

For the first 2 purposes, MUSIC can be used to size the pond and assess its performance as described below. All ponds should be preceded by appropriate pre-treatment to remove coarse sediment.

11.4.2 Water quality ponds

Water quality ponds rely on the settling of suspended solids as the principal treatment mechanism. Vegetation (including submerged macrophytes in a deep pond) can promote nutrient removal and open water can promote UV disinfection. However, these processes cannot be modelled in MUSIC currently.

Pre-treatment is essential upstream of ponds. In MUSIC, the pollutant removal parameters associated with ponds assume that pre-treatment has occurred upstream. Therefore, it is essential to include an appropriate treatment train upstream of a pond in the MUSIC model. This could include a sedimentation basin, or a GPT, capable of removing a substantial proportion of coarse suspended solids.

Ponds must have a minimum of 10% fringing macrophyte coverage in accordance with MUSIC model assumptions.

11.4.3 Storage ponds

If a pond is used to store treated stormwater for reuse, its performance in balancing supplies and demands can be modelled using MUSIC. In this case, the pond may or may not be modelled with EDD.

The permanent pool represents the volume available for reuse and the quantity of water is likely to fluctuate widely depending on supplies and demands.

If a storage pond has a permanent pool below the volume available for reuse, this permanent pool should be either ignored or modelled separately. When modelled separately, EDD can only be represented in the water quality Pond node.

The effectiveness of the pond as a storage system with reuse can be evaluated by checking the node water balance of the Pond node when the model has run.

11.4.4 MUSIC modelling requirements for ponds

Input parameters include:

- identify any High Flow or Low Flow By-passes proposed for the treatment measure. Otherwise adopt the default values of 0 for Low Flow and 100 for High Flow
- input the Surface Area of the pond. It is not the actual surface area, but the area that when multiplied by the Extended Detention Depth will give the volume of extended storage. Where the sides of the basin are battered the Surface Area is the area at half the Extended Detention Depth, that is, the average basin area

- the Extended Detention Depth is the depth between the top of the permanent pool and the lip of the overflow weir
- for MUSIC, set the Permanent Pool Volume as 90% of the actual volume of water. This allows for significant sediment storage. This ensures that the hydraulic retention time during a storm event is not overestimated as that would also overestimate the treatment of contaminants that flow into the wetland. Where this stored water is used for irrigation, this volume cannot be larger than the volume available for reuse which may be limited by extraction point levels or environmental considerations
- Exfiltration is the water lost from the treatment measure into the surrounding soil (Blacktown City Council requires 0 millimetres per hour for ponds, which should be lined to retain water)
- Evaporative Loss as percentage of PET, allows 75% for open water bodies with little to no vegetation
- modify the discharge pipe diameter to ensure a notional detention time long enough to allow settling of the target particle size, typically a minimum of 24 hours where this is a water quality pond. This is assumed to be at the Extended Detention Depth. For complicated discharge relationships, use Custom Storage and Outflow and add the appropriate data.

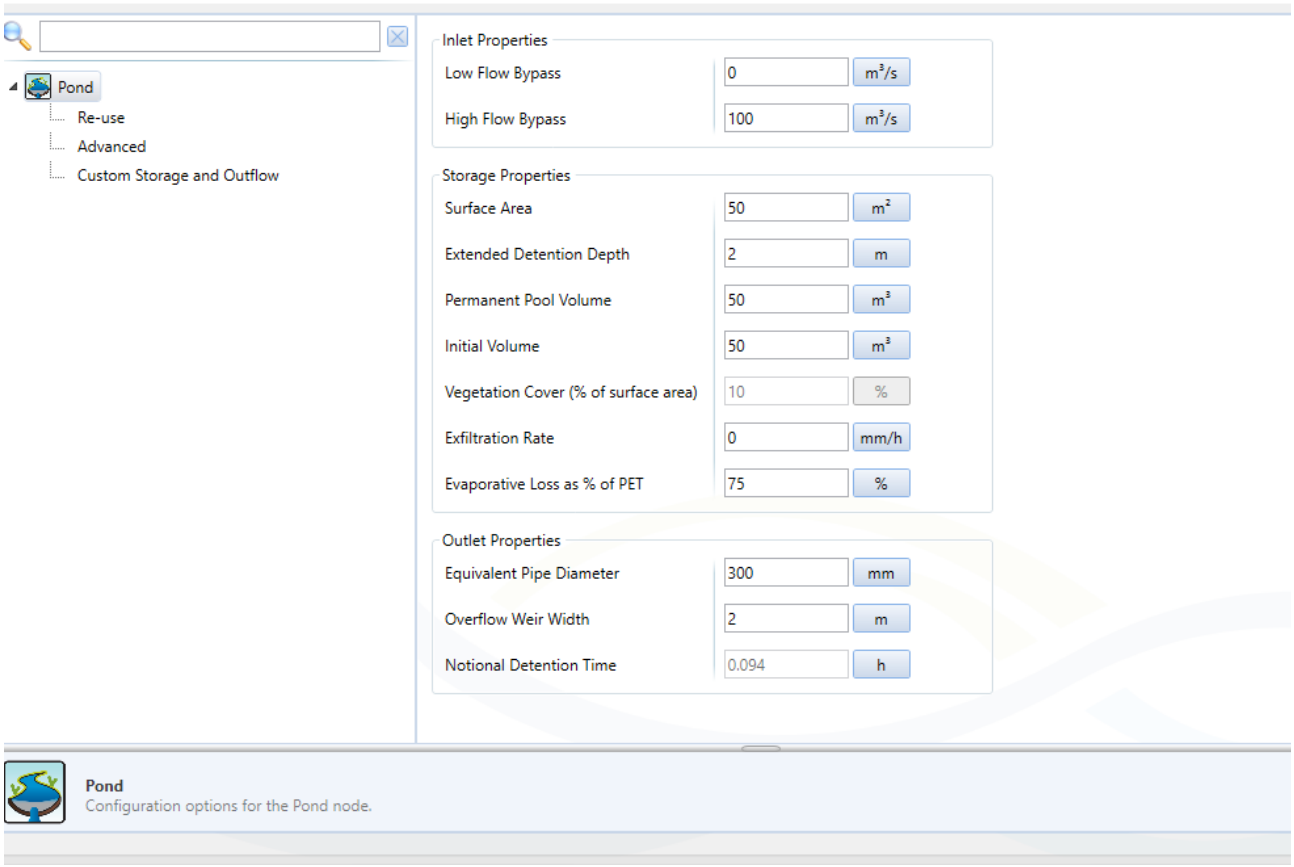


Figure 7. Example of a pond node in MUSIC

- match the Overflow Weir Width to the actual weir length. Choosing weir width that is too small will overestimate the performance of the pond
- the default MUSIC 'k (m/yr)' values of TSS 400, TP 300 and TN 40 are acceptable.
- where stormwater reuse is proposed, click on the 'Re-use tab', and then:
    - tick 'Use stored water for irrigation or other purpose'
    - enter 'Reuse details' to represent the intended demands on water from the storage pond.

11.5 Sedimentation basins

Sediment basins are used to retain coarse sediments from runoff. They operate by reducing flow velocities and encouraging sediments to settle out of the water column.

They are frequently used for trapping sediment in runoff during construction activities and for pre-treatment to measures such as wetlands (for example, an inlet pond). The sedimentation basin represented does not deal with construction activities such as erosion and sediment control, or those assessed using the Blue Book. The sedimentation basins modelled under this section using MUSIC are required for permanent sediment basins or temporary treatment greater than a 2-year duration.

Sediment basins are sized according to the design storm discharge and the target particle size for trapping (generally 0.125 millimetres).

For detailed design parameters for sedimentation basins, refer to Section 11.3.1 for the details of an inlet pond. The empirical principles detailed in that section are to be satisfied before modelling in MUSIC. However, where a sedimentation basin is not required as pre-treatment to a wetland, a higher EDD and other controls can be considered.

Input parameters include:

- identify any specific Low Flow or High Flow By-passes proposed for the treatment measure. Otherwise, adopt the default values of 0 for Low Flow By-pass and 100 for High Flow By-pass
- input the Surface Area of the basin. It is not the actual surface area, but it is the area that when multiplied by the Extended Detention Depth will give the volume of storage. Where the sides of the basin are battered, the Surface Area is the area at half the Extended Detention Depth. That is the average basin area
- the Extended Detention Depth is the depth between the top of the permanent pool (or ground if no permanent pool) and the lip of the overflow weir
- for MUSIC, set the Permanent Pool Volume as ⅔ the actual volume of water which allows for significant sediment storage. This ensures that the hydraulic retention time during a storm event is not overestimated. And it would also overestimate the treatment of contaminants that flow into the sedimentation basin. Pool depths can be up to 2.5 metres but need to allow for batter slopes when calculating volumes
- Exfiltration is the water lost from the treatment measure into the surrounding soil (Blacktown City Council requires 0 millimetres per hour for sedimentation basins, which should be lined to retain water)
- Evaporative loss as percentage of PET, allow 75% for open water bodies
- modify the discharge pipe diameter to ensure a detention time long enough to allow settling of the target particle size
- match the Overflow Weir Width to the actual weir length. Choosing weir width that is too small will overestimate the performance of the sedimentation basin.

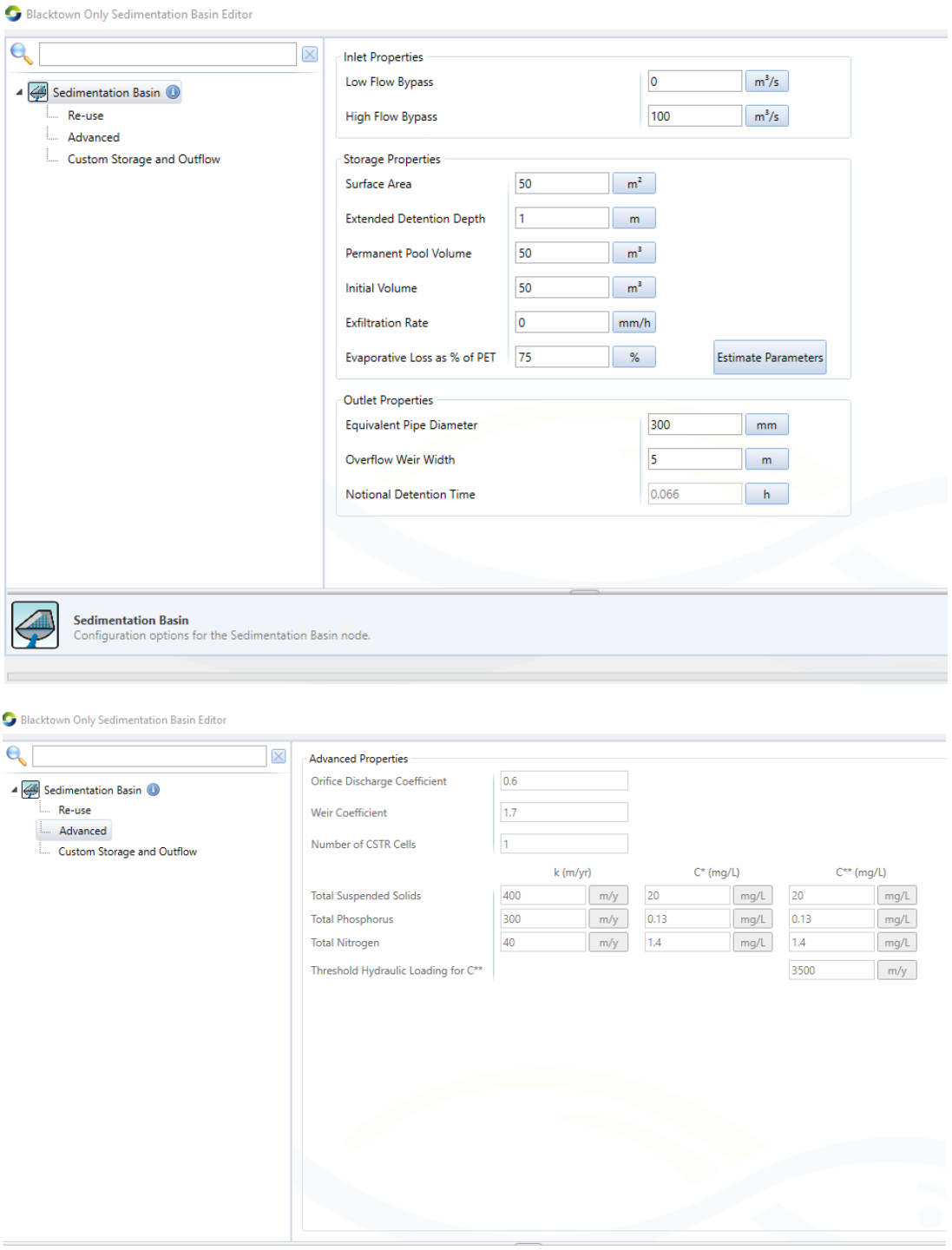


Figure 8. Example of properties of a sedimentation basin in MUSIC

- this treatment measure can be utilised as pre-treatment measure upstream of a wetland or bioretention and allows for a diversion of flows above recommended scour velocities
- re-use in MUSIC will not normally be accepted for a sedimentation basin.
- under the ‘Advanced’ tab set the ‘k (m/yr)’ values to TSS 400, TP 300 and TN 40.

11.6 Detention basins

11.6.1 Conventional detention basins

Detention basins can be above ground or below ground in tanks. See Section 11.6.3 for specific below-ground requirements for certain developments. Blacktown City Council now only uses Version 4 of the Upper Parramatta River Catchment Trust (UPRCT) *On-site Stormwater Detention Handbook*. This is a 2-stage nested orifice control for both environmental low and high level flood flows for use throughout the whole our city. See Section 3.5 for more details. High Early Discharge (HED) pits are no longer permitted in our city.

Areas that are subject to OSD can be identified through the following link: https://www.blacktown.nsw.gov.au/files/content/public/v/30/plan-build/stage-2-plans-and-guidelines/blacktown-planning-controls/blacktown-development-control-plan-2015/osd-map.pdf

Blacktown City Council uses either a deemed-to-comply tool available from www.s3qm.com.au or a separate Blacktown City Council OSD spreadsheet available from Council's WSUD Compliance Officers at wsud@blacktown.nsw.gov.au

The use of the UPRCT spreadsheet Version 4 which was available from the UPRCT is not accepted in our city.

These basins are not seen as water quality devices. Consequently, under the 'Advanced' tab, the k values are all set to 0. They can be used to attenuate the peak flows and extend the hydrograph which may assist any downstream treatment devices by enabling more water to be treated.

As there is no water quality removal with this node, it is useful to model this in MUSIC only where there are water quality devices downstream. It will benefit from reducing the peak flows and extending the hydrograph. If this is not the case, then there is no point in including this as a node in MUSIC. The exception to this is where permanent detention may be used in satisfying the stream erosion index (SEI).

For the layout of conventional detention basins and specifications, see Blacktown City Council's WSUD standard drawings sheets 6101 - 6103.

11.6.2 Detention basins used in conjunction with water quality devices

These detention basins are typically below ground tanks used to store proprietary water quality devices such as filter cartridges. The detention within these systems is used to store or buffer water for treatment before overflowing.

The removal rate of these filter cartridges is represented by a separate Generic node positioned downstream of the Detention Basin node. Any deposition of pollutants within the tank through settlement has already been incorporated into these Generic nodes rates.

Consequently, under the 'Advanced' tab the k values for the Detention Basin node are all set to 0.

11.6.3 Additional design information

- most early drainage designs, incorporating a water quality device as part of the detention basin, have directed the treated flow from the device (underdrain flow) directly back into the basin or to the low flow orifice control. Both proprietary filters and bioretention rely on a water pressure difference (Darcy's Law) to drive the stormwater through the treatment device. Where the water levels rise at the device outlet, such as in a detention basin, there is an immediate reduction in flow. As the downstream water levels continue to rise, the treatable flow rate continues to reduce. When the water levels equalise, all treatment ceases. This does not match the MUSIC model which is based on continuous flow through the treatment device throughout the whole storm. Therefore, the modelled pollutant removal will not reach the targets in practice. This approach is prohibited in our city
- to ensure consistency with the approach adopted in MUSIC, Blacktown City Council requires that the treated underdrain flow (for example bioretention, or proprietary filters) must discharge downstream of the detention discharge control pit. This will ensure ongoing treatment throughout a range of storms. This specific arrangement can be seen in Blacktown City Council's WSUD standard drawing 6102 for proprietary filters. For proprietary filter underdrains, this flow can be significant and the orifice size needs to be reduced to ensure the permissible site discharge can be maintained. For smaller bioretention systems, this bypass is generally so low that it does not impact the remaining discharge operation of the detention basin. However, for larger bioretention basins with filter areas greater than 500 square metres, this bypass can be significant. See Section 11.8.5 for details
- as detailed above, this proprietary filter underdrain flow can be significant. In certain arrangements the underdrain discharge can be so high that it exceeds the restricted 50% AEP (1.5 year ARI) permissible site discharge flow and sometimes even the restricted 1% AEP flow. However, Blacktown City Council will not accept deletion of the 1% AEP orifice and the design needs to be reviewed. Deleting the 50% AEP orifice results in higher discharges in the smaller events and may cause an adverse environmental impact. Before Blacktown City Council will accept deletion of the 50% AEP orifice for permanent on-site stormwater detention, 3 alternatives need to be considered in this order:
    1. if there sufficient fall across the site to permit relocating the filter chamber downstream of the orifice controls
    2. if additional treatment measures, such as bioretention or a Jellyfish or similar, can be added to part of the catchment to reduce the number of cartridges required
    3. if the level of the filter chamber be raised so that the invert of the filter underdrains are set at or above the 50% AEP (1.5 year ARI) weir level. Blacktown City Council will permit such systems to discharge directly to the detention tank. For the purposes of managing Blacktown City Council's OSD Spreadsheet, filters are no longer used for managing water quality as they discharge to the tank and not downstream of the orifice. Blacktown City Council will only agree to delete the 50% AEP orifice, if none of the 3 alternatives are considered viable. Details must be provided.

  Where Blacktown City Council agrees to the deletion of the 50% AEP orifice, the following design steps are required as per Blacktown City Council's WSUD standard drawing 6102:

  Step 1: prepare the Blacktown City Council OSD Spreadsheet allowing for standard filter flow and 1% AEP filter flows, as determined by calculation or through the Ocean Protect spreadsheet. The spreadsheet should say that the 50% AEP orifice is not required

  Step 2: delete the 50% AEP orifice from the drawing while retaining the 50% AEP weir

Step 3: set the pipe with non-return flap from the detention storage back into the filter chamber. This should be either 150 millimetres above the false floor level of the Stormfilter chamber or 250 millimetres above the base level of the AtlanFilter chamber

Step 4: review the size and number of pipes with non-return flaps so that the pipes have about twice the capacity of the standard flow through the filters

Step 5: raise and regrade the floor of the detention storage so that the base grades can be at a minimum of 1% to the pipes with non-return flaps

Step 6: review tank dimensions to contain the required storage

Step 7: recalculate the Blacktown City Council OSD spreadsheet as the average base level will be higher than previously calculated. This will also raise the 50% AEP weir level

Step 8: transfer this new information to the amended drawing.

For below ground detention systems, Blacktown City Council will accept concrete or masonry construction such as tanks or oversized pipes (minimum 450 millimetres diameter at minimum 0.5%) or box culverts (minimum 450 millimetres high). Ensure the oversize pipe capacity is a minimum of 5 times the required capacity. These must be able to be physically inspected and maintained with maintenance access pits at all ends and intermediate access pits at maximum 25 metre intervals.

Provide Maximesh RH3030 for orifice diameters 150 millimetres or less, ensuring the screen has a minimum area of 50 times the orifice area. For orifices 150 millimetres diameter or more, provide Weldlok F40/203 with a minimum area of 20 times the orifice area. No pollutant removal is permitted through use of the Maximesh or Weldlok trash rack within the detention discharge control pit.

Absolute minimum depth in a detention tank is 0.5 metres for residential development and 0.75 metres for business or industrial development.

For the below ground detention tank to improve safety and reduce weight for Work Health and Safety (WHS) provide: a 2 x 0.6 metres x 1.2 metres grated access over the 50% AEP and 1% AEP orifices and screens at depth greater than 1.2 metres.

For the below ground detention tank, all other accesses must be a minimum 0.9 metres by 0.9 metres grate (not sealed). They are to be positioned so the maximum distance from any point in the tank to the nearest grate is not greater than:

- 1.5 metres for clear heights less than 0.7 metres
- 2.0 metres for clear heights less than 1.0 metres
- 3.0 metres for clear heights less than 1.5 metres
- 4.0 metres for clear heights less than 2.0 metres
- 6.0 metres for clear heights greater than 2.0 metres.

Detention tanks are not permitted under habitable floor. However, it may be possible to extend part of a tank under a building subject to both structural design and the distance to grates being satisfied as detailed in the point above.

All detention required for low density residential development (single dwellings or dual occupancies or similar) is to be provided totally underground, except in special circumstances agreed to by Blacktown City Council.

Where subdivision requires detention for the newly created lots, this is to be provided at time of subdivision as a below ground tank. A covenant over the new lots for future OSD will not be accepted.

There is no allowable offset for rainwater tank volumes against the required detention storage volumes in all developments.

For areas not directly impacted by flooding, the OSD spreadsheet and S3QM allows for the input of a known water surface level in a basin or waterway when choosing channel or swale as the OSD discharge location.

The full detention storage and discharge controls are required in areas subject to Riverine or overland flooding. As a concession, due to the restrictive environment, no specific downstream controls are required as a result of the flooding. However, standard backwater levels are required as per S3QM.

The minimum base slope of a below ground tank is 2% as per Blacktown City Council's Engineering guide for development 2005. This reduces the maintenance burden in a confined space.

Blacktown City Council will consider reducing this requirement to a minimum of 1% where measures are incorporated to limit the likely deposition of pollutants into the tank. Such measures could include a GPT upstream treating all flows entering the tank, or a combination of approved pit litter baskets and discharging roof water through a rainwater tank or other treatment. To utilise this option, any flows directly entering the tank from the grates over the tank do not need to be treated. However, a minimum of 95% of the vehicle surface area (driveway and parking areas) must be directed to the device to achieve the 90% annual removal of gross pollutants and hydrocarbons required under a VPA or Section 7.11 water quality offset scheme. Where this bypass area (including the hard stand detention tank area itself) exceeds 5% of the hard stand, an additional treatment device may be required downstream prior to discharge.

Where above ground OSD storage is proposed in carparks or other regular vehicular or pedestrian access areas, the minimum percentage storage below ground is to be:

- 50% for high pedestrian traffic areas such as a shopping centre
- 50% for childcare centres, hospitals and aged care facilities
- 25% for all other development types.

For childcare centres, any OSD storage in the rear yard or children's play area must be fully below ground and fitted with childproof locks. Full above ground OSD, which does not exceed 600 millimetres in depth, is permitted in the front landscape setback of the childcare centre providing it is fully fenced with pool gate access.

For permanent detention basins draining larger catchments, typically greater than 10 hectares (ha), additional safety requirements are required such as:

- set the emergency 1% AEP spillway a minimum of 200 millimetres above the 1% AEP storage level
- design the 1% AEP spillway to convey the 1% AEP flows assuming the control pits are blocked
- set the basin crest level a minimum of 300 millimetres above the emergency spillway
- set the top of berm width a minimum of 2 metres wide

- consideration given of Dam Safety Committee requirements (where population is at risk).

For additional detention requirements see Section 3.5.

11.6.4 MUSIC modelling requirements for detention basins

In MUSIC, the Detention Basin node can be used in 2 ways. Firstly, as a conventional basin used to improve waterway stability and/or reduce flood flows. Secondly, to represent the storage component and operation of some water quality devices such as Stormfilter or AtlanFilters.

MUSIC input parameters include:

- identify any High Flow or Low Flow By-passes proposed for the treatment measure. Otherwise adopt the default values of 0 for Low Flow By-pass and 100 for High Flow By-pass
- input the Surface Area of the basin. It is not the actual surface area, but it is the area that when multiplied by the Extended Detention Depth will give the volume of storage. Where the sides of the basin are battered, the Surface Area is the area at half the Extended Detention Depth. That is the average basin area.
- the Extended Detention Depth is the depth between the design storage level and the average base level of the storage. In determining the Extended Detention Depth and average base level, consider the following:
  - for an above ground detention basin, the minimum grade along the base is 1%
  - for a below ground tank, the minimum grade is 2% (unless noted above)
  - where detention is set above a wetland or bioretention basin, set the base of the OSD to the top of the water quality Extended Detention Depth together with an average of any batter slopes
  - for MUSIC modelling for water quality purposes, where most pollutants are generated in storms less than 1 EY, model a conventional detention basin based on: Version 4 UPRCT with an Extended Detention Depth of the difference in level between the 50% AEP (1 in 2 AEP) weir and the average base level
- for Evaporative Loss as percentage of PET, allow 0% for tanks and 75% for above ground detention systems
- Exfiltration is the water lost from the treatment measure into the surrounding soil. Blacktown City Council requires 0 millimetres per hour for detention basins which should be concrete or lined to retain water
- the Low Flow Pipe Diameter will be determined by using 1 of the 4 methods below:

  Method 1: representing the conventional Version 4 UPRCT detention system using a simplified method by nominating only the low flow orifice diameter. This recognises that most of the pollutant removal occurs up to the 50% AEP event

  Method 2: from the detention calculations using the orifice equation

  Method 3: representing a multi-stage outlet in a conventional detention system by ticking the 'Custom Storage and Outflow' and nominate the stage discharge relationship. A discharge/storage spreadsheet can be imported

  Method 4: for water quality detention basins, the low flow pipe option is chosen which represents the design outflow through the filter cartridge underdrains. This is calculated through the orifice equation to the false floor or base of the tank depending upon filter type multiplied by the number of cartridges or through a spreadsheet available from the supplier

- match the Overflow Weir Width to the actual weir length. Choosing too small a weir width will overestimate the performance of the detention basin
- under the ‘Advanced ’ tab the k values are all set to 0.

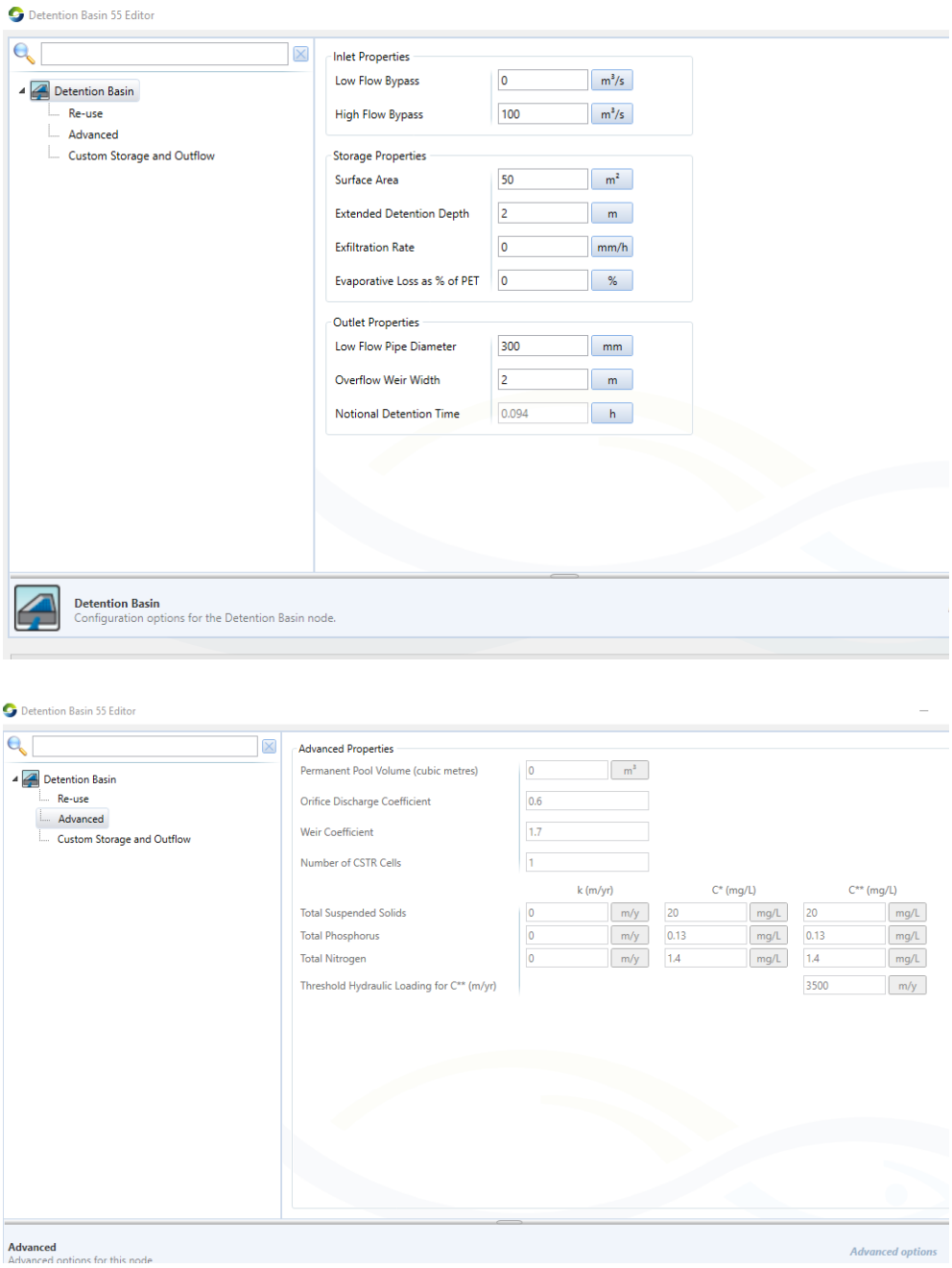


Figure 9. Examples of properties of a detention basin node in MUSIC

11.7 Infiltration systems

Infiltration is not permitted in our city as a treatment system in MUSIC. This is due to the presence of heavy clay as well as significant areas of sodic and saline soils.

11.8 Bioretention systems

11.8.1 General description

Bioretention systems promote the removal of particulate and soluble contaminants by passing stormwater through a filter medium for collection by an underdrain. Well-designed bioretention systems can provide both flow management and water quality benefits. A range of factors affect the treatment performance of the bioretention systems including the type and composition of filter media (for example, loamy sand), together with the type of vegetation used.

All the systems are to be lined with 1.5 millimetres HDPE liner or a minimum 300 millimetres compacted clay or geosynthetic clay liner (GCL). Permanent geotextile is not permitted between any of the layers.

The specifications for bioretention in our city are provided in Blacktown City Council's WSUD standard drawing 6001. Details of a variety of bioretention sizes and layouts are shown on Blacktown City Council's WSUD standard drawings 6002-6009. Bioretention systems are typically densely planted out with sedges and shrubs to help maintain the conductivity of the filter media, promote nutrient removal and create an attractive landscaped form/feature. Large shrubs and some trees are permitted subject to larger filter media thicknesses. Refer to Blacktown City Council's WSUD standard drawing 6007 for allowable plant species and recommended densities.

11.8.2 Temporary bioretention systems

In the growth areas and employment lands where the regional water quality basins have not yet been constructed, temporary water quality basins may be required.

Designers have the option of using bioretention based on Blacktown City Council's WSUD standard drawings 6002-6009, or a temporary bioretention system.

The temporary bioretention typically comprises a 400 millimetres filter layer, a 100 millimetres transition layer and a variable gravel layer depending upon the length of the basin. The gravel thickness is based on the largest subsoil pipe, plus 50 millimetres cover and a subsoil slope of 0.5%.

11.8.3 Hydraulic loading rates for bioretention basins

The biggest single rectification maintenance cost with a constructed bioretention basin is partial, or even total replacement, of the filter media and plants once the filter media becomes clogged. Good design can extend the life of the bioretention basin significantly compared to a poor design. One such design guide is the hydraulic loading rate. In MUSIC, this arrangement would look like the following, assuming the bioretention design in this example is based on a 4 EY flow in Figure 10.

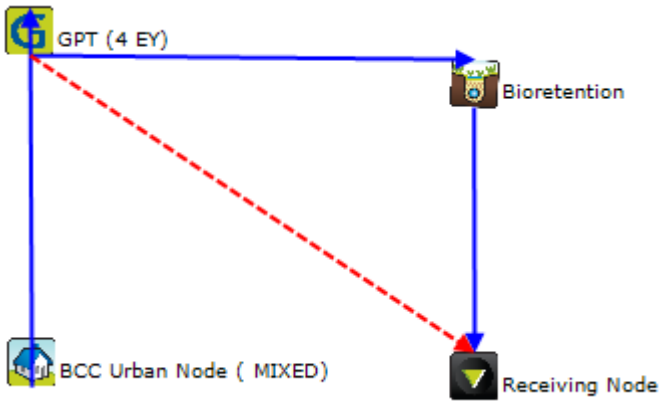


Figure 10. Example of hydraulic loading rate in MUSIC

In this example:

- The GPT has the High Flow By-pass set to the 4 EY (3 month) flow.
- The red dash line from the GPT represents High Flow By-pass. Any flow in excess of the GPT capacity bypasses the bioretention.
- The 4 EY flow is directed to the bioretention.
- Once the model is run:
    - click on the 'Bioretention node'
    - then click on 'Statistics' (drop down menu)
    - then click on 'Mean Annual Loads' (drop down menu)
    - record the inflow Flow in ML/yr noting 1 ML/yr = 1,000 m^3/yr.

Then:

Hydraulic loading rate (m/yr) = Flow (ML/yr) x 1,000 / Bioretention Filter Area (m^2)

Ensure the hydraulic loading rate (HLR) is between 40 and 80 with a desirable target between 50 and 70. This is based on a typical residential catchment. For industrial and business sites with higher TSS loadings, the HLR should be less than 60. For sites with detention storage above the bioretention, the HLR should be between 40 to 60 to offset the additional sediment loading and extra flow through the filter under pressure.

Assuming all other items are fixed, the lower the hydraulic loading rate, the longer before the bioretention will require significant rectification. However, if it is too low, there is a risk the system may not get enough water and plants may experience significant water stress.

11.8.4 Solar access for bioretention systems and thermal impacts

To allow light penetration to the plants during winter, any height above 0.6 m (increasing to 0.9 metres for filter media widths greater than 5 metres) on the north-eastern, northern and north-western sides is to be set to a maximum slope of 1V:1.5H. This can be achieved by using a combination of stepped level planting areas (minimum 900 millimetres wide) and dwarf walls. Blacktown City Council seeks a minimum of 1 hour of sunlight in mid-winter over the filter area.

To reduce thermal loadings or reflected heat to bioretention plants from the south-eastern, southern and south-western sides, the slope should be battered and landscaped at maximum: 1V:3H (private development) or 1V:4H (Blacktown City Council development). Terracing will also be considered with walls typically less than 0.9 metres high and a typical 0.9 metres planting width between to achieve a steeper rise. Tall plants can be used to screen the walls to reduce heat absorption and reflection.

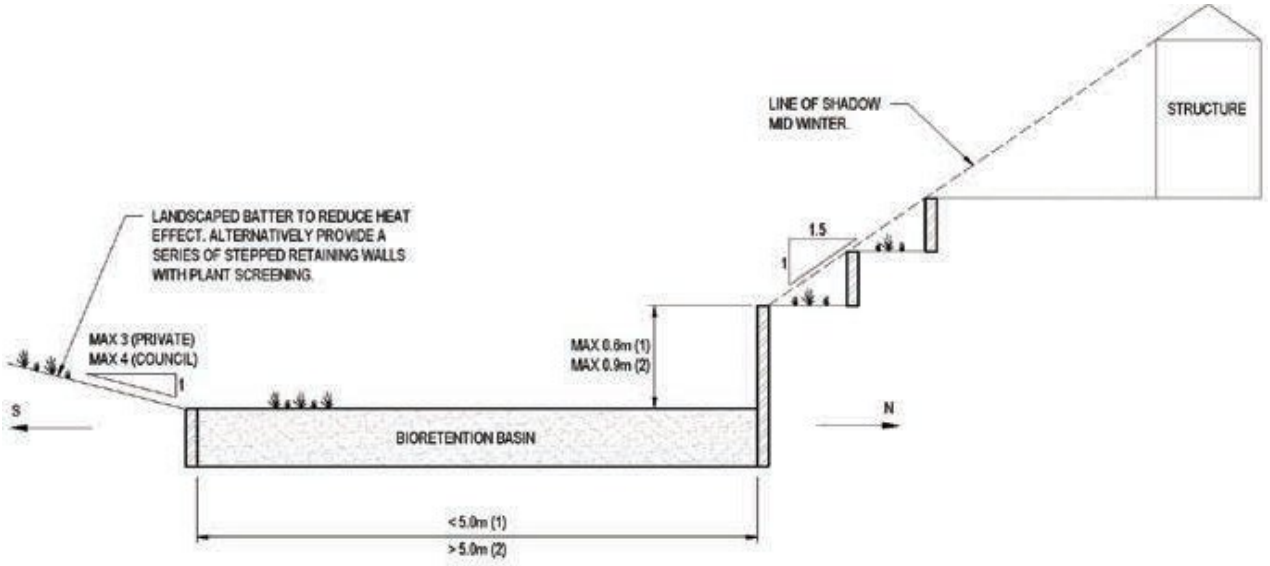


Figure 11. Typical layout of batters and walls to allow sunlight to plants and reduce heat effects

The minimum solar access requirements present difficulties to deep bioretention basins, basins on the southern side of high-rise developments or large industrial buildings, which may be in significant shadow. To assess this impact, shadow diagrams may be required for assessment.

Any part of the bioretention basin filter area that is steeper than this slope or does not meet the minimum 1 hour of sunlight will not be considered as effective filter media area and not included in the MUSIC model. This discounted area can be concreted, or rock lined or similar to act as a flow spreader, or could be planted with shade tolerant plants.This area can be used to provide additional surface area in MUSIC and without contributing to pollutant removal, which will increase performance.

If based on these slope restrictions and/or building shadows, a design cannot achieve sufficient filter area for treatment anywhere on site then the designer should consider that bioretention may not be a practical solution for the development. Figure 12 is an example of a basin width of 4 metres and in MUSIC the effective width is 3.4 metres

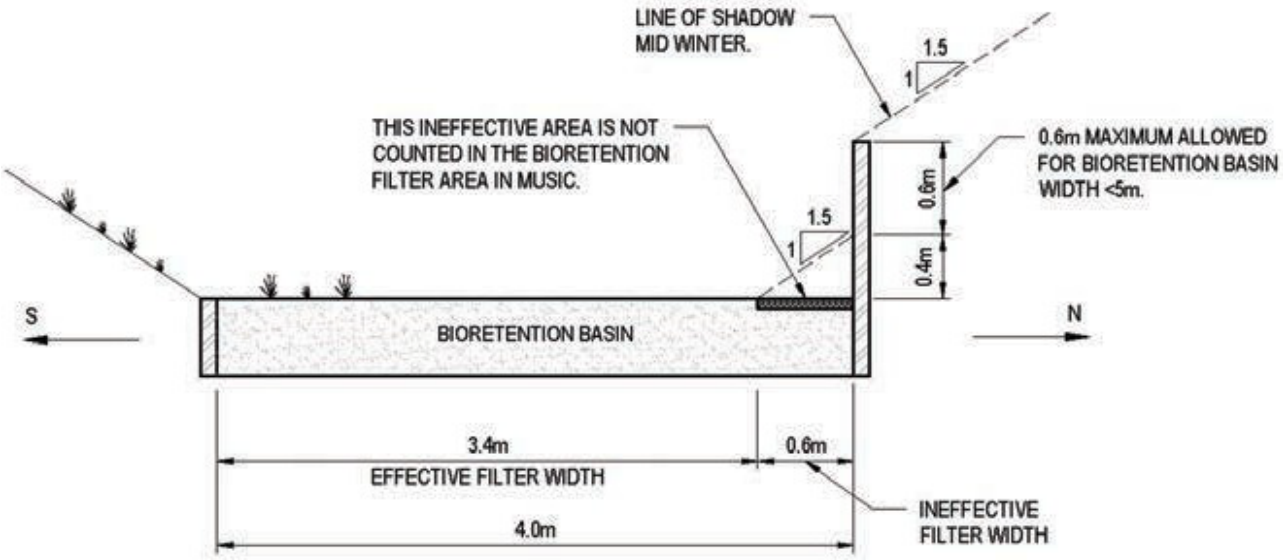


Figure 12. Example of applying the shadow requirements in bioretention basins

11.8.5 Additional design information for bioretention systems

- see Blacktown City Council's WSUD standard drawings 6002-6009.
- high backwater levels can reduce the effectiveness of bioretention basins by reducing or even stopping flows, and ultimately stopping treatment. To reduce the impact of tailwater levels, set the saturated water level at or above:
    - the 50% AEP (1 in 2-year AEP) water level in a natural water body such as a creek or basin immediately downstream of the bioretention basin, or
    - 1 EY in the immediate downstream drainage pit.
- for combined detention with bioretention with filter area greater than 500 square metres, set the top of the filter media to a minimum of the 50% AEP water surface level within the detention basin. Where this contradicts, the bioretention levels determined above the higher level applies
- for temporary basins set the base (invert) of the transition layer at, or above, the 1 EY water level immediately downstream of the basin
- the invert of the GPT discharging to the bioretention is to be set to the top of the extended bioretention detention depth (typically 0.3 metres above the filter media)
- where bioretention is incorporated as part of a detention basin, the subsoil drainage must: discharge downstream of the discharge control pit to maintain a pressure head and ensure ongoing filtration through a range of storms as per the MUSIC model. These treated flows are generally considered low and in small systems, will not affect the performance of the detention basin so can be ignored. In larger basins with filter area of 500 square metres, the orifice size for both 50% AEP and 1% AEP orifice plates must be reduced by manually calculating smaller orifice sizes allowing for a bypass of 0.056 L/s/m^2 of filter area
- three years have been determined as the minimum period required for the bioretention plants to reach maturity and remove pollutants at the required rate. Where bioretention basins are constructed by a developer but will be handed over to Blacktown City Council, the bioretention basin is only to be finalised and planted out once 90% of the upstream development is completed. The developer is then to maintain the bioretention system, including supporting infrastructure (such as GPTs, permeable concrete pipes and silt traps) in addition to the plants themselves for a minimum of 3 years after finalisation. This will be bonded
- the bioretention system is to be lined to prevent exfiltration
- bioretention basins nested into the detention basin must provide either: a separate overflow pit (independent of the subsoil collection pit) that discharges direct to the control pits or provide an overflow weir set to the bioretention EDD with suitable scour protection
- combined permanent bioretention and detention basins will be designed to split the flows so that low (typically treated) flows are directed to the bioretention first. The bioretention will be perched at a higher level than the detention component. The high flow bypass will be directed to the detention basin which can then fill and backflow over the bioretention filter area. Refer to Blacktown City Council's WSUD standard drawing 6002
- in a combined bioretention/detention, the total allowable depth over the filter area is 0.8 metres. This includes the bioretention EDD
- bioretention swales, which have a bed slope greater than 0.5% are not permitted. However, flat linear bioretention systems are permitted provided it can be demonstrated that surface velocities in the 100-year storm will not result in scour of poorly vegetated system (i.e. during establishment)

- bioretention basins must be flat and not on a slope
- roof water or the overflow from a rainwater tank may discharge directly to the bioretention without additional treatment, if required, due to its low sediment content. However, any treatment to remove sediment will be beneficial
- permanent bioretention within private lots, must be connected to a drip or spray watering system at the minimum rate of 0.4 kL/m^2/yr sourced from a non-potable water source such as rainwater and achieve an 80% reuse for that application
- where bioretention basins are constructed adjacent to a building structure or retaining wall, the structure must be designed by a structural engineer registered on NER. The bioretention basin must be self-supporting and allow for replacement of the bioretention media including full excavation down to the underside of the gravel layer
- the sediment forebays or silt traps within the bioretention basin are not to be modelled in MUSIC for any pollutant removal
- permeable concrete or other pipes within the bioretention basin are not to be modelled in MUSIC for any pollutant removal.

11.8.6 Protecting bioretention basins from sediment loading

Bioretention systems are very vulnerable to sediment loading. All discharges (other than roof water as described above) to the bioretention must have pre-treatment to remove as much sediment as possible. A number of alternatives are noted below:

- for bioretention systems with a filter area greater than 100 square metres, Blacktown City Council's preference is a proprietary GPT sized for the 4 EY (3 month) flow (50% of the 1 EY). The GPT should be capable of removing a minimum 45% TSS post development average annual pollutant load reduction. As an alternative for private systems, Blacktown City Council will also consider the use of litter baskets for filter areas up to 200 square metres
- all permanent bioretention systems use a combined silt trap and surcharge pit as detailed on Blacktown City Council's WSUD standard drawing 6006. This is also used as a backup to systems that already include a GPT. This is because Blacktown City Council's experience indicates that there is still a substantial sediment load that can make it through a GPT that will reduce the effective life of the bioretention. A method of draining the residual water within the pit and pipe system needs to be considered within the surcharge pit such as permeable concrete pipes
- for temporary bioretention systems in the growth centres, where the outlet pipes discharge to the filter media surface, provide: a proprietary GPT or a raingarden sediment pit at each discharge point as part of the pipe discharge scour protection. A raingarden sediment pit is typically 400 to 600 millimetres deep. Size the area of the sediment pit proportionally based on the pipe size or flow to be at least twice as long as and as wide as the pipe diameter. For example, a 450 millimetres pipe would typically have a 900 millimetres wide x 1,500 millimetres long sediment pit.

11.8.7 MUSIC modelling requirements for bioretention basins

MUSIC input parameters include:

- identifying whether a by-pass structure will be included within, or upstream of, the treatment measure to control flows, such as a GPT.
- setting the Low Flow By-pass to 0.00.
- setting the High Flow By-pass to 100.00.

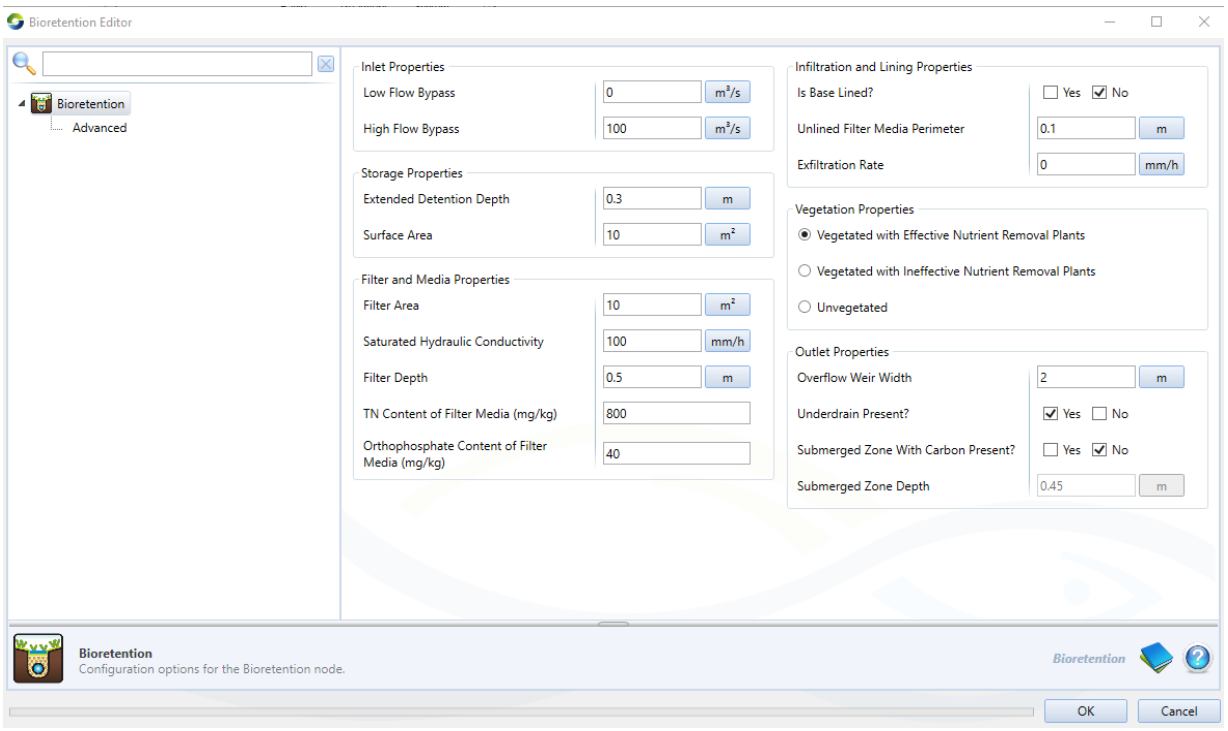


Figure 13. Example of properties of a bioretention node in MUSIC

- identify the Extended Detention Depth (ponding depth) in metres prior to overflowing the control weir of the treatment measure. The maximum Extended Detention Depth is 0.4 metres for Blacktown within private property, but only 0.3 metres for public areas. Blacktown City Council does not accept bioretention swales but will accept linear bioretention basins with the filter media flat
- under the 'Storage Properties' box, provide the Surface Area (m^2) of the surface storage component based upon site constraints. It is not the actual filter media surface area, but it is the area that when multiplied by the Extended Detention Depth will give the volume of storage. Where the sides of the basin are battered the Surface Area is the area at half the Extended Detention Depth. That is the average basin area.
- under the 'Filter and Media Properties' box: Filter Area (m^2) is the area of bioretention filter media available for planting and excludes the areas of pits, sediment traps, flow spreaders, steps and scour protection, plus any area excluded due to sunlight limitations
- Unlined Filter Media Perimeter (m) is set to 0.1 metres (filter is fully lined)
- the Filter Media is a sandy-loam mixture designed to support vegetation/root growth yet still ensure sufficient flow through drainage characteristics
- the maximum Saturated Hydraulic Conductivity permitted in our city is 100 mm/hr
- provide the proposed depth of filter media in metres within the treatment measure. The minimum Filter Depth is 0.4 metres for our city. However, the Filter Depth for most of Council's WSUD standard drawings is 0.5 metres with permeable concrete pipes (this includes regional basins). The following depths are recommended as a minimum within the treatment measure:
    - 0.4 metres for sedges and small shrubs
    - Up to 0.8 – 1.0 metres for tree species

  This will ensure adequate area for root growth is provided within the treatment measure. This depth does not include the transition layer or drainage layer

- TN content of filter media (mg/kg), Blacktown City Council requires this to be less than 1,000 mg/kg. For MUSIC modelling adopt a value of 900 mg/kg
- Orthophosphate content of Filter Media (mg/kg), Blacktown City Council requires less than 80 mg/kg. For MUSIC modelling adopt a value of 40 mg/kg
- Exfiltration is the water lost from the treatment measure into the surrounding soil. Blacktown City Council requires a 0 mm/hr for bioretention basins, which should be lined to retain water.
- please ensure the Base Lined checkbox is ticked as yes
- please select 'Vegetated with Effective Nutrient Removal Plants' in Vegetation Properties. See Blacktown City Council's WSUD standard drawing 6007 for approved plant species.
- Overflow Weir Width (m), as per design
- please ensure the Underdrain Present checkbox is ticked as no. Blacktown City Council requires un-socked PVC slotted pipes within the drainage layer
- please ensure the Submerged Zone checkbox is ticked as no. Although Blacktown City Council does permit submerged or saturated zones for bioretention and is the standard requirement for permanent basins, saturated zones act only as a water source for the plants and not as an extended treatment type.
- the default k-C* values for the bioretention system must not be adjusted without prior written approval from Blacktown City Council.

11.9 Media filtration

There are 3 types of media filtration systems: playing fields, conventional sand filter and permeable pavements. However, only the playing fields and permeable paver approaches are accepted in Blacktown City Council LGA.

11.9.1 Media filtration using conventional underground sand filters

Media filtration usually refers to sand filters that treat stormwater via infiltration through a soil or sand media. Sand filters, unlike bioretention systems, are not vegetated and are usually constructed in underground tanks with much higher filtration rates.

As the sand filters are not vgetated, they can be significantly prone to clogging and require intensive and regular maintenance. These systems present a significant risk due to the frequent confined space entry into a tank for the raking or replacement of sand. The experience of Blacktown City Council is that the owners and operators of such systems significantly underestimate the frequency of maintenance to ensure these systems remain effective. This leads to frequent clogging, ineffective treatment and bypass. Consequently, conventional sand filters are not permitted in our city as a water quality device.

In some circumstances, Council may consider above ground media filtration provided that there is pre-treatment to remove TSS and the media is covered in suitable rock mulch. This will only be considered in the context of a WSUD plan of the whole site and at the discretion of Council.

11.9.2 Permeable pavement filtration

Blacktown City Council accepts permeable pavements provided they are designed in accordance with WSUD standard drawing 6011.

Key modelling considerations include:

- areas of run on water are not permitted. That is, the site will be graded so that only rainfall that falls on the pavers is allowed to percolate into the paver. Another way of expressing this is the ratio of impervious to pervious areas draining to the pavers = 0. This is to ensure the pavers have a minimum 20-yearil life where there are trees overhead and 40 years where there are no trees overhead. Refer to Table 3.1 in *WSUD Source Control Procedures* (Argue, 2008) for more information.
- 40% of the design hydraulic conductivity will be adopted for modelling of flow through the filter media. This is the hydraulic conductivity at the end of the life of the paver and is considered very conservative.
- the pavers will be modelled as a filter media system generally in accordance with the WaterNSW MUSIC modelling guidelines which are:
    - Extended Detention Depth (metres) = 0.00
    - Surface Area (square metres) = area of pavers
    - Exfiltration Rate (mm/hr) = 0.00
    - Filter Median Particle Diameter (mm) = see Table 9
    - the Filter Area should be set to the open area of the paving material. Some pavers are solid concrete with hexagonal openings in each corner. Others are resin bound and have massive opening areas. This should be a maximum of 40% for resin bound pavers and equal to the opening area for open pavers, which are often around 5% to 10% of the paved area. Provide supporting information from the paver supplier with any application
    - set the Saturated Hydraulic Conductivity to the smallest median particle diameter (D_{50}) in the pavement profile from the base, or sub-base layers. See examples in Table 9. The maximum saturated hydraulic conductivity for a sand media filter in Blacktown is 360 mm/hr. Blacktown City Council requires certification from the filter media supplier, or engineer, that the filter media has a minimum hydraulic conductivity as defined by ASTM F1815-06 (actual, not predicted) of 4 times the rate specified in MUSIC
    - set Filter Depth to equal depth of base and subbase layers but noting that coarse gravel layers do not improve water quality and should not be included. Media should be finer than 1 millimetre which is the median particle diameter for a coarse sand transition layer which is not to be modelled
    - set the Weir Length equal to the side of the paved area grade to
    - under the Advanced tab set the k values to TSS 400, TP 300 and TN 40.

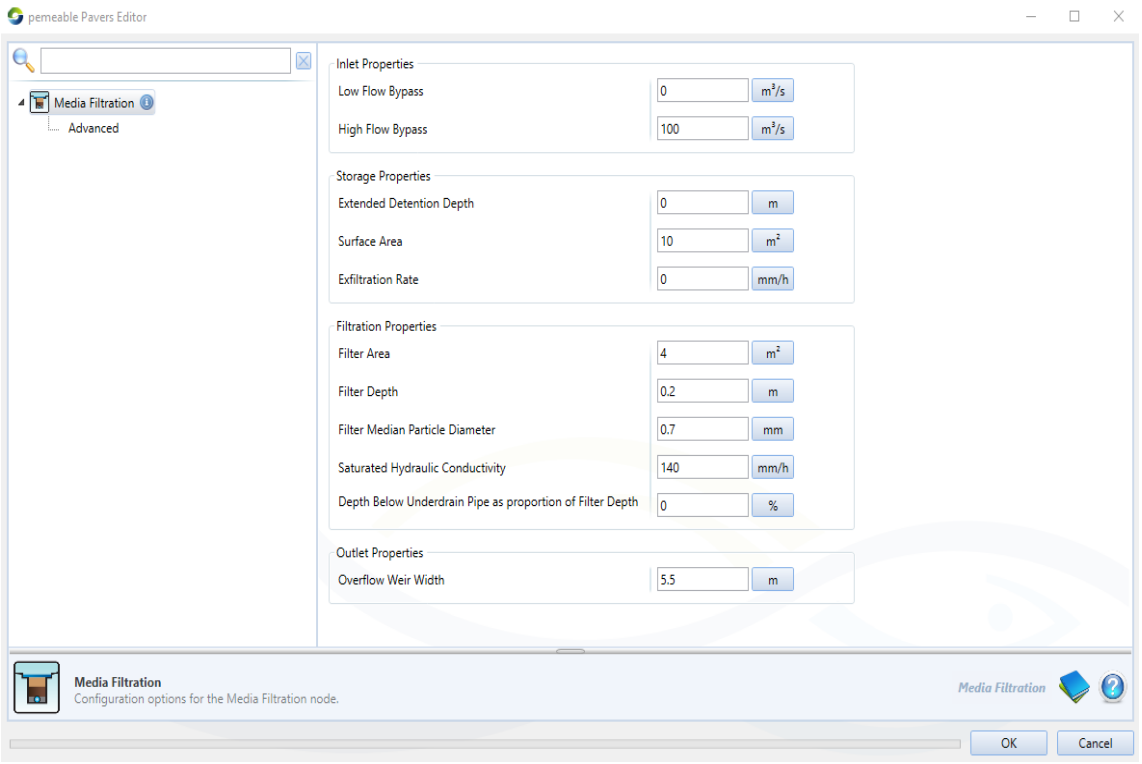


Figure 14. Example of properties of permeable pavers in MUSIC

Soil type	Median particle size D50 (mm)	Saturated hydraulic conductivity (mm/hr)
Coarse sand	1.0	360
Sand	0.7	140
Fine sand	0.45	70

Table 9. Filter median particle size and saturated hydraulic conductivity

11.10 Green roofs

Green roofs are a way of reducing directly connected impervious areas from an urban development and aid in reducing the urban heat island effect.

Blacktown City Council will only accept well proven proprietary green roof systems that include well protected waterproof membranes and robust drainage layers. All waterproof membranes must be tested for leakage during construction and certification from the installer must be provided.

To encourage more extensive use of green roofs and utilise the criteria and modelling procedure outlined below, Blacktown City Council requires a minimum of 25% green coverage for a roof. The roof area must comprise a minimum of 90% of the total site area.

Blacktown City Council also requires that green roofs are irrigated to a nominal depth of 400 millimetres per annum (0.4 kL/m^2/ per year) unless: specialist xeriscaping is undertaken by a landscape design professional with green roof experience and they specify a reduced level of irrigation.

To apply the method under this section, the design must:

- have a waterproof membrane that is tested for leakage

- be able to structurally support the green roof and weight of plants and media
- have a minimum soil depth of 400 millimetres and be a loam soil complying with the definition of loam soil in AS4419 (certificates will be required)
- have a minimum soil depth different to that shown above where certified by an experienced green roof landscape designer
- consider inclusion of a rockwool or similar water absorbent product to reduce frequency of runoff and support vegetation growth through wicking
- have a robust underlying drainage system
- have an automated irrigation system.

There is no default Green Roof node in MUSIC so Blacktown City Council has developed an appropriate generic node that reflects the above requirements.

The following approach will be adopted for MUSIC:

Step 1: split the roof into 2 source nodes. The first being that part of the roof that is impervious and is a traditional roof and the second being that part of the roof represented by the green roof

Step 2: model the source node representing that part of the roof that is to remain impervious as an impervious node using the BCC other impervious areas land use. It must be directed to a rainwater tank or downstream treatment or receiving node. It is to be 100% impervious

Step 3: model the green roof as a 100% impervious node with a BCC other impervious areas land use and direct it to a generic node with the parameters noted in Table 10.

Pollutant	Input	Output
Flow	100	32
Total Suspended Solids	100	100
Total Phosphorus	100	43
Total Nitrogen	100	26.4

Table 10. Green roof parameters

11.11 Design of gross pollutant traps (GPTs)

GPTs typically remove rubbish and debris and can also remove sediment and hydrocarbons from stormwater runoff depending upon the type of device.

These treatment measures can be very effective in the removal of solids conveyed within stormwater which are typically larger than 5 millimetres in size. Some devices can remove finer sediments. Different devices remove different amounts of gross pollutants, TSS, TP, TN or hydrocarbons.

All proprietary GPT Nodes for use in the treatment train have to be pre-approved by Blacktown City Council. Blacktown City Council currently has MUSIC nodes available for a range of devices. Designers need to obtain the MUSIC nodes or check on MUSIC-*link*. These nodes will set the removal rates for the pollutants within MUSIC.

11.11.1 Design standards

Design event

- 4 EY as part of a treatment train or upstream of a bioretention basin or wetland design.
- 2 EY for independent devices.

For GPTs approved for removing oils and hydrocarbons to Blacktown City Council targets, see Section 11.16.

GPTs must be sized appropriate to the GPT, TSS, TP and TN loading within the catchment allowing for a maximum 2 EY (6 month) cleaning cycle.

In the absence of better hydrologic information, consider the minimum design flows as:

- Q4EY (m^3/s) = 0.07 x A (ha)
- Q2EY (m^3/s) = 0.09 x A (ha)

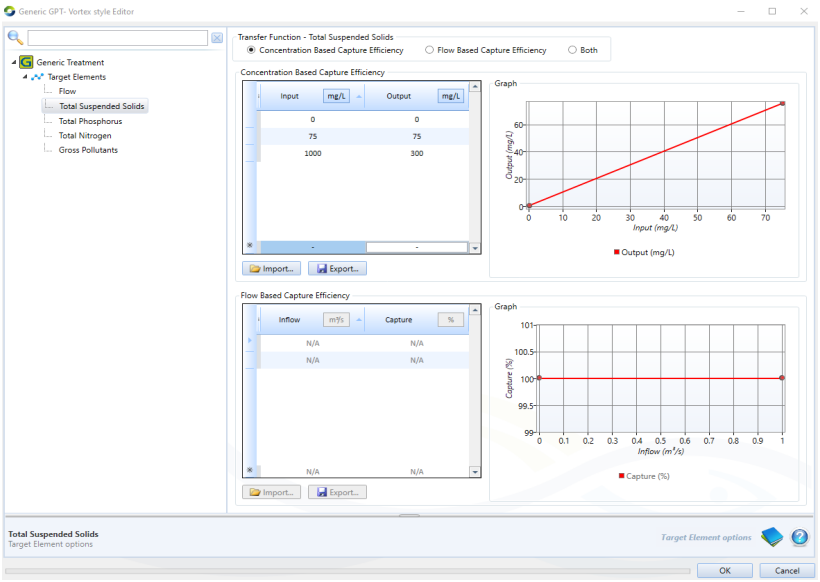


Figure 15. Example of TSS removal in a Vortex style CDS node

Storage volume

The actual device must be capable of storing twice the minimum expected loading. The storage capacity should be clearly defined and should not impede the screen area or affect the flow rate capacity.

Miscellaneous requirements

GPTs upstream of a regional bioretention basin should achieve a minimum 45% removal of overall TSS when run in MUSIC.

Where GPTs are placed adjacent to a street with a standard 9 metres carriageway, the carriageway is to be widened to 10 metres (while maintaining the 3.5 metres nature strip) to provide parking for an eductor truck used to clean the GPT.

Set the invert of the GPT discharging to a bioretention basin to the top of the extended bioretention detention depth (typically 0.3 metres above the filter media).

Using the approved Blacktown GPT node:

- calculate the required high flow by-pass for the treatment measure:
  - 4 EY as part of a treatment train or upstream of a bioretention basin or wetland design
  - 2 EY for independent devices

  This will depend upon the proprietary device or diversion weir. In some cases, the allowable flow through the device approved by Blacktown City Council may be less than that claimed by the manufacturer
- the concentration based capture efficiency should be checked to comply with Blacktown City Council approved pollutant removal rates.

11.12 Buffers

Buffers are not permitted as a pollutant removal device within MUSIC in our city.

11.13 Swales

Vegetated swales are open vegetated channels that can be used as an alternative stormwater conveyance system to conventional kerb and channel along roads and associated underground pipes. The interaction of surface flows with the vegetation facilitates an even distribution and slowing of flows. This encourages particulate pollutant settlement. Swales can be incorporated into streetscape designs and can add to the aesthetic character of an area.

Swales are both source and conveyance controls. However, they are not to be confused with grassed channels. For swales to affect water quality treatment devices they need to have low hydraulic loading rates and have equally low velocities. As such, swales are typically located alongside roads where lateral inflow occurs. Swales are not to be used as an end of line treatment device. That is, a grass channel where depth of flow is typically too deep to allow water to come into contact with the grass. These situations are unlikely to comply with the maximum permissible velocity criteria.

Swales are encouraged as general good practice. However, to provide pollutant removal in MUSIC they need low velocities, no erosion, continuous vegetation cover and good maintenance.

11.13.1 Design standards

Hydraulic design criteria

Use the Manning's equation with n = 0.04 to meet the following velocity requirements:

- maximum permissible slope of 1%
- provide calculations on drawings:
  - 1 EY then maximum V_{swale} < 0.5 m/s
  - 10% AEP then maximum V_{swale} < 1.0 m/s
- refer to Blacktown City Council's WSUD standard drawing 6010.

Vegetation

Thick dense turf only (typically mown 50 to 100 millimetres high), no macrophytes or native grasses. Macrophytes and grasses are appropriate in bioswales as they have no bed slope.

Irrigation

To ensure continuous vegetation cover, a permanent watering system is required that delivers at the minimum rate of 0.4 kL/m^2/year. Irrigation should cover the wide root system. Consider the health implications and risks of spray irrigation. If using harvested stormwater, subsurface irrigation may be advisable.

Check dams and drop structures

When considering steep swales, the hydraulic assessment will determine maximum allowable grades. Where the site grades cause the velocities to exceed the nominated values, it may be possible to set up a series of flatter sections with drop structures and still allow modelling in MUSIC. Where this cannot be achieved, Blacktown City Council will not accept the use of swales as they are likely to erode and have poor water quality outcomes.

See the use of check dams on steep swales in Blacktown City Council's WSUD standard drawing 6010 sheet 2.

11.13.2 MUSIC modelling requirements for swales

How to model in MUSIC

Having satisfied the criteria above, use the following approach in MUSIC when modelling a swale with lateral inflow into the swale. That is, inflow coming in from the side of the swale such as a roadside swale:

Step 1: divide the swale into 3 equal segments

Step 2: create a source node that would drain to the top segment and direct this to the second or middle swale segment

Step 3: create a second source node to represent the catchment draining to the middle swale segment and direct this to the last swale segment

Step 4: create a source node that would drain to the lowest segment and direct this to join the model after the swale so that it receives no treatment in the swale.

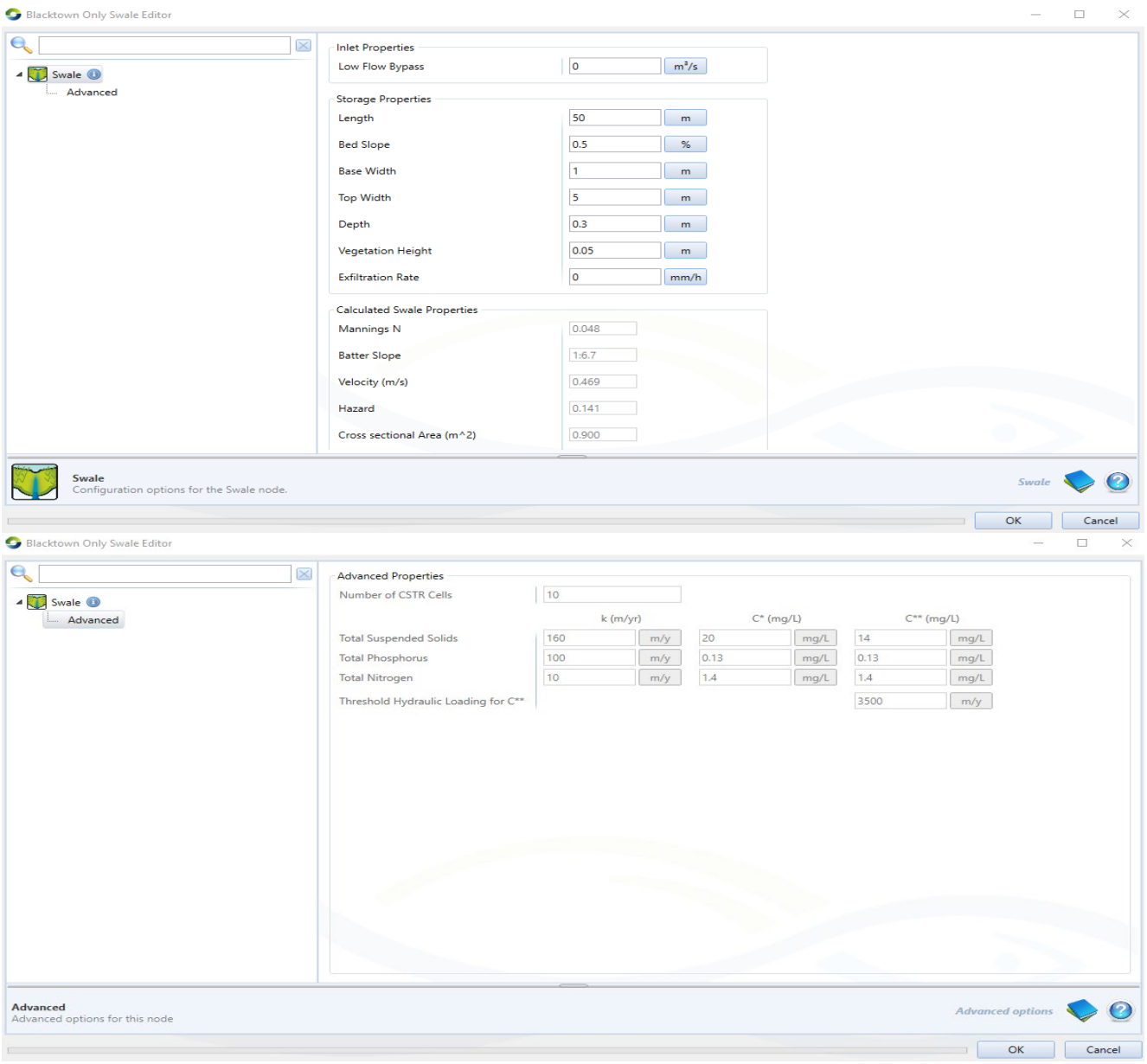


Figure 16. Swale node for MUSIC

This approach would see the top catchment treated in the middle and lower segments and the middle catchment only treated in the lower segment, while the lower segment does not receive treatment.

To create the Swale node, input the following parameters into the Blacktown Swale MUSIC node

- the Low Flow By-pass is set to 0
- length will be determined by whether the full flow is conveyed along the full length of the swale or equally distributed. See guidelines and comments above
- the swale characteristics of Bed Slope, Base Width, Top Width and Depth are all required to match the hydraulic assessment previously calculated
- Vegetation Height: Assume 0.05 metres
- Exfiltration is to be 0 millimetres per hour for swales
- under the Advanced tab set the 'k (m/yr)' values to TSS 160, TP 100 and TN 10.

11.14 Rainwater and stormwater reuse

11.14.1 General description and considerations

Rainwater and stormwater can serve multiple purposes. Rainwater and stormwater harvesting systems:

- are designed to provide an alternative source of water for non-potable uses such as:
    - toilet flushing, hot water, laundry washing, vehicle washing, irrigation, industrial wash down or industrial process water. Rainwater tanks must only accept runoff from a non-trafficable roof
    - rainwater and stormwater that has come into contact with the ground generally requires significant additional treatment and effective disinfection prior to reuse. Industrial process water using stormwater is subject to specific assessment. Stormwater tanks accept runoff from a variety of sources resulting in reuse water of a lower quality than rainwater. Such sources include surface flows, trafficable roofs with areas such as courtyards, barbeques and landscaped areas, green roofs and overflows from rainwater tanks
- are a treatment measure, as some settling occurs in the storage tank, and when the water is used, pollutants are removed along with the water
- reduce the flows from the site in storms to protect the local waterways by reducing the frequency and duration of such flows. It also protects and enhances natural watercourses and their associated ecosystems and ecological processes and minimise stream erosion. As such, they can contribute to the reduction of the SEI. They are not intended, nor should they be seen as a component of detention for major storm or flood events. Rainwater tanks do not provide any offset to detention storage requirements
- maximise amenity by integrating water cycle management measures into the landscape and urban design
- ensure the principles of ecological sustainable development are applied in consideration of economic, social and environmental values in water cycle management.

11.14.2 Design of rainwater and stormwater tanks

Residential development

There are no minimum reuse targets (except as noted below) as the development is subject to BASIX. However, the tank will form part of the treatment train for removal of pollutants. Only rainwater tanks will be permitted for this type of development.

The exception to the above is where medium to high density residential development uses water quality swales and bioretention as part of the treatment train. Swales and permanent bioretention within private allotments must have a non-potable watering system for the filter media. This will ensure optimum plant health and effective pollutant removal and shall allow for a depth of irrigation of 400 millimetres annually over the whole bioretention area including the batters. The irrigation water would ideally be supplied from a non-potable source such as a rainwater tank.

Industrial and business developments

The tank size will be determined to generally meet the 80% non-potable reuse requirement under Part J. This includes the business component of any mixed use or shop top housing (including landscape watering).

Note that the determination of the non-potable reuse demand is a function of the total site demand for non-potable water. This would include the demand for toilet flushing, laundry water and outdoor irrigation. It would not include cooling water. The following section describes the methods that are to be used to work out the non-potable demand on a site and how to then model this in MUSIC to demonstrate compliance with our development controls.

The effectiveness of the rainwater tank at meeting the demands upon it can be evaluated by clicking on the ‘Rainwater Tank’ of Scenario Results after running MUSIC. Right click on ‘Node Water Balance’ tab to review the percentage Reuse Demand Met result in the table.

In areas where recycled water is available from Sydney Water, a rainwater tank is still required for business and industrial development. However, the tank top-up or solenoid bypass is to be supplied by recycled water. Similarly, when a rainwater tank is provided as part of the treatment train or in meeting the SEI target, the rainwater tank top-up must be supplied by recycled water in residential development where recycled water is available.

First flush systems are used to remove the initial flush of pollutants that are washed off the roof at the start of the storm and should be provided on all tank arrangements. This does not need to be modelled in MUSIC for residential development (due to small roof area).

For larger first flush systems for industrial and some business development, the impacts of the first flush system on the MUSIC model performance can be modelled.

MUSIC modelling of first flush systems is optional for roof areas less than 1 hectare draining to the tank but must be modelled for roof areas greater than 1 hectare draining to the tank.

Consider the first flush system as a detention tank set upstream of the rainwater tank in the model, with the tank volume equivalent to the volume of first flush device and the low flow pipe diameter equal to the dribble hole size or outlet pipe size.

Under the ‘Advanced’ tab, set k = 0 for TSS, TP and TN. The overflow is directed as primary flow to the rainwater tank and the low flow pipe flows directed as secondary flow to the next device bypassing the rainwater tank.

11.14.3 Non-potable reuse rates for modelling reuse tanks in MUSIC

The following demand rates are provided as a guide for MUSIC modelling purposes. Internal use is a daily demand and irrigation is an annual demand in MUSIC.

For **residential development (excluding home units or multistorey dwellings)**, allow for rainwater reuse per dwelling based on the area of lots as follows:

Lots	Allowance
lots > 730 m^2	allow 0.1 kL/day internal use & 55.0 kL/year as PET- Rain
lots > 520 & < 730 m^2	allow 0.1 kL/day internal use & 45.0 kL/year as PET- Rain
lots > 320 & < 520 m^2	allow 0.1 kL/day internal use & 32.0 kL/year as PET- Rain
lots < 320 m^2	allow 0.1 kL/day internal use & 25.0 kL/year as PET- Rain
row houses, villas and townhouses	allow 0.1 kL/day internal use & 20.0 kL/year as PET- Rain
apartments or home units	allow 0.0 kL/day internal use (that is, zero daily demand) & 0.4 kL/year/m^2 of watered landscape areas (excluding turf) as PET- Rain (where used)

For general **industrial and business developments (as defined in Part J)**, including schools, child-care centres, places of worship and not-for-profit organisations, allow for reuse as follows:

- **Toilets and urinals**: for internal rainwater reuse, allow 0.1 kL/day per toilet or urinal (if using S3QM, this is called a high-use toilet) in industrial/business developments including any disabled toilet. However, where the site is only occupied for 5 days per week, the daily usage rate is to be proportioned by 5/7. Similarly, where there is an additional afternoon or night shift, increase the rate proportionally depending upon staffing numbers. For schools, allow for rainwater usage of 0.06 kL/day per toilet or urinal
- **Irrigation**: for irrigation of landscaped areas only (excluding turf), allow 0.4 kL/yr/m^2 as PET-Rain under Annual Demand for sprinkler systems or subsurface irrigation. Higher rates may be required for specific landscape requirements. Such demand rates will not be accepted by Blacktown City Council for in the MUSIC model. However, the tank size can be increased to allow for such demand. The exception is playing fields where specific watering rates from an irrigation specialist is provided
- **General washdown** industrial/business:
    - for small units less than1,000 square metres, provide 1 non-potable tap per industrial unit
    - for larger units, provide 1 non-potable tap per 50 metres of building perimeter
    - allow a daily usage of 0.005 kL/day per tap.
- **Car washing (hand):** detail the estimated number of vehicle washes required. This could be allocated daily or monthly depending upon whether the demand is constant throughout the year. For hand washing of cars, allow:
    - 0.02 kL/wash for systems that collect and recycle wash water (if possible)
    - 0.03 kL/wash for systems that discharge direct to the sewer or other approved treatment system
- **Automatic vehicle washing or other**: other internal usage may involve automatic car washes, truck washing, or other industrial usage. Specific data will need to be provided from suppliers or others to justify these reuse rates

- **Motels and hotels**: specific internal usage requirements:
    - allow for a daily demand of 0.025 kL/day for toilet flushing in a private room or motel/hotel suite. If using S3QM, this is called a low-use toilet
    - allow for a daily demand of 0.1 kL/day for toilet flushing in a public or staff area, including any disabled toilet or for rooms with multiple bed occupancies (such as in a nursing home or hospital). If using S3QM, this is called a high-use toilet
- **Cooling towers**: only potable water is recommended for use in cooling towers. Cooling tower demand is not counted as non-potable demand
- **Hot water usage**: rainwater (not stormwater) for hot water use is technically feasible providing the water reaches the correct temperature to kill the pathogens. The Committee on the Uniformity of Plumbing and Drainage Regulations (CUPDR) recognises that rainwater can be used for hot water
- **Boarding houses**: exempt from the 80% reuse requirement as all boarding houses are subject to BASIX
- provide a calculation sheet to detail how the final non-potable usage rates have been determined.

11.14.4 Tank size allowances

The effective rainwater tank volume used in MUSIC is smaller than the physical tank size to allow for:

- anaerobic zone
- off-take points
- mains water top-up levels
- overflow levels.

For physical tank sizes, allow for a 10% reduction in effective tank volume for MUSIC before overflow. For example, where a 10.0 kL tank is physically specified on the drainage plan, it is to be modelled as 9.0 kL in MUSIC.

Where a tank is supplying a high landscape demand such as a playing field, the percentage reduction may need to be higher than above. An irrigation specialist must assess and ensure there is sufficient buffer in the tank for the next irrigation cycle, as mains water refill rates may be significantly lower than the irrigation flow rate.

For residential development, the physical tank size is as required for BASIX but reduced as above for MUSIC. Where rainwater tank sizes proposed by the designer are larger than those specified in BASIX, amend the BASIX certificate to match.

When assessing low density residential subdivisions, allow for a rainwater tank size of 2.25 kL supplied, but modelled as 2.0 kL in MUSIC per dwelling. Also allow for a surface area of rainwater tank of 1.7 square metres per dwelling. Consider only 50% of the roof area draining to the tank.

11.14.5 Achieving 80% non-potable demand for business and industrial development

For business and industrial development, the target is 80% non-potable reuse. Residential development generally has no minimum percentage reuse (except as noted previously). To design a tank for reuse involves balancing the supply and demand and selecting an appropriate tank size. The simple guidelines are as follows:

- increasing tank size increases percentage reuse
- increasing roof area increases percentage reuse
- increasing demand decreases percentage reuse.

In some business developments it may not be possible to meet the 80% non-potable reuse target using rainwater alone due to high demand and/or small roof areas. The first response is to provide waterless urinals to reduce the demand. If the target is still not met, provide a rainwater tank and a separate stormwater tank and split the non-potable demands. Typically, all toilet flushing is met through rainwater, when the other uses, such as landscape watering, is met through treated stormwater (where fit for purpose). The design of the stormwater tank and its constraints are detailed in Section 11.14.2. As the stormwater tank has a much larger catchment area than the rainwater tank collecting only roof water, the use of smaller stormwater and rainwater tanks can result in a more economical solution than a single much larger rainwater tank.

In some situations, even after splitting the non-potable demand it may still not be possible to achieve the 80% non- potable reuse for the rainwater tank. In most of these situations the demand and roof area are fixed and the only variable is the tank size. To meet a reasonable proportion of reuse, run a series of MUSIC models varying only the rainwater tank size while maintaining the roof area and demand.
Plot the various MUSIC tank sizes in kL on the X axis and corresponding percentage reuse on the Y axis.

In the example in Figure 17 below, Blacktown City Council would accept a rainwater tank size in MUSIC of 65 kL (73 kL constructed). Beyond 65 kL the curve flattens off and the increase in percentage reuse slows significantly even where the tank size becomes much larger.

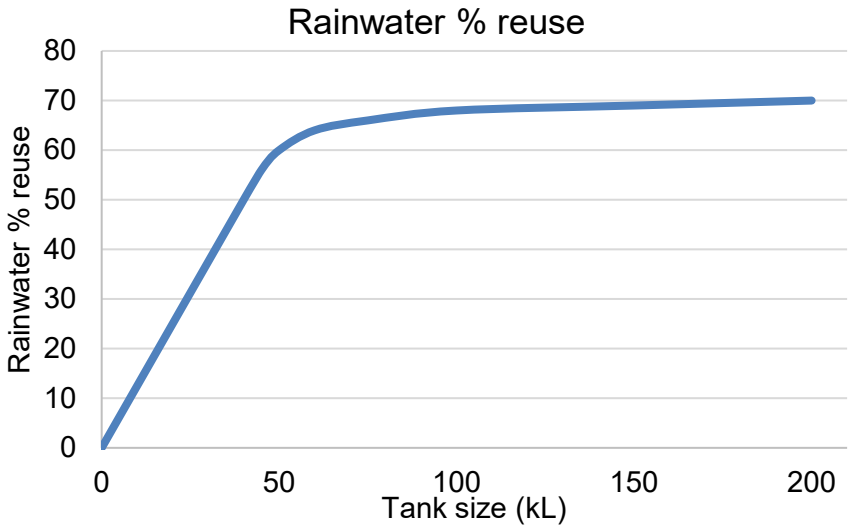


Figure 17. Rainwater percentage reuse

11.14.6 Stormwater tank modelling constraints

Rainwater tanks provide the highest quality of non-potable water. However, stormwater tanks can be effectively used to assist in meeting the 80% non-potable reuse target when roof water draining to a rainwater tank alone cannot meet the demand.

Stormwater tanks differ from rainwater tanks in that they may collect water from a variety of sources including driveways, parking areas and landscaped areas as well as rainwater tank overflows. These varied sources can adversely affect the quality of water and increase the range and levels of pollutants that may be captured.

Two key reference documents include:

1) *National Water Quality Management Strategy, Australian Guidelines for Water Recycling: Managing Health and Environmental Risks (Phase 2) Stormwater harvesting and reuse* (July 2009), Natural Resource Management Ministerial Council, the Environment Protection and Heritage Council, and the National Health and Medical Research Council.
2) *Managing urban stormwater: harvesting and reuse* (April 2006), Department of Environment and Conservation NSW.

Consequently, stormwater reuse is permitted where it can be demonstrated that the water will be fit for purpose and comply with NSW Environment Protection Authority's Stormwater harvesting and reuse guidelines. This may require disinfection or limiting exposure.

Where stormwater reuse is proposed, a risk assessment shall be undertaken and submitted as part of the DA documentation. Stormwater reuse is best suited for developments that have comprehensive secondary filtration of stormwater.

The water draining to the tank must be pre-treated to reduce the risk from the varied pollutant sources draining to it. This reflects the multiple barriers approach in the Australian Guidelines for Recycled Water (stormwater harvesting and reuse).

Once the stormwater is treated in a best practice stormwater treatment train, which includes primary (GPT) and secondary treatment (fine filtration), pumped and filtered stormwater from the tank should be subject to: further and finer filtration and possibly disinfection, depending upon end usage. To minimise the risk of exposure, watering through drip irrigation is not only more efficient, it is preferred to aerial spraying. The filter has to be fine enough to protect the drip nozzles from clogging.

If the treated stormwater is to be used for toilet flushing in a residential development, it would need to be effectively disinfected. The design of such systems may be subject to the *Water Industry Competition Act* (WIC Act).

An engineer registered with NER must certify that the non-potable water from the stormwater tank and:

- has as a minimum, met the criteria under Tables 4.5 and 6.4 of *Managing Urban Stormwater: Harvesting and Reuse* by the Department of Environment and Conservation NSW December 2006. Also, the requirements in *National Water Quality Management Strategy (NWQMS), Australian Guidelines for water recycling: managing health and environmental risks (phase 2)* (2009) including an assessment as prescribed in the guidelines
- has assessed whether higher standards may be appropriate for certain uses subject to risk assessment
- is appropriate for its intended use
- has considered all the requirements of the backflow prevention and cross connection control guidelines contained in the latest CUPDR publications
- ensure all the stormwater reuse pipes are coloured with a purple stripe or similar to differentiate with the purple rainwater reuse pipe. This is to avoid cross contamination between the 2 different reuse systems and with the mains water supply pipes (see Figure 18)
- established an annual cross contamination check to ensure that the stormwater reuse pipe has not accidentally been connected with the rainwater reuse or potable water

- clearly detailed on the non-potable water supply and irrigation plan where additional water quality treatment may be required to meet the targets.



Figure 18. Example of different types of purple pipes.

11.14.7 Improving the cost effectiveness of rainwater tanks

There are significant ways that developments can reduce costs affiliated with their rainwater tank systems as listed below:

- **Use an above ground tank in lieu of an underground concrete tank**: below ground concrete tanks last longer (approximately 80 years) compared to above ground tanks. Above ground tanks may need to be replaced twice in an 80-year period. The much higher capital costs of concrete tanks, including the cost of excavation and removal of soil, is still substantially higher than the multiple steel or other above ground tanks. Such comparison assumes there is sufficient space on site to locate the above ground tank together with appropriate screening for larger business tanks
- **Supply rainwater for additional uses**: when the modelling achieves the minimum 80% reuse requirement for standard reuse for business and industrial development, or any tank sized for residential purposes, then any additional non-potable uses will further reduce potable water demand. For example, cold water supply to laundries or watering turf areas which is not normally considered in the modelling. This will provide additional savings on the business owner's Sydney Water bill
- **Increase the roof area draining to the rainwater tank:** maximising the roof area draining to the tank will generally result in smaller tank sizes or more efficient systems. Where redevelopment covers only part of the site, the efficiency of the rainwater tank is improved if additional roof area from elsewhere on-site can be directed to the tank
- **Use of solar panels to offset electricity usage**: a solar panel system can be designed to offset electricity costs for the tank pump
- **Reduce operation and maintenance costs**: use landscaper or general building maintenance staff for regular maintenance tasks such as cleaning first flush systems, filter systems and silt traps rather than using specialist contractors.

11.14.8 Special requirements for private hospitals, aged care centres and nursing homes

Internal reuse of rainwater is not recommended for these facilities.

11.14.9 Modelling rainwater and stormwater tanks in MUSIC

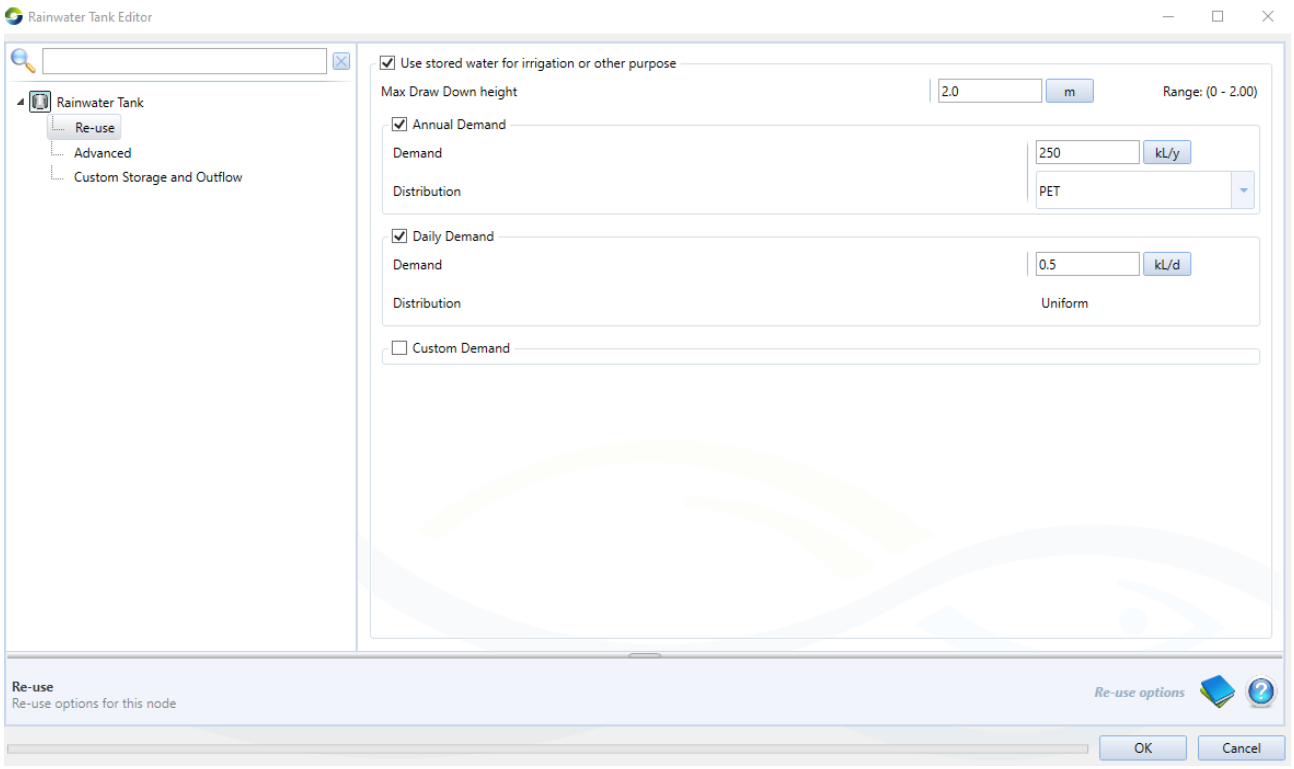


Figure 19. Example of properties of a rainwater tank in MUSIC

Input parameters for either rainwater or stormwater tanks

- identify any High Flow or Low Flow By-passes proposed for the treatment measure. Generally, the default values are retained. A Low Flow By-pass may be appropriate where there is a diversion to a large first flush system such as on an industrial development (see also the notes on modelling for a first flush system described in Section 11.14.2). A High Flow By-pass would apply where there is a 2-stage downpipe system designed to operate at differing flow depths in the gutter so that only the flows up to a certain flow are directed to the tank and the overflow bypasses. Such applications may apply in larger industrial buildings
- input the Tank Volume for MUSIC (with percentage reduction from the physical size as noted above in Section 11.14.4)
- the Depth Above Overflow can be determined from the detailed design or a reasonable estimate of overflow pipe size
- the Surface Area can be determined from the available information where the tank size is known, or a reasonable estimate
- the Overflow Pipe Diameter determined from the detailed design or a reasonable estimate
- set the Initial Volume as the Total Volume
- click on the Re-use tab
- tick ‘Use stored water for irrigation or other purpose’

- for irrigation, using PET-Rain is recommended. It is defined as an annual demand (kL/yr) and scaled according to the daily PET value minus the daily rainfall data contained in the Meteorological template used to create the model rainfall (that is, when PET exceeds rainfall, reuse will occur, or more simply you don't water the garden when it is raining.) Daily demand (kL/day) refers to more constant internal usage such as toilet flushing, laundry use, some industrial processes or vehicle washing. Monthly distribution would only apply to a specific industrial reuse or possibly vehicle washing (this parameter is not generally used)
- details of general allowable rates are indicated above.

11.14.10 Additional stormwater tank MUSIC modelling requirements

The stormwater tank is to be modelled in MUSIC using the Rainwater Tank node. Consequently, the characteristics of a stormwater tank in MUSIC mirror that of a rainwater tank. The designer needs to ensure that when the 'Use stored water for irrigation' or the 'Other purpose' box is checked that the demands are appropriate and fit for purpose.

Where a stormwater tank is used in MUSIC, it does so as part of a treatment train. From the parameters detailed in Section 11.14.9:

- the upstream node, representing the pre-treatment devices suggested in Section 11.14.6, are to bypass the overflow or High Flow By-pass around the stormwater tank. The treated (for example pipe or underdrain) flow is only directed to the stormwater tank as a secondary link
- in MUSIC, when the pre-treatment device is an offline system, an additional Generic node upstream of the device is added to represent a diversion weir. At the Generic node, direct the High Flow By-pass around the proprietary device and stormwater tank. Only the Low Flow is directed to the proprietary device as a secondary flow link with all the flow from the device then directed to the stormwater tank.

11.15 Generic node

This node allows the user to simulate the treatment performance of water quality measures not listed within the default parameters. The use of these nodes for specific treatment devices is not permitted without approval from Blacktown City Council. A range of approved Generic nodes are available directly from Blacktown City Council for a range of existing proprietary devices or through MUSIC-*link*. The methodology of using the current approved proprietary devices is detailed in Chapter 12.

The use of the Generic node is permitted when used for a flow transfer function without any treatment. This may be to represent a diversion weir. Such nodes are also used when determining the SEI as detailed in Chapter 13.

11.16 Hydrocarbons and oils

Part J requires greater than 90% removal of the average annual load of hydrocarbons. Total petroleum hydrocarbons (TPH) is a surrogate for hydrocarbons.

Higher risk developments include:

- service stations
- fuel depots
- car and truck washing facilities
- high turn-over uncovered car parks which have over 50 car spaces

- mechanical workshops and car sales yards that have outside parking for more than 20 vehicles.

Blacktown City Council may deem other types of development as high risk on the basis they would generate elevated TPH loads.

Lower risk developments include all other developments that are required to comply with the hydrocarbon water quality target.

There is currently no defined way for MUSIC to assess the removal of hydrocarbons, or oils from stormwater. Blacktown City Council relies on deemed-to-comply solutions to achieve the target.

11.16.1 General removal requirements for lower risk development

The location of the device is important to achieve the required removal rate. Locating the device upstream of a detention basin may not be appropriate if significant and substantial areas (including the detention area) bypass treatment. A minimum of 95% of the area exposed to vehicle hydrocarbons (driveways and parking areas) must be treated by the device.

Blacktown City Council adopts 5 methods for deemed-to-comply solutions for hydrocarbon removal in low risk developments. These methods are described below.

Method 1: Gross pollutant trap (GPT) with hydrocarbon trap

Design the proprietary GPT for 2 EY with a hydrocarbon trap. The GPT needs to be designed specifically to retain floatable oil and hydrocarbons without significant agitation that would emulsify the hydrocarbons and oils. The use of oil absorbent pillows is not an acceptable solution.

Acceptable devices include:

- HumeCeptor (Humes)*
- Ecoceptor Class 3 (Atlan Stormwater)
- Flowceptor (Atlan Stormwater)
- Spillceptor (Atlan Stormwater)
- EcoSep (Rocla)*
- Vortechs (Ocean Protect)
- Vortsentry HS (Ocean Protect)
- ESK (Ocean Protect)*
- A device approved under ENE 858.1 (2002)*.

*Where these devices only treat hydrocarbons (and in some instances, fine sediments), these may be coupled with either a GPT or approved litter baskets. They must demonstrate compliance with both hydrocarbon and gross pollutant load reduction criteria as specified in Part J for business or industrial developments greater than 2,000 m^2 in area that elect to pay for off-site stormwater treatment.

Method 2: Oil baffle (excluding Jellyfish)

Provide a separate treatment for removal of gross pollutants. For the removal of hydrocarbons, the oil baffle is designed to limit the velocity (V) under, through and over the diversion weir, or under the baffle, to 0.4 m/s to reduce the risk of floatable or bound oils or hydrocarbons escaping the device based on V = 0.4 = Q / A. The area A (m^2) is determined by multiplying the effective gap between the baffle and the weir (overflow), or between the baffle and the floor of the device (G_{eff}) in m and the length of the weir (L) in m such that A = G_{eff} x L.

Step 1: used to determine the flow (Q) in m^3/s for the oil baffle depending upon where in the treatment train the device is located. 4 alternatives are considered:

- online, but upstream of an OSD system then Q = 1% AEP peak flow rate



- online, but downstream of an OSD system then Q = permissible site discharge (PSD) from the 50% AEP OSD outlet control



- online, but not part of an OSD system then Q = 5% AEP peak flow rate



- offline, minimum Q = 2EY flow if used with a diversion weir upstream.



Step 2: used to determine the effective gap between the baffle and the weir, or between the baffle and the floor of the device (G_{eff}) in metres. This will be based on the device chosen and available levels on site. In the standard configuration the impermeable oil baffle is set 0.25 m upstream of the weir. G_{eff} is considered the lesser of the gap between the baffle and the weir (that is, 0.25 m as above), or the gap between the underside of the baffle and the pit base level (G) in m reduced by allowing for sediment accumulation (typically 0.05 m). Then: $G_{eff} = G - 0.05 <= 0.25$ m

Step 3: used to determine the length of the weir L the equation becomes: $L = Q / (0.4 \times G_{eff})$ (m) where Q is in m^3/s and at the maximum of $G_{eff} = 0.25$ m. Then the minimum length of weir L = 10 x Q (m), but longer lengths for lower G_{eff}

Note:

- in some configurations, a weir is not used to regulate the flows and in this arrangement the length (L) above refers to the length of baffle across the tank without bends or corners
- in principle, the lower the bottom of the baffle relative to the overflow level, the more effective the removal should be. The absolute minimum is 0.2 metres, but typically 0.3 to 0.45 metres is used in conjunction with other devices
- where AtlanFilters (Section 12.3) or StormFilters (Section 12.16) are used, the oil and floatables baffle forms part of the filter chamber. See relevant sections for baffle dimensions. The outlet pipe must be below the base of the filter chamber
- where a stand-alone device is used (not part of a filter chamber) the outlet pipe invert can be the same or slightly lower than the inlet pipe invert

- set the top of the baffle a minimum of 150 millimetres above the design hydraulic grade line (HGL) level (flow depth) as determined in Step 1 above. Alternatively, where possible, extend to the underside of the pit.

Baffle configuration. The configuration of the oil baffle arrangement is detailed in Figure 20 and Figure 21. The design HGL is based on Step 1.

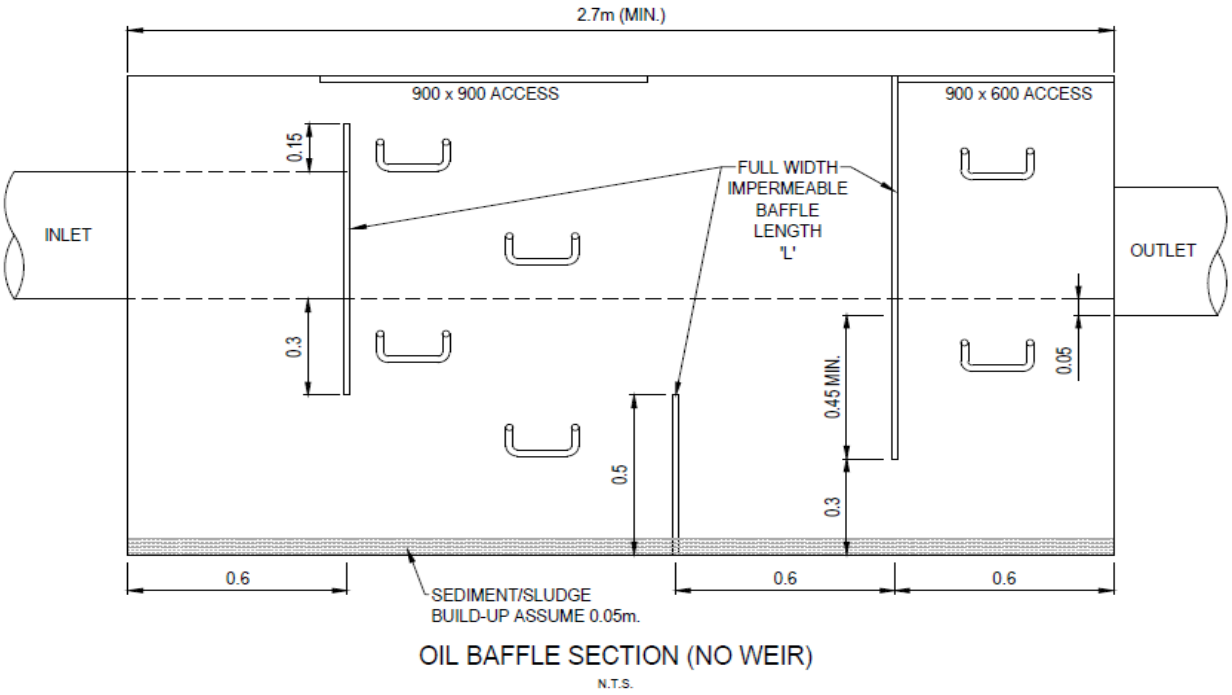


Figure 20. Oil baffle configuration (no weir)

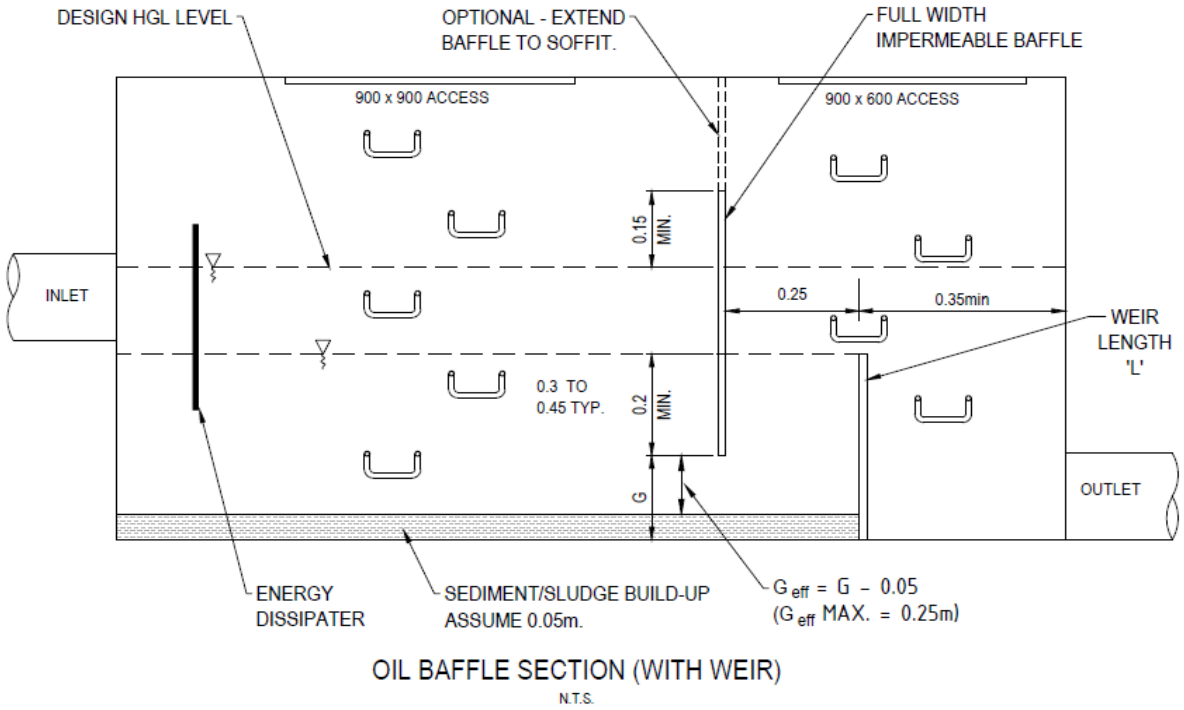


Figure 21. Oil baffle configuration (with weir)

Method 3: Oil baffle for Jellyfish

There are 2 empirical ways to achieve the target hydrocarbon removal for the Jellyfish unit supplied by Ocean Protect. These are specifically detailed in Section 12.13.4 for the unit itself. In this method, provide separate upstream treatment for removal of gross pollutants (typically OceanGuards or a GPT) to protect the Jellyfish device.

Method 4: Bioretention with shallow or no OSD

The plants within the bioretention system can be effective in treating and removing hydrocarbons from stormwater providing the water can be retained. This treatment is limited as once the bioretention system overflows (typically more frequently than 6 EY) any floatable hydrocarbons will bypass treatment and the target will not be achieved. The general approach below is designed for stand-alone bioretention systems or those combined with shallow on-site stormwater detention on private property using small bioretention systems:

Step 1: model the bioretention system in the MUSIC model with a 300 millimetres Extended Detention Depth and design with a 400 millimetres Extended Detention Depth. This provides some additional storage for oil removal prior to overflow

Step 2: install sediment pits at each inflow point to the bioretention system. These should be located where pipe inlets discharge to the filter media surface, or the silt traps within the pits for a saturated bioretention system (as per Blacktown City Council's WSUD standard drawings 6003 and 6006 to collect sediment to protect the bioretention and pick up some bound hydrocarbons that adhere to the sediment particles

Step 3**:** for the overflow pit:

- position the overflow pit away from any walls to ensure flows can access all 4 sides
- provide a baffle arrangement around the overflow pit to contain floatable hydrocarbons
- provide a level concrete area around the overflow pit set at the filter media level and a minimum of 500 millimetres wide
- provide a galvanised or stainless-steel baffle or similarly durable strong material around the overflow pit or pits
- set the baffle 200 millimetres off the concrete pit walls all around and extending from 200 millimetres below the 400 millimetres Extended Detention Depth to a minimum of 200 millimetres above the bioretention Extended Detention Depth. Alternatively, it should be 100 millimetres above the 1% AEP storage level in a combined system
- support the baffle from the concrete level area or brackets off the pit
- size the minimum overflow pit W * W using the following, allowing for a velocity of 0.4 m/s, and ignoring the dead zone in the corners

  W (m) = 3.5 * Q

  allow Q_{2EY} for standalone and $Q_{1\%\ AEP}$ for OSD where Q is in m^3/s and W = 0.6 m (minimum).

Where this is applied to combined bioretention and OSD the baffle is to extend to 100 millimetres above the 1% AEP storage level. This requirement makes this method difficult for a deep OSD which will severely restrict maintenance access to the pit.

Method 5: Fully covered car parks and driveways

Fully covered means that little to no part of the driveways or carparks (less than 30 square metres) are exposed to rainfall. Periodic maintenance and washdown of these covered carpark areas may direct some hydrocarbons to the drainage system, however, the risk is considered low.

For most reasonable sized covered carparks (less than or equal to 50 car spaces), a simple baffle pit or standard trapped gully pit similar to the proprietary silt arrestor with self-trapping gas seal required by Sydney Water, is provided upstream of the pump pit. Direct all surface flow from the carpark to this pit before overflowing to the pump pit. Where feasible, do not mix groundwater seepage flows and carpark surface runoff that may contain hydrocarbons.

For larger sized covered carparks (greater than 50 car spaces), provide a device approved under Section 11.16.1 (Method 1 or 2). However, simpler methods of gross pollutant removal may be considered.

11.16.2 Removal requirements for higher risk developments

Within high risk developments there are 2 zones: A high risk zone and a lower risk zone.

Zone 1: High risk contamination zone – vehicle refilling area and tank fill points

These areas are to discharge to the sewer system and are to be subject to an approved trade waste agreement. Sydney Water is responsible for setting terms of trade waste agreements including permissible total petroleum hydrocarbons (TPH) concentrations in wastewater. Typically, compliance with ENE 858.1 (2002) is required.

Zone 2: Lower risk contamination zone – parking, footpath and trafficable areas

This zone applies to all areas of site that discharge to the stormwater system and exclude roof areas (providing they can be separately managed or discharged) and any high-risk contamination zone areas which are to discharge to sewer.

Discharge from lower risk contamination zones is to be limited to a maximum allowable discharge concentration of 5-parts per million (ppm) of total petroleum hydrocarbons (TPHs).

Current acceptable approaches include:

- **installation of a device which complies with ENE 858.1 (2002).** This could be part of the treatment train for full on-lot treatment, or with a GPT to trap 90% of annual average load of gross pollutants where full off-site treatment is required. Some devices which comply with ENE 858.1 (2002) may also trap gross pollutants
- **a Sydney Water approved coalescing plate separator** sized to treat the first 15 millimetres of rainfall runoff. Provide a diversion weir upstream with baffle to divert the hydrocarbons and oils into the tank with a non-return flap. The Coalescing Plate Separator must be sized to treat and empty the entire first flush tank volume within a maximum of 12 hours. To remove gross pollutants and other target pollutants the Coalescing Plate Separator must be used in conjunction with a generic gross pollutant trap or as part of the treatment train
- **Flowceptor** (Atlan Stormwater) or similar device with a coalescer and an overflow system sized so that the device does not go into bypass more frequently than twice per year
- **Spillceptor** (Atlan Stormwater) or ESK (Ocean Protect) or EcoSep (Civil Mart - Rocla) or similar device with a coalescer, but without an overflow system sized for a 5% AEP flow rate as there is no bypass

- **Spillceptor** (Atlan Stormwater) or ESK (Ocean Protect) or EcoSep (Civil Mart - Rocla) or similar device with a coalescer and no overflow system, but with an upstream diversion pit. Size the device and the diversion pit so that the device does not go into bypass more frequently than twice per year. The diversion pit is to contain a diversion weir and baffle sized as per Method 2 above for low risk zones (Section 11.16.2), or for small systems an approach similar to Method 4 can be considered.

12 Stormwater quality improvement devices

Proprietary stormwater quality improvement devices (SQIDs) play an important role in Blacktown City Council's stormwater management strategy. We encourage innovation and competition and most importantly, validating device performance is critical to delivering improvements in water quality.

Rigorous field testing must be undertaken to stress test a SQID and to help quantify its pollutant removal capability. This in turn facilitates the way it is modelled as part of a treatment train in MUSIC or within S3QM. Proprietary device MUSIC performance nodes can be found on Blacktown City Council's default nodes model that can be downloaded from MUSIC-*link*.

12.1 Approved stormwater quality improvement devices for use in Blacktown City

To date, based on rigorous peer reviews of independently collected monitoring data, a number of devices have been approved. These devices are described in detail in this chapter, together with the approved performance rates. The list of currently approved SQIDs are shown in Table 11.

For full details of the approval associated with these devices, including the approved model numbers and pollutant removal rates, refer to section 12.2. All devices have been approved for use on private property. Some devices are also approved for use on public land where they will become Council assets once adopted.

The WSUD checklists are to be used for each device at each stage of the development (i.e. application, before issues of construction certificate, and before issues of an occupation or subdivision certificate) are also shown in table 11.

Company name	SQID name	Asset type	Approved pollutants removed					Use on Council land	WSUD checklist series
			TN	TP	TSS	GP	Hydro-carbons and oils		
Atlan Stormwater (formerly SPEL)	AtlanBasin	Modular bioretention with replaceable filter media	Y	Y	Y	Y	N	N	131 231 431
www.atlan.com.au	AtlanFilter (previously SPEL Filter and Bay Filter)	Replaceable filter media	Y	Y	Y	Y	Y – must be installed with oil baffle	N	121 221 421
	Ecoceptor*	GPT	N	N	N	Y	Y	N	130 230 430
	Flowceptor (formerly Stormceptor)	GPT	N	Limited	Limited	Y	Y	N	130 230 430
	FlowGuard	Replaceable filter media	Y	Y	Y	N	N	N	124 224 424
	Spillceptor (formerly Puraceptor)	GPT	N	Limited	Limited	Y	Y	N	130 230 430
	StormSack	Litter Basket	Y	Y	Y	Y	N	N	140 240 440
	Vortceptor	GPT	N	Limited	Limited	Y	Y – must be installed with oil baffle	Y	130 230 430

Company name	SQID name	Asset type	Approved pollutants removed					Use on Council land	WSUD checklist series
			TN	TP	TSS	GP	Hydro-carbons and oils		
Humes www.holcim.com.au/humes	HumeCeptor	GPT	Limited	Limited	Limited	Limited	Y	N	130 230 430
	HumeFilter	Replaceable filter media	Y	Y	Y	Y	N	N	125 225 425
	HumeGard	GPT	Y	Y	Y	Y	N	N	130 230 430
Ocean Protect	ESK*	Oil and water separator	N	N	Limited	Y	Y	N	NA
www.oceanprotect.com.au	Filterra	High flow media filtration	Y	Y	Y	Y	Y – extend EDD to 400 mm	N	132 232 432
	Jellyfish	Cartridge	Y	Y	Y	Y	N	N	122 222 422
	OceanGuard	Litter basket	Y	Y	Y	Y	N	N	141 241 441
	OceanSave	GPT	N	Limited	Limited	Y	Y – must be installed with oil baffle	Y	130 230 430
	StormFilter	Cartridge	Y	Y	Y	Y	Y – must be installed with oil baffle	Y	120 220 420

Company name	SQID name	Asset type	Approved pollutants removed					Use on Council land	WSUD checklist series
			TN	TP	TSS	GP	Hydro-carbons and oils		
Ocean Protect	VortSentry HS*	GPT	N	N	N	N	Y	N	130 230 430
www.oceanprotect.com.au	Vortechs*	GPT	N	N	N	N	Y	N	130 230 430
Civil Mart - Rocla	CDS	GPT	N	Limited	Limited	Y	Y – must be installed with oil baffle	Y	130 230 430
https://civilmart.com.au/our-brands/rocla/	Ecosep*	GPT	N	N	N	N	Y	N	130 230 430
	Enviropod 200	Litter basket	Y	Y	Y	Y	N	N	142 242 442

Table 11. List of currently approved stormwater quality improvement devices for use in Blacktown City Council - as of June 2026

Important notes:

Some models of the above devices are not approved for use in Blacktown. Approved models are shown in following sections.

*No additional information related to this device.

Council does not endorse any particular stormwater quality improvement device; our approval is based on each product meeting the required specifications.

Council reserves the right to remove any device from its approved list of proprietary products at any time.

12.2 AtlanBasin (Atlan Stormwater)

12.2.1 General description

The AtlanBasin is a modular bioretention system. It is a prefabricated, self-contained unit using horizontal flow bioretention technology and multistage pre-treatment removing nitrogen, phosphorus, total suspended solids and gross pollutants. It relies on horizontal flow and a treatment train.

The AtlanBasin can be used to treat car parks, roads, residential, business and light industrial catchments.

Conditions of use:

1. Council limits the hydraulic loading rate to 250m/year. This can be calculated by dividing the volume of stormwater arriving at the AtlanBasin and dividing it by the surface area of the media
2. all installations shall include an approved 'Atlan StormSack' litter basket within the device to ensure that gross pollutants are captured
3. installations shall not be configured in series (i.e. one 'AtlanBasin' is not permitted to flow into another 'AtlanBasin')
4. installations in parallel are acceptable though it is preferred to have the 'AtlanBasin' located as a source control. When installed in parallel, splitter pits would need to be used to ensure a maximum of 10 Litres/second per module. Splitter pits shall be designed in accordance with Council's WSUD standard drawings
5. the only approved models for use in Blacktown are shown in Table 12.

Model No.	Configuration	Treatable flow rate (L/S)
MWS-L-3-6	In-line or off-line	2
MWS-L-4-13	In-line or off-line	4
MWS-L-4-17	In-line or off-line	6
MWS-L-4-21	In-line or off-line	10

Table 12. AtlanBasin Model numbers and treatable flow rates

MUSIC modelling

Use a generic node in MUSIC with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) to be set at the treatable flow rate shown in Table 12 above.

The percentage pollutant removal rates used to construct generic treatment nodes are shown in Table 13.

Pollutant	Removal (%)
Total Suspended Solids	86
Total Phosphorous	65
Total Nitrogen	50
Gross Pollutants	99

Table 13. Basin pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency method. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	140
Total Phosphorous	10	3.5
Total Nitrogen	50	25
Gross Pollutants	1000	10

Table 14. AtlanBasin Generic Node input and output values

12.2.2 Inspections and testing

The AtlanBasin shall be inspected annually to observe the time taken to drain following a storm event. If the media is observed to not fully drain to its lowest level within 2 hours of the cessation of flow, then the media shall be investigated and replaced.

Saturated hydraulic conductivity of the filter media shall be measured at the start of the third year of operation using a single ring infiltrometer as follows: 150 millimetres minimum diameter pipe driven to the full depth of the media. If the hydraulic conductivity falls below 3000 mm/hour the media shall be rectified to restore a minimum Ksat of 3560 mm/hour. This test shall be observed by Council's WSUD Compliance Officer. If required, a second test following media rectification shall also be observed by Council's WSUD Compliance Officer.

Vegetation shall be inspected annually and if less than 80% coverage it shall be supplemented with additional planting. This will also be observed by Council's WSUD Compliance Officer.

12.3 AtlanFilter (Atlan Stormwater)

12.3.1 General description

The AtlanFilter is a cartridge filter system that incorporates an up-flow treatment process where water passes through the media, removing nitrogen, phosphorous and total suspended solids. A weir is required downstream to provide sufficient head to activate the cartridge. Flow is regulated via siphon. On completion of a storm or filter cycle, the filter backwashes and dislodges sediments from the filtration media cartridge.

Note: See also Section 12.19 for specific requirements for the maintenance of the filter cartridges.

12.3.2 Characteristics

An impermeable baffle is required 250 millimetres upstream of the AtlanFilter to retain floatable pollutants including hydrocarbons. The baffle is to extend to the underside of the tank or 250 millimetres above the design storage. Limit the velocity under and through the baffle to 0.4 m/s allowing for a 50 millimetres sludge layer over the floor.

The weir length (L) in the table below is calculated on the flow Q (m^3/s) under and through the baffle based on the following:

- online, as part of an OSD system then Q = 1% AEP
- online, but not part of an OSD system then Q = 5% AEP
- offline, minimum 2 EY (6 month) flow (75% of 1 EY) if used with a diversion weir upstream.

The upstream **diversion** weir height is a minimum of the AtlanFilter weir level, plus the depth of flow over the AtlanFilter weir (using the weir equation for flow Q), plus the pipe and pit losses to the diversion weir.

g

Cartridge name	Cartridge height (mm)	Weir height H above floor (mm)	Flow through each at weir height TF (L/s)	Baffle depth D below weir level (mm)	Length of weir L (m)
Half-height SF.1.5-EMC	660	550	1.5	250	16.6 x Q
Full-height SF.3.0-EMC	860	850	3.0	450	10 x Q

Table 15. AtlanFilter characteristics and controls

In certain circumstances, Blacktown City Council may consider slightly lower weir heights where the leg of the filter is cut down due to a height restriction. For example, half height weir reduced to 500 millimetres or full height.

A typical section through the AtlanFilter chamber is shown below in Figure 22.

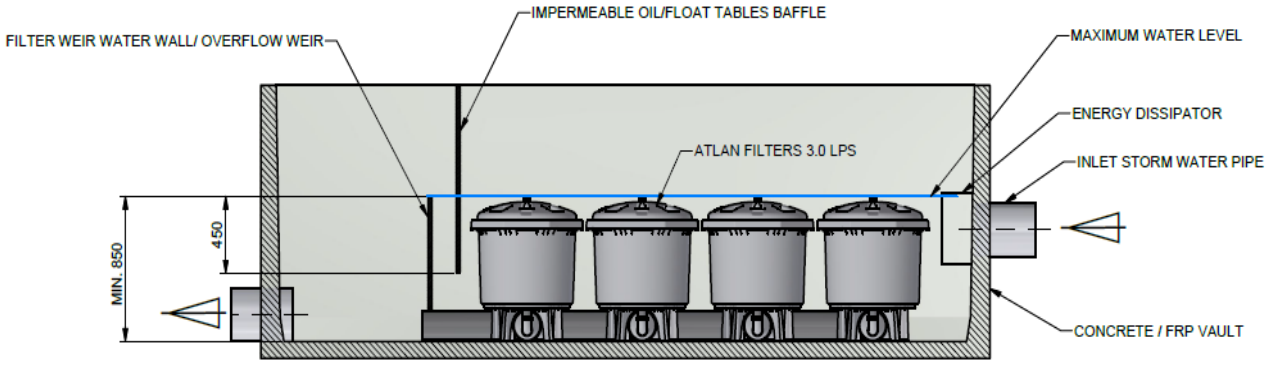


Figure 22. Section through a full-height AtlanFilter chamber

12.3.3 MUSIC modelling requirements for AtlanFilters

The AtlanFilter system is controlled through 2 nodes: a Detention Node followed by a Generic Node. Consider 'No.' in the notes below as representing the number of cartridges.



Figure 23. Example of MUSIC setup for AtlanFilter

1. Detention Node with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) leave default as 100.00
- Surface Area (square metres)
- both 1.5 EMC and 3.0 EMC = Area of detention basin node is 1 square metres per cartridge
- Extended Detention Depth (metres) = weir height from table above for the cartridge height (either 850 millimetres or 550 millimetres)
- Exfiltration Rate (mm/hr) leave default as 0.00
- Evaporative Loss as percentage of PET leave default as 0.00
- Low Flow pipe Diameter (mm) adopt an outflow pipe diameter that yields a notional detention time in the detention basin node of 0.1 hours (plus or minus 10%)
- Overflow Weir Width (metres) set to the weir length L from Table 15
- there is no reuse permitted using this node
- under the 'Advanced' tab the k values are all set to 0.

Note that Council will not accept a different configuration here. It is because it will result in a different hydraulic loading rate than that which has been tested and approved under Stormwater Australia's *Stormwater Quality Improvement Device Evaluation Protocol* (SQIDEP).

2. Generic node with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) = No. of cartridges proposed x treatable flow rate
- either 1.5 L/s/cartridge for half height or 3 L/s/cartridge for full height cartridges/ 1,000 (Table 15).

The percentage pollutant removal rates to be adopted in the generic node are as follows:

Pollutant	Removal rate (%)
Total Suspended Solids	85
Total Phosphorous	74
Total Nitrogen	59
Gross Pollutants	95

Table 16. AtlanFilter pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the Concentration Based Capture Efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	150
Total Phosphorous	10	2.6
Total Nitrogen	100	41
Gross Pollutants	1000	50

Table 17. AtlanFilter Generic node input and output values

12.3.4 Specific requirements for AtlanFilters with on-site stormwater detention

To ensure compliance with the MUSIC model, the acceptable approach across our city is for the treated underdrain flow (for example proprietary filters) to discharge downstream of the detention discharge control pit. This includes areas where water quality treatment is incorporated into a detention basin itself. This will ensure ongoing treatment throughout a range of storms. For proprietary filter underdrains, this flow can be significant and the orifice size needs to be reduced to ensure the permissible site discharge can be maintained.

A standard configuration of AtlanFilter chamber with detention is shown on Blacktown City Council's WSUD Standard Drawing 6102. When using Council's OSD spreadsheet, select Yes for question:'Will filter cartridges be used to manage water quality?'.

Note that to ensure the AtlanFilter operate through all storms as in MUSIC, the underdrains discharge downstream of the orifice controls. Without this, if the tailwater level rises to the weir height, the cartridges will stop working as they rely on a siphon. The 50% AEP (1.5 year ARI) orifice is to be reduced in size (perhaps even not included) to ensure the 50% AEP permissible site discharge is maintained.

Then:

$Q_{50\%\ AEP\ Orifice}$ = $PSD_{50\%\ AEP}$ – Flow through Each Cartridge at Weir Height x No. Cartridges

Use the orifice equation (C_d = 0.61) with $Q_{50\%\ AEP\ Orifice}$ to determine the reduced orifice size at the 50% AEP control pit weir level. This process is undertaken automatically with the use Council's OSD spreadsheet when the design filter flow rate is entered at design flow from filter cartridges (L/s).

Similarly, the 1% AEP orifice is to be reduced in size to ensure the 1% AEP permissible site discharge is maintained. Then:

$Q_{1\%\ AEP\ Orifice}$ = $PSD_{1\%\ AEP}$ – Flow through Each Cartridge at 1% AEP water level x No. Cartridges

Use the orifice equation (C_d = 0.61) with $Q_{1\%\ AEP\ Orifice}$ to determine the orifice size at the 1% AEP emergency overflow weir level. This process is undertaken automatically with the use of Council's OSD spreadsheet however it is necessary to assess the flow through the cartridges at the higher water level and input this into the spreadsheet at Filter Cartridge flow with 1% AEP Head (L/s).

As the flow through each AtlanFilter is controlled by an effective orifice then with increasing depth and pressure head the flow through the cartridge will increase (C_d = 0.61).

Half-height-1,5 EMC – Allow for an orifice diameter of 37.64 millimetres set 230 millimetres above the floor.

Full-height- 3.0 EMC – Allow for an orifice diameter of 41.15 millimetres set 230 millimetres above the floor.

12.3.5 On-site stormwater detention problems with AtlanFilters and solutions

In some designs the calculated 50% AEP (1.5 year ARI) orifice size will reduce below Blacktown City Council's minimum of 25 millimetres. Where the 50% AEP orifice size is calculated below 25 millimetres, delete the 50% AEP orifice completely subject to the alternatives/limitations below and direct all low (1.5-year ARI) flows through the underdrain. Reconfigure the floor of the detention tank to direct the low point to the non-return flap into the AtlanFilter chamber so that all low flows are redirected back into the filter chamber. See Blacktown City Council's WSUD standard drawing 6102.

However, where the 50% AEP orifice is deleted due to higher underdrain flows, this might result in adverse environmental impacts. To avoid deleting this, Blacktown City Council will require the investigation of possible partial alternate WSUD treatment. Deleting the 50% AEP orifice will only be accepted where it can be demonstrated that no reasonable alternatives exist.

For specific steps about how to redesign the basin/tank to delete the 50% AEP orifice, see Section 11.6.3.

In some designs with a large number of filter cartridges, our OSD spreadsheet may indicate that applying the high underdrain flow from the filters in the 1% AEP event will result in the 1% AEP orifice not being required. This is not acceptable and suggests that more filters have been modelled than is reasonable. To ensure a 1% AEP orifice is implemented and that Council's permissible site discharge is not exceeded, the number of filter cartridges shall be reduced to ensure a minimum 25 millimetres 1% AEP orifice by either:

- treating part of the site using alternative treatment measures
- splitting the filters and relocating the filter cartridge chamber downstream into multiple tanks subject to suitable levels to ensure the discharge is above the 1% AEP storage level.

12.4 Flowceptor (Atlan Stormwater)

The Flowceptor GPT utilises the processes of physical screening and floatation/sedimentation to separate the litter and coarse sediment from stormwater runoff. It is also a Class 1 treatment device and will remove all total petroleum hydrocarbons (TPH) to below 5 ppm. The Flowceptor pollutant removal rates are detailed below. Flowceptors are not permitted in a series. As a GPT, the Flowceptor should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month) (75% of the 1 EY) depending upon the location. The device is sized to ensure the treatment flow rate matches or exceeds the design flow. The treatment flow rate is input into the node as the high flow by-pass in m^3/s.

The MUSIC node can be obtained through MUSIC-*link* and clicking on Blacktown City Council. When clicking on the Flowceptor node click 'Notes' to determine the various high flow by-pass flows for the various device sizes.

MODELS	WEIGHT (KG)	SERIES	TFR LPS	VOLUME LTRS
FLO.100/15.C1.2C.	120	100	2	800
FLO.100/20.C1.2C.	190	100	4	1300
FLO.100/25.C1.2C.	260	100	6	1950
FLO.200/30.C1.2C	300	200	4	2700
FLO.200/40.C1.2C	330	200	7	3000
FLO.200/50.C1.2C.	380	200	8	3800
FLO.200/70.C1.2C.	410	200	11	4900
FLO.300/80.C1.2C.	650	300	12	6500
FLO.300/100.C1.2C.	700	300	14	7250
FLO.300/130.C1.2C.	800	300	19	8900
FLO.300/160.C1.2C.	900	300	23	10600
FLO.300/220.C1.2C.	1100	300	31	13150
FLO.300/250.C1.2C.	1200	300	35	14600
FLO.400/300.C1.2C.	1700	400	43	23200
FLO.400/400.C1.2C.	2000	400	56	26150
FLO.400/500.C1.2C.	2400	400	71	29900
FLO.400/600.C1.2C.	2700	400	82	33900
FLO.400/700.C1.2C.	3000	400	102	40100
FLO.400/750.C1.2C.	3300	400	108	41500
FLO.400/850.C1.2C.	3500	400	122	46600
FLO.400/1000.C1.2C.	4100	400	137	5100
FLO.400/1100.C1.2C.	4600	400	148	53300
FLO.500/800.C1.2C.	4300	500	111	55000
FLO.500/1000.C1.2C.	5200	500	139	58500
FLO.500/1200.C1.2C.	6200	500	167	71900
FLO.500/1500.C1.2C.	7400	500	208	85000
FLO.500/1600.C1.2C.	8740	500	222	92900
FLO.500/1800.C1.2C.	9380	500	250	99800
FLO.500/2000.C1.2C.	10190	500	278	108500
FLO.500/2300.C1.2C.	319	500	320	122300
FLO.500/2700.C1.2C.	375	500	375	137300
FLO.500/2900.C1.2C.	403	500	403	147700

Table 18 Flowceptor treatable flow rates

(Source: Flowceptor Product brochure from Atlan Stormwater https://atlanstormwater.com/au/flowceptor/)

The percentage removal rates are as follows:

Pollutant	Point 1% Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 25 mg/L	thereafter 55%
Total Phosphorous	0% removal up to 0.15 mg/L	thereafter 15%
Total Nitrogen	Zero	
Gross Pollutants	90%	

Table 19. Flowceptor pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table above) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	25	25	1,000	300
Total Phosphorous	0.15	0.15	10	8.5
Total Nitrogen	0	0		
Gross Pollutants	10	1		

Table 20. Flowceptor GPT Node input and output values

12.5 FlowGuard (Atlan Stormwater)

12.5.1 General description

The FlowGuard is a device intended to treat runoff from areas, typically car parking or driveways or hardstand areas, which are unable to be directed to a more traditional filter device such as AtlanFilter. Council is unlikely to only accept a FlowGuard in the absence of other elements of a treatment train.

FlowGuard uses its own filter media. It is designed to fit within a precast or prefabricated modular channel such as 300 millimetres wide trench grate or channel drain.

FlowGuard has a 2-stage treatment mechanism which first settles sediment in a sedimentation chamber followed by filtration through its filter media which sits within a filter bag.

12.5.2 Conditions of approval and MUSIC modelling

1. FlowGuard shall be configured so that the hydraulic loading rate on the device does not exceed 0.5 litres per lineal metre of FlowGuard trench module
2. approved pollutant removal rates are included in Table 21 below.

Pollutant	Removal rate %
Gross Pollutants	0 (not verified)
Total Suspended Solids	88
Total Phosphorus	69
Total Nitrogen	67

Table 21. Approved pollutant removal rates (from SQIDEP)

When Atlan FlowGuard is proposed and its performance is to be modelled in MUSIC, the MUSIC model shall adopt the SQIDEP verified method, summarised as follows:

- a High Flow By-pass is to be set equal to the 0.5 L/s X the number of lineal metres of FlowGuard proposed
- pollutant removal rates for the generic node are set to the values shown in Table 22 below. In the MUSIC node, the pollutant removal rates are represented by the following point for the concentration based capture efficiency method. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Inflow	Outflow
Gross Pollutants	15	15
Total Suspended Solids	1000	120
Total Phosphorus	5	1.55
Total Nitrogen	50	16.5

Table 22. FlowGuard MUSIC Generic Node input and output values

12.5.3 Maintenance, inspection and testing

Maintenance of the device shall be carried in accordance with Council's Operation and Maintenance Manual and as follows:

- the sedimentation box shall be emptied every 4 months
- the filter media shall be replaced no later than every 12 months or earlier if required
- replacement certificates shall be provided to Council's WSUD Compliance Officer every 12 months.

12.6 Spillceptor (Atlan Stormwater)

The Spillceptor GPT (formerly Puraceptor) utilises the processes of physical screening and floatation/sedimentation to separate the litter and coarse sediment from stormwater runoff. It is also a Class 1 treatment device and will remove all total petroleum hydrocarbons (TPH) to below 5 ppm. The Spillceptor contains an Automatic Closure Device (ACD). This device comprises a water buoyant ball that is sensitive to any change in the water density as a consequence of hydrocarbon, or oil build up. Thereby it automatically closes, preventing pollutants from discharging to drains and waterways.

Spillceptors are not permitted in series. As an online system, the device must be designed for a minimum 5% AEP flow as there is no internal bypass system within the Spillceptor due to the ACD. As an offline system with an upstream diversion weir and baffle, the Spillceptor should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month) depending upon the location and treatment train. The device is sized to ensure the treatment flow rate matches or exceeds the design flow. The treatment flow rate is input into the node as the high flow by-pass in m^3/s.

The MUSIC node can be obtained through MUSIC-*link* and clicking on 'Blacktown City Council'. When clicking on the Spillceptor node, click 'Notes' to determine the various High Flow By-pass flows for the various device sizes.

The percentage pollutant removal rates are as follows:

Pollutant	Point 1% Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 25 mg/L	thereafter 55%
Total Phosphorous	0% removal up to 0.15 mg/L	thereafter 15%
Total Nitrogen	Zero	
Gross Pollutants	90%	

Table 23. Spillceptor pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	25	25	1000	450
Total Phosphorous	0.15	0.15	10	8.5
Total Nitrogen	0	0		
Gross Pollutants	10	1		

Table 24. Spillceptor GPT Node input and output values

Models	Treatment Flow Rate	Weight (KG)	Dimensions (mm)								Maximum Inlet & Outlet Pipe Size (mm) Configuration*	Manholes				Max. Spill at Shut Off	Max. Working Capacity
			A	B	A&B	C	D	L	OD	ID		Qty	Size	Qty	Size		
100 Series Tanks - 900 mm Internal Diameter																	
SPI.002.C1.2C	2 LPS	120	400	820	1220	800	420	1700	930	900	100	2	450 ID	-	-	250	800
200 Series Tanks - 1200 mm Internal Diameter																	
SPI.004.C1.2C	4 LPS	330	460	1100	1560	1080	480	2600	1350	1200	150	2	600 ID	-	-	1,000	2,050
SPI.006.C1.2C	6 LPS	400	425	1135	1560	1095	465	3035	1350	1200	150	2	600 ID	-	-	1,300	2,550
SPI.008.C1,2C	8 LPS	450	460	1100	1560	1060	500	3800	1350	1200	150	2	600 ID	-	-	1,900	3,200
SPI.010.C1.2C	10 LPS	500	450	1110	1560	1060	500	4600	1350	1200	150	2	600 ID	-	-	2,500	3,900
SPI.013.C1.2C	13 LPS	550	446	1114	1560	1040	520	5800	1350	1200	150	3	600 ID	-	-	3,000	4,800
SPI.015.L.C1.2C	15 LPS	600	425	1135	1560	1065	495	6500	1350	1200	150	3	600 ID	-	-	3,400	5,400
300 Series Tanks - 1850 mm Internal Diameter																	
SPI.015.S.C1.2C	15 LPS	650	620	1630	2250	1600	650	3000	1950	1850	300	1	600 ID	1	900 x 600	3,500	5,500
SPI.020.C1.2C	20 LPS	850	625	1625	2250	1585	665	4000	1950	1850	300	1	600 ID	1	900 x 600	3,900	7,300
SPI.030.C1.2C	30 LPS	1100	660	1590	2250	1575	675	4860	1950	1850	300	1	600 ID	1	900 x 600	5,500	10,800
SPI.040.8.C1.2C	40 LPS	1180	550	1600	2150	1500	650	5900	1950	1800	300	1	600 ID	1	900 x 600	8,000	13,400
SPI.040.C1.2C	40 LPS	1240	650	1600	2250	1550	700	6540	1950	1850	300	2	600 ID	1	900 x 600	9,000	14,400
SPI.050.L.C1.2C	50 LPS	1400	650	1600	2250	1520	730	8500	1950	1850	300	2	600 ID	1	900 x 600	10,000	18,000
SPI.060.L.C1.2C	60 LPS	1550	650	1600	2250	1500	750	10,000	1950	1850	300	2	600 ID	1	900 x 600	11,200	21,600
SPI.070.L.C1.2C.	70 LPS	1700	650	1600	2250	1500	750	11,600	1950	1850	300	2	600 ID	1	900 x 600	12,400	25,200
400 Series Tanks - 2480 mm Internal Diameter																	
SPI.050.S.C1.2C	50 LPS	1400	720	2230	2950	2150	800	4680	2600	2480	375	1	600 ID	1	900 x 600	9,000	18,000
SPI.060.S.C1.2C	60 LPS	1560	550	2400	2950	2220	730	5500	2600	2480	375	1	600 ID	1	900 x 600	10,700	21,600
SPI.070.S.C1.2C	70 LPS	1710	750	2200	2950	2150	800	6550	2600	2480	375	3	600 ID	1	900 x 600	12,400	25,200
SPI.080.C1.2C	080 LPS	2000	600	2350	2950	2250	700	7500	2600	2480	375	3	600 ID	1	900 x 600	14,900	29,600
SPI.090.C1.2C	090 LPS	2300	715	2235	2950	2150	800	8400	2600	2480	375	3	600 ID	1	1200 x 600	16,200	32,400
SPI.100.C1.2C	100 LPS	2550	710	2240	2950	2150	800	9000	2600	2480	375	3	600 ID	1	1200 x 600	17,700	35,700
SPI.110.C1.2C	110 LPS	2650	700	2250	2950	2150	800	9600	2600	2480	375	3	600 ID	1	1200 x 600	18,300	38,200
SPI.120.C1.2C	120 LPS	2750	570	2400	2970	2300	670	10,230	2600	2480	375	3	600 ID	1	1200 x 600	21,700	43,200
SPI.150.C1.2C	150 LPS	3360	670	2280	2950	2150	800	13,420	2600	2480	375	4	600 ID	1	1200 x 600	27,700	54,000
SPI.180.C1.2C	180 LPS	3580	650	2300	2950	2150	800	15,400	2600	2480	375	5	600 ID	1	1200 x 600	32,500	64,800
SPI.200.C1.2C	200 LPS	4150	555	2395	2950	2230	720	16,500	2600	2480	375	5	600 ID	1	1200 x 600	36,200	72,000

Table 25. Spillceptor treatment flow rates and construction specifications

(Source: Spillceptor Product brochure from Atlan Stormwater https://atlanstormwater.com/au/atlan-spillceptor/)

12.7 StormSack (Atlan Stormwater)

The StormSack is a catch pit insert that captures and retains litter, debris and other pollutants as runoff enters the storm drain system. StormSacks come with a 200-micron bag size and are used upstream of tertiary treatment devices to reduce the pollutant loading. Each StormSack with a bag depth greater than 300 millimetres has a high flow by-pass of 11 L/s (0.011 m^3/s). Where a number of StormSacks are used in similar catchments, it is common to direct the upstream catchments to a single node with the high flow by-pass increased in proportion to the number of StormSacks. StormSacks are not permitted in series.

Where StormSacks are used upstream of a detention system, the underside of the bag should be set a maximum of 250 millimetres below the 50% AEP weir level.

The constraints are listed below in Table 26.

Bag depth below overflow (mm)	Pit size (mm x mm)	Maximum mixed catchment area (m^2)	Maximum roof catchment area (m^2)	MUSIC High Flow By-pass (m^3/s)
> 300	900 x 900	550	850	0.011
> 300	900 x 600	440	680	0.011
> 300	600 x 600	330	510	0.011

Table 26. StormSack constraints > 300 mm

To treat the surface flow, StormSacks with bag depths greater than 300 millimetres, require a minimum clear depth of 500 millimetres below the grate to any inlet or outlet pipe obvert. StormSacks treating surface flows and upstream pipe flows require a minimum clear depth of 500 millimetres from the invert of the upstream pipes to be treated to the obvert of the outlet pipe. Where these pits are treating upstream pipe flows the inverts of all pipes in and out of the pit are to be shown.

Bag depth below overflow (mm)	Pit size (mm x mm)	Maximum mixed catchment area (m^2)	Maximum roof catchment area (m^2)	MUSIC High Flow By-pass (m^3/s)
170 to 300	900 x 900	340	510	0.005
170 to 300	900 x 600	285	410	0.005
170 to 300	600 x 600	200	310	0.005

Table 27. StormSack constraints > 170 mm, but < 300 mm

To treat the surface flow, StormSacks with bag depths greater than 170 millimetres but less than 300 millimetres, require a minimum clear depth of 400 millimetres below the grate to any inlet or outlet pipe obvert.

StormSacks treating surface flows and upstream pipe flows require a minimum clear depth of 350 millimetres from the invert of the upstream pipes to be treated, to the obvert of the outlet pipe. Where these pits are treating upstream pipe flows the inverts of all pipes in and out of the pit are to be shown.

StormSacks are not permitted in 450 millimetres x 450 millimetres pits in our city.

The StormSack MUSIC node can be obtained through MUSIC-*link* and clicking on Blacktown City Council. The percentage average annual pollutant removal rates are as follows:

Parameter	Removal rate %
Total Suspended Solids	61
Total Phosphorous	28
Total Nitrogen	45
Gross Pollutants	90

Table 28. StormSack pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency.

For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	100	390
Total Phosphorous	10	7.2
Total Nitrogen	100	55
Gross Pollutants	1,000	100

Table 29. StormSack GPT Node input and output values

Visit www.atlanstormwater.com/au/ for further information.

12.8 Vortceptor (Atlan Stormwater)

The Vortceptor GPT utilises the processes of indirect screening and floatation/sedimentation to separate the litter, pollutants and coarse sediment from stormwater runoff. This particular system uses a fibreglass body. The Vortceptor is generally an offline system with the driving head produced by an upstream diversion weir. This is a vortex system that minimises blockage of the indirect screen. This device can only be cleaned with an eductor truck.

The device can be designed to direct a proportion of the treatable flow (from 0% up to 100% of the treatable flow) to an alternate destination such as a bioretention basin and return the balance to the main flow. Vortceptor devices are not permitted in series, although these devices may be used in parallel – for example, twin, side by side setups to increase the overall treatable flow rate.

As a GPT, the device should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month or 75% of the 1 EY) depending upon the location. See also Section 11.11. The device is sized to ensure the treatable flow rate matches or exceeds the design flow. The treatable flow rate is input into the node as the high flow by-pass in m3/s. Treatable flow rates are detailed in Table 32.

The MUSIC node for the Vortceptor device can be obtained through MUSIC-*link* and clicking on ‘Blacktown City Council’. The percentage pollutant removal rates are as follows:

Pollutant	Point 1 % Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 75 mg/L	thereafter 70%
Total Phosphorous	0% removal up to 0.5 mg/L	thereafter 30%
Total Nitrogen	Zero	
Gross Pollutants	98%	

Table 30. Vortceptor pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	75	75	1,000	300
Total Phosphorous	0.5	0.5	10	7
Total Nitrogen	50	50		
Gross Pollutants	100	2		

Table 31. Vortceptor GPT Node input and output values

All Vortceptor devices in Blacktown must contain an oil baffle.

The following models only are approved for use in Blacktown.

Vortceptor Model No.	Configuration	Treatable flow rate (L/s)	Sump storage (m^3)	Floatables (m^3)
SVI.025	In-line	24	0.5	
SVI.055	In-line	50	1.29	
SVI.055.M	In-line	50	1.29	
SVO.096	Off-line	95	2.31	
SVO.140	Off-line	140		
SVO.180	Off-line	180	2.31	0.56
SVO.220	Off-line	220	4.19	1.34
SVO.360	Off-line	350	4.19	1.34
SVO.530	Off-line	530		
SVO.800	Off-line	800		
SVO.810	Off-line	800	10.13	5.65
SVO.1200	Off-line	1200	10.13	5.65
SVO.1600	Off-line	1600	10.13	5.65

Table 32. Vortceptor treatable flow rates

Visit www.Atlan Stormwater.com.au for further information.

12.9 HumeCeptor (Humes)

HumeCeptor, as opposed to other GPT's, is a source control product targeting petroleum hydrocarbons, oils, grease, total suspended solids, heavy metals and nutrients. The device does have the capacity to capture small pieces of litter such as cigarette butts and pieces of paper. Consequently, a HumeCeptor alone does not satisfy the gross pollutant 90% annual removal rates required under Part J for developments subject to a VPA or Section 7.11 of the Environmental Planning and Assessment Act. It must have an approved litter basket or other approved supplemental device to remove the gross pollutants.

To support the use of a HumeCeptor STC model in MUSIC, the Humes calculator PCSWMM for HumeCeptor available from Humes is to be run.

Step 1: project details nominate 80% TSS removal

Step 2: where the device is located downstream of a detention system, a minimum of 7 storage points are required

Step 3: use the Parramatta North (Masons Hill) rainfall data

Step 4: particle size distribution highlight MUSIC

Step 5: TSS loading highlight buildup/washoff

Step 6: run the simulation

Step 7: design summary STC model is to achieve a minimum of 80% TSS removal.

A copy of the HumeCeptor Design Summary report is to be supplied with the submission. The HumeCeptor is to have a 100 m^3/s high flow by-pass.

The MUSIC nodes for the HumeCeptor can be obtained through MUSIC-*link* and clicking on 'Blacktown City Council'. The percentage pollutant removal rates are as follows:

The percentage pollutant removal rates are as follows:

Pollutant	Point 1 % Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 15 mg/L	thereafter 80%
Total Phosphorous	0% removal up to 0.10 mg/L zero	thereafter 30%
Total Nitrogen	0% removal up to 1 mg/L	Thereafter 15%
Gross Pollutants	75%	

Table 33. HumeCeptor pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	15	15	1000	200
Total Phosphorous	0.10	0.10	5	3.5
Total Nitrogen	1	1	10	8.5
Gross Pollutants	20	5		

Table 34. HumeCeptor GPT Node input and output values

12.10 HumeFilter UPT (Humes)

12.10.1 General description

The HumeFilter Universal Pollutant Trap (UPT) is a stormwater treatment device featuring a multistage treatment method, including primary screening, media filtration, and membrane filtration. Water first enters the device where sediment and solids may fall to the sump floor. Water then passes through a screen, then through the granular activated carbon layer and finally through the pleated cartridge.

The device is designed to be off-line but has internal bypass capacity. HumeFilters shall not be arranged in series. The approved models for use in Blacktown are shown in Table 35 including the approved treatable flow rates.

HumeFilter Model No.	Inlet Pipe Diameter	Treatable flow rate (L/s)
UPT1200	150 mm uPVC @ 0.5% grade	12
UPT1800	225 mm uPVC @ 0.4% grade	30
UPT2400	225 mm uPVC @ 1.0% grade	55
UPT3000 uPVC	300 mm uPVC @ 0.8% grade	100
UPT3000 RCP	300 mm RCP @ 1.1% grade	100
UPT3600 uPVC	375 mm uPVC @ 0.55% grade	160
UPT3600 RCP	375 mm RCP @ 0.8% grade	160

Table 35. HumeFilter Model numbers and key design information

MUSIC modelling

Use a generic node in MUSIC with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) to be set at the treatable flow rate shown in Table 35 above.

The percentage pollutant removal rates are as follows:

Pollutant	Removal (%)
Total Suspended Solids	89
Total Phosphorous	75
Total Nitrogen	50
Gross Pollutants	90

Table 36. HumeFilter pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	110
Total Phosphorous	5	1.25
Total Nitrogen	50	25
Gross Pollutants	1000	100

Table 37. HumeFilter Generic Node input and output values

12.11 HumeGard (Humes)

The HumeGard GPT utilises the processes of physical screening and floatation/sedimentation to separate the litter and coarse sediment from stormwater runoff. It incorporates an upper bypass chamber with a floating boom that diverts treatable flows into a lower treatment chamber for settling and capturing coarse pollutants from the flow. HumeGards are not permitted in series.

As a GPT, the HumeGard should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month or 75% of the 1 EY) depending upon the location. The device is sized to ensure the treatment flow rate matches or exceeds the design flow. The treatment flow rate is input into the node as the high flow by-pass in m^3/s.

Blacktown City Council does not consider HumeGard as a device that removes the 90% of the average annual load of hydrocarbons as requested by Council.

The approved models for use in Blacktown are shown in Table 40.

The MUSIC node can be obtained through MUSIC-*link* and clicking on 'Blacktown City Council'. The pollutant removal rates are as follows:

Pollutant	Removal (%)
Total Suspended Solids	37
Total Phosphorous	30
Total Nitrogen	21
Gross Pollutants	90

Table 38. HumeGard pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	630
Total Phosphorous	10	7
Total Nitrogen	100	79
Gross Pollutants	1000	100

Table 39. HumeGard GPT Node input and output values

HumeGard® model	Pipe diameter or box culvert width (mm)	Treatment flow rate (L/s)	Total pollutant capacity (m^3)	Length (mm)	Width (mm)	Height (mm)
HG12	300	85	3	2,000	1,758	2,500
HG12A	375	100	3	2,000	1,758	2,500
HG15	450	130	3	2,000	1,758	2,500
HG15A	525	150	3	2,000	1,758	2,500
HG18	600	600	3	2,100	2,100	2,115
HG24	600 - 750	1,050	8	2,500	2,700	2,740
HG27	750 - 900	1,110	7	2,500	3,000	2,715
HG30	750 - 825	1,330	12	2,500	3,350	3,365
HG30A	900	1,160	11	2,500	3,350	3,365
HG35	900	1,540	12	2,500	3,850	3,390
HG35A	1,050	1,370	11	2,500	3,850	3,390
HG40	900	1,910	16	2,850	4,350	3,390
HG40A	1,050	1,750	14	2,850	4,350	3,390
HG40B	1,200	1,580	12	2,850	4,350	3,390
HG45	1,200	1,960	19	2,900	4,900	3,915
HG45A	1,350	1,780	19	3,200	4,900	3,915
HG50 and above	Custom					

Table 40. HumeGard treatable flow rates

(Source: Table 2 of the HumeGard GPT Technical Manual Issue 4 by Humes, March 2017)

12.12 Filterra (Ocean Protect)

12.12.1 General description

The Filterra System is a high-flow biofiltration / bioretention technology. It is designed for high volume/flow treatment and to achieve high pollutant removal. It has a small footprint which allows it to be used on highly developed sites such as landscaped areas, parking lots and streetscapes.

Applications include treating car parks, industrial, business and residential developments. Filterra can be configured to treat small or large catchments.

Filter media for the Filterra device is made by Ocean Protect to meet a quality standard. A critical aspect of Filterra is the use of mulch on the top of the media which is sacrificial and needs to be replaced every year. Replacement of the mulch will protect the media, in some cases indefinitely.

Both the Filterra Bioscape insitu basin configuration and the Filterra precast system are configured similarly. They typically feature an extended detention depth of 150–400 millimetres, an offline configuration with an inlet forebay and a weir designed specifically to suit the treatment flow rate capacity, flow distribution, and a bubbler network. Key layers include a 75 mm mulch layer, a 530 mm media layer, and a 150–250 mm drainage layer, concluded with appropriate plant selection and an overflow outlet structure.

MUSIC modelling

Filterra is to be modelled in MUSIC using a detention basin node upstream of a generic node as shown in Figure 24 below:

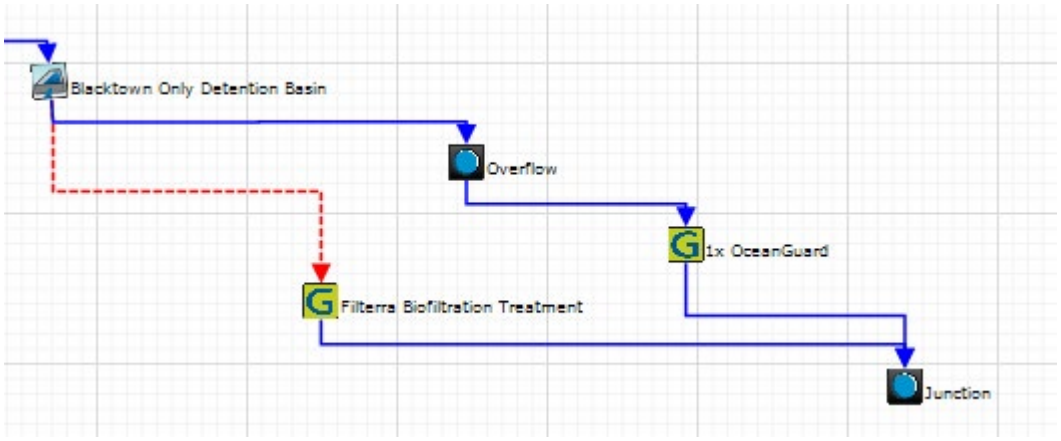


Figure 24. Example of MUSIC setup for Filterra

The upstream detention basin node models the aboveground extended detention storage for the device and is to be configured as follows:

- Low Flow By-pass set to 0 m^3/s
- High Flow By-pass set to 100 m^3/s
- the surface area of the detention basin is the surface area of the Filterra (area of filter shall be greater than 0.3% of catchment area)
- Extended detention depth would vary between 150 millimetres and a maximum of 1 metre and be the depth of storage proposed over the Filterra
- the Exfiltration rate of the detention basin (which models stormwater flowing down through the Filterra media shall be 3560 mm/hr)
- Low flow outlet shall be 1 millimetres which is the minimum. This typically creates a very high notional detention time
- Overflow weir width shall be the proposed overflow weir width.

Set the primary flow link to the overflow and set the red secondary flow link from the detention basin node to the generic node so that infiltrated water (water exfiltrated from the detention basin) only is directed toward the generic node. To do this tick the box which states ‘Infiltration’ in the red secondary flow link.

Assuming there is an OceanGuard litter basket placed in the downstream overflow pit, model the inclusion of the litter basket as per this Handbook and the proposed configuration.

Following the detention basin node, use a generic node in MUSIC with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00.
- High Flow By-pass (m^3/s) to be set at 100.

The percentage pollutant removal rates used to construct generic treatment nodes are shown in Table 41.

Pollutant	Removal (%)
Total Suspended Solids	79.5
Total Phosphorous	80.4
Total Nitrogen	45.9
Volumetric Loss	2

Table 41. Filterra pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following point for the concentration based capture efficiency method. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	205
Total Phosphorous	10	1.96
Total Nitrogen	10	5.41
Gross Pollutants	10	9.5

Table 42. Filterra Generic Node input and output values

12.12.2 Inspections and testing

Saturated hydraulic conductivity of the filter media (not the mulch) shall be measured at the start of the third year of operation using a single ring infiltrometer as follows: 150 millimetres minimum diameter pipe driven to the full depth of the media. If the hydraulic conductivity falls below 3000 mm/hr, the media shall be rectified to restore a minimum K_{sat} of 3560 mm/hr. This test result shall be provided to Council's WSUD Compliance Officer. If required, a second test result following media rectification shall also be provided to Council's WSUD Compliance Officer.

Vegetation shall be inspected annually and if less than 80% coverage (or if for example was designed with one tree), it shall be supplemented with additional planting or planting replaced. This will also be observed by Council's WSUD Compliance Officer

12.13 Jellyfish (Ocean Protect)

12.13.1 General description

The Jellyfish filter is designed as an offline filter structure. A weir is required upstream to provide pressure head to drive stormwater through the system. The unit removes coarse sediment, particulate-bound pollutants, oils and debris. Large, heavy particles fall to the sump (sedimentation) and low specific gravity pollutants rise to the surface (floatation). Each unit comprises a series of high flow and drain-down filter carts, similar in appearance to vertical tentacles. The drain-down carts assist with the self-cleaning backwash.

See also Section 12.19 for specific requirements for the maintenance of the filter cartridges.

12.13.2 MUSIC modelling requirements for Jellyfish

The Jellyfish is represented by a Generic node with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) = (As per Table 45 below)/1,000

The percentage pollutant removal rates are as follows:

Pollutant	Removal Rate (%)
Total Suspended Solids	88
Total Phosphorous	53.6
Total Nitrogen	44.7
Gross Pollutants	75

Table 43. Jellyfish Generic Node input and output values

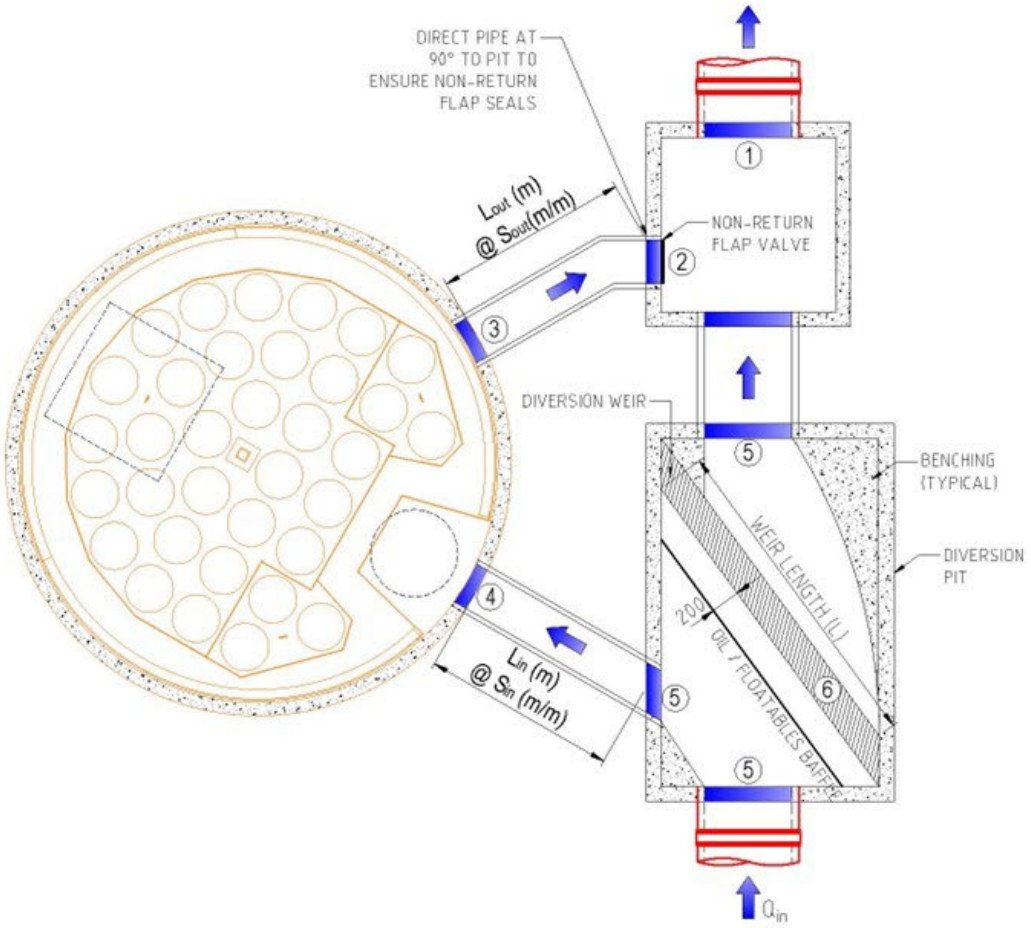
In the MUSIC node the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	1000	115
Total Phosphorous	10	4.64
Total Nitrogen	100	55.3
Gross Pollutants	1000	250

Table 44. Jellyfish Generic Node input and output values

12.13.3 Characteristics

Set up the configuration of the off-line standard Jellyfish and diversion weir as detailed below. The flap prevents debris washing back up the system and clogging the Jellyfish.



LEVELS

These levels are typically based on the minimum design 1 EY HGL at ①

② = Minimum 1 EY HGL at ① + 0.050m (Pipe Invert)
③ = Pipe Invert at ② + L_{out} x S_{out}
④ = Pipe Invert at ③ + 0.150m
⑤ = Pipe Invert at ④ + L_{in} x S_{in}
⑥ = 0.310m + higher of invert at ⑤ or obvert at ③

NOTES

1. L_{weir} (m) = 17.8 x Q_{in} (m^3/s)
2. The oil / floatables baffle is to extend from 0.17m below the weir level to 0.1m above the design water level or pit soffit within the diversion pit.
3. Access points are required for the diversion pit to allow obstruction removal from both sides of the weir. For larger pits provide access points on both sides of the weir.
4. Provide step irons for depths > 0.9 m
5. Final weir level at ⑥ to be confirmed by manufacturer prior to obtaining Construction Certificate.

Figure 25. Standard Jellyfish arrangement for Blacktown City Council

The various unit names and treatable flow rates (L/s) approved by Council are detailed in Table 45.

Unit ID	High-flow cartridges	Drain-down Cartridges	Flow rate (L/s)
JF1200-1-1	1	1	7.5
JF1200-2-1	2	1	12.5
JF2250-3-1	3	1	17.5
JF2250-4-1	4	1	22.5
JF2250-5-1	5	1	27.5
JF2250-6-1	6	1	32.5
JF2250-7-2	7	2	40
JF2250-8-2	8	2	45
JF2250-9-2	9	2	50
JF2250-10-2	10	2	55
JF3250-11-2	11	2	60
JF3250-12-2	12	2	65
JF3250-13-3	13	3	72.5
JF3250-14-3	14	3	77.5
JF3250-15-3	15	3	82.5
JF3250-16-3	16	3	87.5
JF3250-17-3	17	3	92.5
JF3250-18-3	18	3	97.5
JF3250-19-4	19	4	105
JF3250-20-4	20	4	110
JF3250-21-4	21	4	115
JF3250-22-4	22	4	120
JF3250-23-4	23	4	125
JF3250-24-4	24	4	130
JF3250-25-5	25	5	137.5
JF3250-26-5	26	5	142.5
JF3250-27-5	27	5	147.5
JF3250-28-5	28	5	152.5

Table 45. Jellyfish treatable flow rates

Jellyfish can be designed as a low drop version with a 230 millimetre weir. However, the treatable flow rates would need to be adjusted/reduced. For the same model, the flow rates are calculated as half of those in Table 45.

12.13.4 Achieving the 90% average annual load reduction in total hydrocarbons

To achieve this target, Blacktown City Council requires the collection of total hydrocarbons for a minimum 2 EY (6 month or 75% of 1 EY) flow. Utilising either of the methods detailed below enables a Jellyfish treatable flow rate less than the 2 EY flow. To protect the Jellyfish from being blocked with gross pollutants, additional treatment is required upstream using OceanGuards, or an approved GPT.

Method 1: provide a GPT with hydrocarbon trap upstream of the Jellyfish

Design the GPT in accordance with Section 11.16.1 Method 1, or

Method 2: provide a Jellyfish diversion pit with weir and oil baffle

Provide an inline (Jellyfish) diversion pit with a minimum 310 millimetres weir to divert only the treatable flow rate to the Jellyfish unit. The weir is designed to overflow the design flow Q in the pipe, ignoring the Jellyfish treatable flow rate (assuming the Jellyfish flow is blocked).

An impermeable baffle is required 200 millimetres upstream of the Jellyfish weir to retain floatable pollutants including hydrocarbons and direct these to the Jellyfish unit.

The baffle is to extend from 170 millimetres below the weir level, up to 100 millimetres above the design AEP water level. This will allow an emergency overflow over the top of the weir as required.

Limit the velocity under and over the baffle to 0.4 m/s with no sludge layer over the pit floor. That is, L (weir length) = 17.8 x Q (m) where Q under and over the baffle is based on the following:

- online as part of an OSD system, then Q = 1% AEP
- online, but not part of an OSD system then Q = 5% AEP
- online, but downstream of an OSD system then Q = 1% AEP permissible site discharge (PSD).

Note:

- ensure the inlet pipe and Jellyfish outlet pipe are clear of the baffle to avoid short circuiting
- this approach is unacceptable for the low drop version Jellyfish as baffle depth is ineffective, and in this case, the GPT in line with Section 11.16.1 Method 1 must be provided.

12.14 OceanGuard (Ocean Protect)

The OceanGuard is a catch pit insert that captures and retains litter, debris and other pollutants as runoff enters the storm drain system. OceanGuards come in several bag sizes, 200 micron (white) and 1,600 micron (black). The white 200 micron is the only one permitted in our city and is used upstream of tertiary treatment devices to reduce the pollutant loading to these devices. Where a number of OceanGuards are used in similar catchments, it is common to direct the upstream catchments to a single node with the high flow by-pass increased in proportion to the number of OceanGuards. OceanGuards are not permitted in series in MUSIC.

Where OceanGuards are used upstream of a detention system, the underside of the bag should be set (for type 2 or 3 OceanGuards) to a maximum of 250 millimetres below the 50% AEP weir level.

Only the models outlined below are approved for use in Blacktown.

Type M2, L2, L3, XL2 or XL3 OceanGuards

The constraints are detailed in Table 46.

OceanGuard type	Bag depth (mm)	Overall depth (mm)	Pit size (mm x mm)	Maximum mixed catchment area (m^2)	Maximum roof catchment area (m^2)	MUSIC High Flow By-pass (m^3/s)
M2	300	450	600 x 600	500	750	0.02
L2	300	450	900 x 900	1,000	1,500	0.02
L3	600	700	900 x 900	1,100	1,650	0.02
XL2	300	450	1,200 x 1,200	1,400	2,100	0.02
XL3	600	700	1,200 x 1,200	1,500	2,250	0.02

Table 46. OceanGuard constraints (type M2, L2, L3, XL2 or XL3)

Type M2, L2, or XL2 OceanGuards treating only surface flows require a minimum clear depth of 500 millimetres below the grate to any inlet or outlet pipe obvert. OceanGuards treating surface flows and upstream pipe flows require a minimum clear depth of 500 millimetres from the invert of the upstream pipes to be treated, to the obvert of the outlet pipe. Where these pits are treating upstream pipe flows, the inverts of all pipes in and out of the pit are to be shown. Provide a minimum clearance of 750 millimetres for types L3 and XL3.

Type M1, L1 or XL1 OceanGuards

Subject to Blacktown City Council approval, and only in exceptional circumstances where severe height limitations prevent the use of the depth ID 2 or 3 bag and no alternatives exist, will a type 1 filter bag (170 millimetres high) be considered. The filter bad would be have an overall depth of 270 millimetres. The constraints are detailed below in Table 47.

OceanGuard type	Bag depth	Pit size (mm x mm)	Maximum mixed catchment area (m^2)	Maximum roof catchment area (m^2)	MUSIC High Flow By-pass (m^3/s)
M1	170	600 x 600	200	300	0.01
L1	170	900 x 900	400	600	0.01
XL1	170	1,200 x 1,200	560	840	0.01

Table 47. OceanGuard constraints (type M1, L1 or XL1)

Type M1, L1 or XL1 OceanGuards treating only surface flows require a minimum clear depth of 320 mm below the grate to any inlet or outlet pipe obvert. OceanGuards treating surface flows and upstream pipe flows require a minimum clear depth of 270 mm from the invert of the upstream pipes to be treated, to the obvert of the outlet pipe. Where these pits are treating upstream pipe flows the inverts of all pipes in and out of the pit are to be shown.

Type S OceanGuards

Type S OceanGuards for 450 millimetres x 450 millimetres pits are not accepted in our city for MUSIC modelling. The current pollutant removal rates for this device from MUSIC for the 200-micron filter bag are:

Pollutant	Removal (%)
Total Suspended Solids	54
Total Phosphorous	30
Total Nitrogen	21
Gross Pollutants	95

Table 48. OceanGuard pollutant removal rates

Generic nodes shall be constructed using the following input and output values:

For each parameter the base input and output values (not shown in the table) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	100	46
Total Phosphorous	10	7
Total Nitrogen	10	7.9
Gross Pollutants	10	0.5

Table 49. OceanGuard GPT node input and output values

In the MUSIC node the pollutant removal rates are represented by the following points for the concentration based capture efficiency.

12.15 OceanSave (Ocean Protect)

The OceanSave GPT utilises the processes of indirect screening and floatation/sedimentation to separate the litter, pollutants and coarse sediment from stormwater runoff. The unit is generally an off-line system with the driving head produced by an upstream diversion weir. This is a vortex system that minimises blockage of the indirect screen. Some configurations of the device can only be cleaned with an eductor truck.

The device can be designed to direct a proportion of the treatable flow (from 0% up to 100% of the treatable flow) to an alternate destination such as a bioretention basin and return the balance to the main flow. OceanSave devices are not permitted in series, although devices may be used in parallel – for example, twin, side by side setups to increase the overall treatable flow rate.

As a GPT, the device should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month or 75% of the 1 EY) depending upon the location. See also Section 11.11. The device is sized to ensure the treatable flow rate matches or exceeds the design flow. The treatable flow rate is input into the node as the high flow by-pass in m3/s. Treatable flow rates are detailed in Table 52.

The MUSIC node for the OceanSave device can be obtained through MUSIC-*link* and clicking on ‘Blacktown City Council’. The percentage pollutant removal rates are as follows:

Pollutant	Point 1% Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 75 mg/L	thereafter 70%
Total Phosphorous	0% removal up to 0.5 mg/L	thereafter 30%
Total Nitrogen	Zero	
Gross Pollutants	98%	

Table 50. OceanSave pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	75	75	1,000	300
Total Phosphorous	0.5	0.5	10	7.0
Total Nitrogen	50	50		
Gross Pollutants	100	2		

Table 51. OceanSave GPT node input and output values

All OceanSave devices in Blacktown must contain an oil baffle.

Only the models below are approved for use in Blacktown.

OceanSave	Diameter (m)	Typical depth below invert (m)	Maximum Flow (L/s)	Recommended single pipe size* (mm) diameter	Water quality flow rate (L/s)	Sump storage capacity (m^3)	Oil storage capacity (L)
OS-0606	1.2	1.3	228	375	28	0.8	150
OS-0809	1.5	1.6	368	450	68	0.8	250
OS-1112	2.25	2.0	685	600	155	2.5	600
OS-1515	2.25	2.8	690	600	290	4.4	650
OS-2318	3.25	3.4	1380	750	580	11.9	2000
OS-2324	3.25	4.0	1290	750	900	9.5	2000

Table 52. OceanSave treatable flow rates

12.16 StormFilter (Ocean Protect)

12.16.1 General description

StormFilters are filter cartridges containing a blend of zeolite, perlite and activated carbon. An alternate StormFilter blend called Psorb is not currently approved for use in Blacktown. A StormFilter weir is required downstream to provide sufficient head to activate the syphon in the cartridge. The cartridges come in 3 options, each with an orifice at the base that controls the flow to the piped underdrains below. These pipes are covered in concrete called a false floor (typically 50 to 160 millimetres thick) to provide a safe and smooth surface for cleaning, support the underdrain pipes and to remove standing water. See also Section 12.19 for specific requirements for the maintenance of the filter cartridges.

12.16.2 Characteristics

An impermeable baffle is required 250 millimetres upstream of the StormFilter weir to retain floatable pollutants including hydrocarbons. The baffle is to extend to the underside of the tank roof or 250 millimetres above the design storage. Limit the velocity under and through the baffle to 0.4 m/s allowing for a 50 millimetres sludge layer over the false floor.

The weir length (L) in Table 53 is calculated on the flow Q (m^3/s) under and through the baffle based on the following:

- online as part of an OSD system, then Q = 1% AEP
- online, but not part of an OSD system then Q = 5% AEP
- offline, minimum 2 EY (6 month) flow (75% of 1 EY) if used with a diversion weir.

The upstream diversion weir height is a minimum of the StormFilter weir level, plus the depth of flow over the StormFilter weir (using the weir equation for flow Q), plus the pipe and pit losses to the diversion weir.

Cartridge name	Cartridge height (mm)	Weir height H above false floor (mm)	Flow through each at weir height TF (L/s)	Baffle depth D below weir level (mm)	Length of weir L (m)
310	600	390	0.7	200	17.8 x Q
460	600	540	1.1	300	13.2 x Q
690	840	770	1.6	400	10 x Q

Table 53. StormFilter characteristics and controls

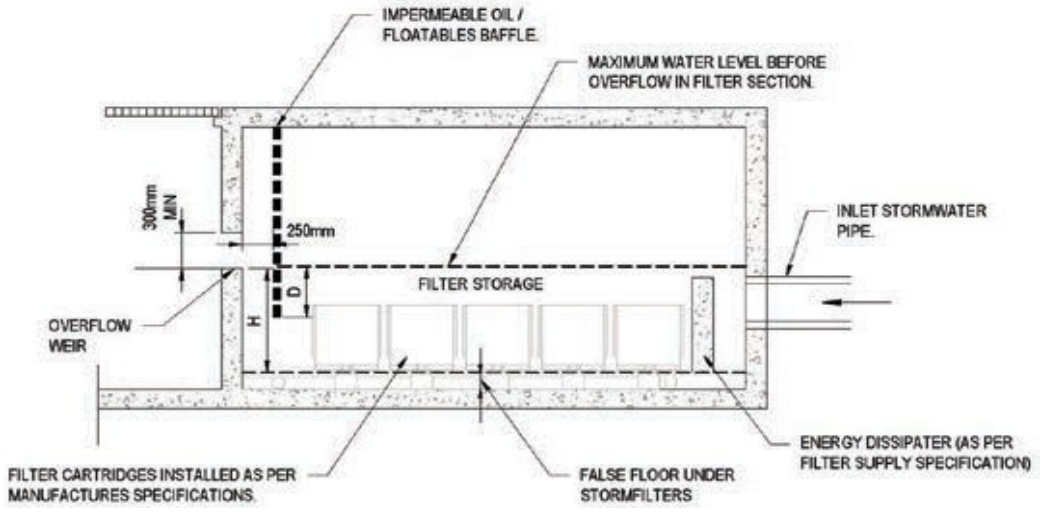


Figure 26. StormFilter chamber arrangement

12.16.3 MUSIC modelling requirements for StormFilters

The StormFilter system is controlled through 2 nodes: a Detention node followed by a Generic node. Ocean Protect has a spreadsheet that can assist with populating each of these nodes. Consider 'No.' in the notes below as representing the number of cartridges.

1. Detention node with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) leave default as 100.00
- Surface Area (m^2) = area of Stormfilter chamber upstream of the weir – (No. x 0.177)

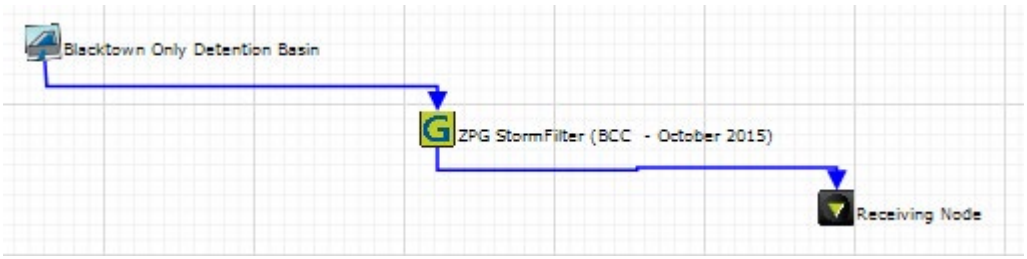


Figure 27. Example of MUSIC setup for StormFilters

- Extended Detention Depth (m) = Weir height from Table 53 for the cartridge type
- Exfiltration Rate (mm/hr) leave default as 0.00
- Evaporative Loss as percentage of PET leave default as 0.00
- Low Flow Pipe Diameter (mm) calculated using orifice equation for total filter flow at weir height (Refer to the Ocean Protect Calculator)
- Overflow Weir Width (m) set to the weir length L from the Table 53. There is no reuse permitted using this node
- under the Advanced tab the k values are all set to 0.

Note:

- the minimum area of StormFilter chamber upstream of weir = No. x 0.36 m^2
- the maximum area of Stormfilter chamber upstream of weir = 20 m^2/ha which is equivalent to 51.3 m^2/ha (310 mm high cartridge), 37.0 m^2/ha (460 mm) and 26.0 m^2/ha (690 mm)

2. Generic node with the following characteristics:

- Low Flow By-pass (m^3/s) leave default as 0.00
- High Flow By-pass (m^3/s) = No. x TF (Table 53)/1,000

The percentage pollutant removal rates are as follows:

Pollutant	Removal (%)
Total Suspended Solids	85
Total Phosphorous	59
Total Nitrogen	33
Gross Pollutants	95

Table 54. StormFilter pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	100	15
Total Phosphorous	10	4.1
Total Nitrogen	10	6.7
Gross Pollutants	1000	50

Table 55. StormFilter node input and output values

12.16.4 Hydraulic loading rates for StormFilters

Over time, the cartridges will accumulate pollutants causing a restriction to flow and ultimately blockage in the cartridge. As a guide to ensure the cartridges remain as functional as possible for the design life, an assessment can be made using MUSIC of the TSS removed by the Stormfilter tank (using mean annual loads not treatment train effectiveness). This should be divided by the number of cartridges in the tank to obtain a TSS removal (kilograms) per cartridge. The rates per cartridge should fall below the rates below depending upon cartridge type. Only TSS is considered in this instance and depending upon the catchment characteristics, TP and TN removal capability may still be depleted prior to clogging with TSS:

- 310 – 14 kilograms
- 460 – 22 kilograms
- 690 – 32 kilograms.

Where the above rates are exceeded, it is suggested that additional cartridges are added. This is so that the hydraulic loading rate falls below these rates, even where the TSS targets have already been achieved in MUSIC.

12.16.5 Specific requirements for StormFilters with on-site stormwater detention

Most early drainage designs incorporating water quality devices as part of the detention basin have directed the treated flow from the device (underdrain flow) directly back into the basin or to the low flow orifice control. Both proprietary filters and bioretention rely on a water pressure difference (Darcy's Law) to drive the stormwater through the treatment device. Where the water levels rise at the device outlet, such as in a detention basin, there is an immediate reduction in flow. As the downstream water levels continue to rise, the treatable flow rate continues to reduce. When the water levels equalise, all treatment ceases. This does not match the MUSIC model which is based on continuous treatment throughout the whole storm duration. The pollutant removal will not reach the targets in practice. This approach is prohibited in our city.

To meet the requirements of the MUSIC model, the acceptable approach within our city where water quality treatment is incorporated into a detention basin itself, is for the treated underdrain flow (for example proprietary filters) to discharge downstream of the detention discharge control pit. This will ensure ongoing treatment throughout a range of storms. For proprietary filter underdrains, this flow can be significant and the orifice size needs to be reduced to ensure the permissible site discharge can be maintained.

A standard configuration of StormFilter chamber with detention is shown on Blacktown City Council's WSUD standard drawing 6102. When using Council's OSD spreadsheet, select Yes for question:'Will filter cartridges be used to manage water quality?. To ensure the StormFilters operate through all storms as in MUSIC, the underdrains discharge downstream of the orifice controls. Without this, if the tailwater level rises above the false floor, the cartridge discharge will start to slow and will stop working completely when it reaches the StormFilter weir height. The 50% AEP (1 in 2 AEP) orifice is to be reduced in size to ensure the 50% AEP permissible site discharge is maintained.

Then:

$Q_{50\%\ AEP\ Orifice}$ = $PSD_{50\%\ AEP}$ – Flow through Each Cartridge at Weir Height x No. Cartridges

Use the orifice equation (C_d = 0.61) with $Q_{50\%\ AEP\ Orifice}$ to determine the reduced orifice size at the 50% AEP control pit weir level. Note that this process is undertaken automatically with the use of Council's OSD spreadsheet when the total design filter flow rate is entered at Design flow from Filter Cartridges (L/s).

Similarly, the 1% AEP orifice is to be reduced in size to ensure the 1% AEP permissible site discharge is maintained. Then:

$Q_{1\%\ AEP\ Orifice}$ = $PSD_{1\%\ AEP}$ – Flow through Each Cartridge at 1% AEP water level x No. Cartridges

Use the orifice equation (C_d = 0.61) with $Q_{1\%\ AEP\ Orifice}$ to determine the reduced orifice size at the 1% AEP emergency overflow weir level. This process is undertaken automatically with the use of Council's OSD spreadsheet. However, it is necessary to assess the flow through the cartridges at the higher water level and input this into the spreadsheet at filter cartridge flow with 1% AEP Head (L/s).

The flow through each StormFilter is controlled by an orifice, then with increasing depth and pressure head, the flow through the cartridge will increase. Ocean Protect can provide a spreadsheet to determine the flow through the cartridges at the 1% AEP storage level.

12.16.6 On-site stormwater detention problems with StormFilters and solutions

In some designs, the calculated 50% AEP (1.5 year ARI) orifice size will reduce below Blacktown City Council's minimum of 25 millimetres. Where this occurs, delete the 50% AEP orifice completely (subject to the alternatives and limitations below) and direct all low flows through the underdrain. Reconfigure the floor of the detention tank to direct the low point to the non-return flap into the StormFilter chamber, so that all low flows are redirected back into the filter chamber. Essentially, the controlled flow through the Stormfilter underdrains acts as the low flow (50% AEP) control in place of the standard orifice. See Blacktown City Council's WSUD standard drawing 6102.

However, where the 50% AEP orifice is deleted due to higher underdrain flows, this results in adverse environmental impacts. To avoid this, Blacktown City Council will require the investigation of possible partial alternative WSUD treatments. Deleting the 50% AEP orifice will only be accepted where it can be demonstrated that no reasonable alternatives exist.

For specific steps as to how to redesign the basin/tank to delete the 50% AEP orifice, see Section 11.6.3.

In some designs that use a large number of StormFilters, our OSD spreadsheet may indicate that applying the high underdrain flow from the filters in the 1% AEP event will result in the 1% AEP orifice not being required. This is not acceptable and suggests that more filters have been modelled than is reasonable. To ensure a 1% AEP orifice is required, it will be necessary to reduce the number of filter cartridges to ensure a minimum 25 millimetres 1% AEP orifice by either:

- treating part of the site using bioretention which has generally nil impact on the orifice size
- splitting the filters into multiple tanks with some more remote from the detention and higher such that the discharge is above the 1% AEP storage level
- treating part of the site using other alternative devices such as a Jellyfish
- relocating the filter cartridge chamber downstream, subject to suitable levels.

12.17 CDS GPT (Civil Mart – Rocla)

The Rocla Continuous Deflective Separation (CDS) GPT utilises the processes of indirect screening and floatation/sedimentation to separate the litter, pollutants and coarse sediment from stormwater runoff. The CDS unit is generally an off-line system with the driving head produced by an upstream diversion weir. This is a vortex system that minimises blockage of the indirect screen.

The CDS device can be designed to direct a proportion of the treatable flow (from 0% up to 100% of the treatable flow) to an alternate destination such as a bioretention basin and return the balance to the main flow. CDS devices are not permitted in series, although CDS devices may be used in parallel – for example, twin, side by side setups to increase the overall treatable flow rate.

As a GPT, the CDS should be designed for a minimum 4 EY (3 month) flow up to the 2 EY (6 month or 75% of the 1 EY) depending upon the location. See also Section 11.11. The device is sized to ensure the treatable flow rate matches or exceeds the design flow. The treatable flow rate is input into the node as the high flow by-pass in m3/s. Treatable Flow Rates are detailed on the CDS nodes from MUSIC-*link*.

The MUSIC nodes for each CDS device can be obtained through MUSIC-*link* and clicking on Blacktown City Council. The percentage pollutant removal rates are as follows:

Pollutant	Point 1 % Removal (concentration based)	Point 2 % Removal (concentration based)
Total Suspended Solids	0% removal up to 75 mg/L	thereafter 70%
Total Phosphorous	0% removal up to 0.5 mg/L	thereafter 30%
Total Nitrogen	Zero	
Gross Pollutants	98%	

Table 56. CDS pollutant removal rates

In the MUSIC node, the pollutant removal rates are represented by the following points for the concentration based capture efficiency. For each pollutant, the base input and output values (not shown in the table below) are to be 0 for both input and output values, i.e. 0,0.

	Point 1		Point 2	
Pollutant	Input	Output	Input	Output
Total Suspended Solids	75	75	1,000	300
Total Phosphorous	0.5	0.5	10	7.0
Total Nitrogen	50	50		
Gross Pollutants	100	2		

Table 57. CDS GPT node input and output values

All CDS devices in our city must contain an oil baffle. The CDS0506 (Nipper) and CDS3000 series come with an inbuilt oil baffle. However, it must be specified for all other units including the CDS0708.

12.18 Enviropod 200 (Civil Mart – Rocla)

The Envriopod 200 is an effective, easily maintained catch pit insert that captures and retains litter, debris and other pollutants as runoff enters the storm drain system. Enviropod 200 baskets come in 200 micron (white). The 200-micron litter basket is the only one permitted in our city and is used upstream of secondary treatment devices to reduce the pollutant loading to these devices.

Where Enviropod 200 litter baskets are used upstream of a detention system, the underside of the bag should be set a maximum of 250 millimetres below the 50% AEP weir level.

Only the following models are approved for use in Blacktown.

Model Number	Pit Size	MUSIC High-Flow By-pass rates (L/s)
EP6060-200	600 x 600	20
EP6090-200	900 x 600	20
EP9060-200	900 x 900	20
2 x EP9060-200	1200 x 1200	20

Table 58. Approved Enviropod models

Enviropod 200 litter baskets are 500 millimetres deep from the seal to the base of the bag. Where depth is an unavoidable constraint Council will approve shorter bags. A clear depth of 500 millimetres below the grate to the obvert of the pipe is required to ensure blockage will not occur. Where pits are treating upstream pipe flows, the invert levels of all pipes into and out of the pit are to be shown on the design drawings. Refer to the manufacturer for additional information.

MUSIC high flow by-pass rates are to be set at 20 l/s. Where a number of Enviropod 200 baskets are used in similar catchments it is common to direct the upstream catchments to a single node with the high flow by-pass increased in proportion to the number of Enviropod 200 baskets. Enviropod 200 baskets are not permitted in series in MUSIC.

The current pollutant removal rates for this device from MUSIC for the 200-micron filter bag are:

Pollutant	Removal (%)
Total Suspended Solids	54
Total Phosphorous	30
Total Nitrogen	21
Gross Pollutants	95

Table 59. Enviropod pollutant removal rates

Generic nodes shall be constructed using the following input and output values:

For each parameter the base input and output values (not shown in the table) are to be 0 for both input and output values, i.e. 0,0.

Pollutant	Input	Output
Total Suspended Solids	100	46
Total Phosphorous	10	7
Total Nitrogen	10	7.9
Gross Pollutants	10	0.5

Table 60.Enviropod GPT node input and output values

12.19 Specific requirements for maintenance of filter cartridges

12.19.1 Failure mechanisms for filter cartridges

The filter cartridges are generally designed so that the flow rate through the cartridges is controlled by the orifice imbedded into the cartridge and not the cartridge itself. This allows for the filter to do its job over an extended period by capturing pollutants without generally affecting the flow rate.

Each filter has an inbuilt backwash function to further prolong the life of the cartridge. However, over time the filter cartridge will continue to trap pollutants and eventually it will reach a point where the blockage will affect the flow rate through the cartridge. This will affect the ability to meet Blacktown City Council's load-based water quality objectives. The critical blockage time for the cartridge may vary for the different manufacturers. This reduction in flow rate represents clogging by TSS that will reduce effective removal of all pollutants. However, specific elements within the filter composition that targets removal of TP or TN may be expended well before blockage by TSS. This requires good maintenance plans by the manufacturer to prevent this occurring.

12.19.2 Minimum performance standard

Blacktown City Council's minimum requirement is that the actual flow rate through the filter cartridges cannot be lower than 90% of the design flow rate. It is assumed that most cartridges will perform reasonably well for an initial period after the date of installation. This initial period is considered as 3 years. After 3 years, a flow test is to be undertaken to justify the continued use of the existing filter cartridges using Section 12.16. 8 or 12.16.9. All filter cartridges must be replaced after 5 years.

It is also recognised that there may be some high pollutant generating sites that may require more frequent maintenance or replacement.his should be assessed and managed by the maintenance contractor and/or the original cartridge supplier where known.

12.19.3 Flow test method – individual cartridge constant head

To test whether the individual cartridge is still maintaining the minimum required flow rate for either a StormFilter, AtlanFilter, Jellyfish, or subsequent Blacktown City Council approved filter cartridge, undertake the following:

Step 1: the flow test method is to be performed by a maintenance contractor. The contractor must possess the qualifications and licences required to undertake these works safely and efficiently in accordance with Blacktown City Council and New South Wales policy and legislation

Step 2: provide a Blacktown City Council report to record the Flow Test results (click on this link to access the report or email wsud@blacktown.nsw.gov.au). Detail the address, date, time, specific chamber (if more than 1), number of and cartridge manufacturer, size, design flow rate and total number of cartridges in each tank. Provide geotagged and dated digital photos or videos indicating the site, cartridge(s) chosen and test rig operation

Step 3: select from the filter chamber a random sample of 1 cartridge per 25 cartridges or part thereof. Where there are multiple chambers treating different catchments, select from each chamber a random sample of 1 cartridge per 25 cartridges or part thereof

Step 4: use a specially designed rig that enables the filter cartridge to be positioned in with a standard configuration with an overflow set to the design weir/head high for that particular cartridge type

Step 5: fill the test rig to overflowing using a hydrant standpipe or water supply tanker. Maintain sufficient flow rate to ensure continual overflow

Step 6: provide a calibrated measuring container of exact known volume (L) below the pipe underdrain, or Jellyfish overflow pipe

Step 7: ensure the water from the weir overflow cannot enter the container

Step 8: empty any water from the container and place under the pipe flow and determine the time (in seconds) needed to fill the container

Step 9: calculate the actual flow rate through the filter cartridge in L/s by dividing the volume of the container (L) by the time (s). Where multiple cartridges are being tested consider the average flow rate

Step 10: compare this actual (or average) flow rate with the standard flow rate as supplied by the manufacturer

Step 11: where the actual (or average) flow rate is less than 90% of the design total flow rate, select an additional random sample of 1 cartridge per 25 cartridges or part thereof. Where experience has shown that the first set of sample cartridge(s) tested are likely to be a reasonable representation of the remainder of the cartridges, then the contractor can proceed direct to Step 13

Step 12: repeat the test as above for the new samples

Step 13: where the actual (or average) flow rate of the new sample is less than 90% of the design total flow rate, replace all the cartridges within that chamber

Step 14: the maintenance contractor must report the results and findings of the test to the WSUD Compliance Officer, at wsud@blacktown.nsw.gov.au within 1 week of performing the test.

12.19.4 Flow test method – individual cartridge constant flow (flow meter)

To test whether the individual cartridge is still maintaining the minimum required flow rate for either a StormFilter, AtlanFilter, Jellyfish, or subsequent Blacktown City Council approved filter cartridge, undertake the following:

Step 1: the flow test method is to be performed by a maintenance contractor. The contractor must possess the qualifications and licences required to undertake these works safely and efficiently in accordance with Blacktown City Council and New South Wales policy and legislation

Step 2: provide a Blacktown City Council report to record the Flow Test results (email wsud@blacktown.nsw.gov.au). Detail the address, date, time, specific chamber (if more than 1), number of and cartridge manufacturer, size and design flow

Step 3: select from the filter chamber a random sample of 1 cartridge per 25 cartridges or part thereof. Where there are multiple chambers treating different catchments, select from each chamber a random sample of 1 cartridge per 25 cartridges or part thereof

Step 4: use a specially designed rig that enables the filter cartridge to be positioned in it with a standard configuration including weir height

Step 5: fill the test rig using a hydrant standpipe or water supply tanker and allow the flow to discharge through the filter underdrain or Jellyfish pipe overflow

Step 6: maintain the flow rate so that the water level in the test rig remains constant. Record this actual flow rate using the calibrated flow meter. Where multiple cartridges are being tested, consider the average flow rate

Step 7: compare this actual (or average) flow rate with the standard flow rate as supplied by the manufacturer. Where the constant water level is not at the standard height then consult the manufacturer for the non- standard design flow rate

Step 8: where the actual (or average) flow rate is less than 90% of the design total flow rate select an additional random sample of 1 cartridge per 25 cartridges or part thereof. Where experience has shown that the first set of sample cartridge(s) tested are likely to be a reasonable representation of the remainder of the cartridges then the contractor can proceed direct to Step 10

Step 9: repeat the test as above for the new samples

Step 10: where the actual (or average) flow rate of the new sample is less than 90% of the design total flow rate replace all the cartridges within that chamber

Step 11: the maintenance contractor must report the results and findings of the test to the WSUD Compliance Officer at wsud@blacktown.nsw.gov.au within 1 week of performing the test.

12.19.5 Maintenance contract

The maintenance contract is to include at a minimum:

1. A requirement that either the filter cartridges
   a. are replaced no later than 3 years after the date of installation, or
   b. must pass a flow test every year, 3 years after the date of installation in accordance with Sections 12.19.3 or 12.19.4 above.

Note: If the flow test fails it must be replaced immediately. Further, all filters must be replaced a maximum 5 years after installation.

2. A requirement that the cartridges must be replaced by a cartridge of the same kind (for example, same branding, size and type) as per approved conditions.
3. The flow test report and photos are to be submitted to Blacktown City Council's WSUD Compliance Officer at wsud@blacktown.nsw.gov.au within 1 week of each test activity.

12.20 Requirements for approval of new stormwater quality improvement devices

Proprietors are to note that for Council to consider the approval of any new stormwater quality improvement device (SQID) for use in Blacktown City, it must first obtain approval through Stormwater Australia as per the *Stormwater Quality Improvement Device Evaluation Protocol* (SQIDEP) approval process. For more information on the SQIDEP process, refer to Stormwater Australia.

In addition to obtaining approval through SQIDEP, Council will assess additional items not currently assessed through SQIDEP as per Table 61 below.

Table 61. Items/factors needed for Blacktown City Council assessment of SQIDs

Item/factor	Description
Dissolved inorganic nitrogen results	Analysis of long-term water quality data collected by Council indicates the presence of higher levels of dissolved inorganic nitrogen in our waterways. This has been confirmed by both grab sampling and auto sampling. Therefore, for a test site and field test data to be acceptable to Council, it must have a minimum of 25% dissolved inorganic nitrogen. It is strongly recommended that the applicant undertake pre-testing to determine if their chosen site is suitable and truly representative. This is to ensure that devices which claim to remove nitrogen are not simply removing high proportions of particulate nitrogen which would overstate nitrogen removal on a typical Blacktown development site.
Test site design	The applicant is to ensure that the test site is representative of real-world applications and therefore that the test site design conditions would be transferred directly to an approval. For example, if there was a 3 m3 storage vault with 3 filter cartridges in it, Blacktown City Council would require future installations to have the same ratio of vault volume to number of filter cartridges. Blacktown City Council urges caution with test site design to ensure it is representative and does not commercially jeopardise the product. This is entirely at the risk of the proprietor.

Item / factor	Description
	The applicant is to ensure that the test site and its contributing catchment area is well matched to the tested device to ensure that the device is properly stress tested. SQIDEP requires at least 2 qualifying events to exceed the treatable flow rate of the SQID and equally, events that are too small to properly stress test the device will be rejected. The purpose of field validation of a SQID is to demonstrate how it performs over a prolonged period of time and when subject to multiple storms. If an applicant is monitoring across 2 sites, there is to be a minimum of 7 tests undertaken at each test site over a minimum 12-month period.
Performance	SQIDEP allows for a number of different metrics to be used to measure SQID performance. To provide a consistent approach to SQID approval, Blacktown City Council requires the performance of the SQID to be measured using the average of the Efficiency Ratio (ER) and Median Concentration Reduction Efficiency (CRE). Blacktown City Council may apply discount factors to claimed performance. The discount factor will consider a broad range of considerations such as: applicability of overseas test data, reliability of overseas test data and accreditation of overseas testing laboratory, completeness of an application, duration of monitoring, quality of data collected, risk of failure and other relevant factors. Discount factors are not typically applied to a final approval as the final approval should be based on results from a minimum of 15 local storm events. However, where there is an element of testing that is not 'representative' a discount factor may be applied to a final approval.
Maintenance	All maintenance that occurs during the monitoring period is to be witnessed by an independent experienced engineering hydrologist (water quality specialist) peer reviewer and recorded. The frequency of maintenance and the tasks that occurred during the approval process testing period will form part of the SQID approval. This information will be used to inform the WSUD Compliance Program inspection and maintenance frequencies, and the responsibilities of asset owner(s) accordingly. Council may condition an inspection and maintenance frequency on the device based on industry standards. Alternately, Council may specify other criteria as it sees fit (such as flow tests) to form the basis of when maintenance should occur. Should the applicant wish to vary the specified maintenance interval, long term performance data will need to be provided to the satisfaction of Council. The QAPP shall provide approved methods to assure no tampering will occur during the testing process.
Conflict of interest declaration	All monitoring results submitted to Council must be collected, tested and analysed by an independent contractor/s. Results will not be accepted from proprietors where a conflict of interest (perceived or actual) has been identified.

To obtain approval for use of the SQID on Council (public) land, the following information must also be submitted for assessment with an independent peer reviewed approval of all claims.

Item / factor	Description
Lifecycle costs	The applicant must provide an indicative cost of the SQID including the initial installation costs, ongoing operation and maintenance costs and device renewal.
Maintenance	The applicant must provide details of the maintenance requirements for each component of the device including frequency. Details should include any specialised equipment that needs to be used as well as key safety factors that must be taken into consideration. For Council to approve a SQID for use in the public domain, the SQID must be capable of being maintained using current standard maintenance practices and equipment. For example, gross pollutant traps must be capable of being cleaned via a grab arm (in addition to eduction cleaning) which reduces maintenance costs.
Sustainability	The applicant must provide for each key component of the SQID: • the level of embodied carbon contained within the device compared with other common materials – for example fibre glass and concrete • material used to create devices and its components, including their source and sustainability credentials. This applies to both filter media (where applicable) and the device structure/housing. Unsustainable sources of filter media will not be supported • manufacturing location and method of transport to Blacktown City and site • longevity of device and its components • waste classification of device and its components (can it be reused or recycled).

Changes to existing approvals

Where an applicant is seeking variations to an already approved device, including approval for new models, new approval will need to be obtained.

Costs for assessment and approval of new SQIDs and amendments to approved SQIDs

Assessing a SQID for compliance with Blacktown City Council's requirements is a time-consuming task that provides a commercial benefit to the applicant. Blacktown City Council will recover costs associated with the assessment, approval and updates required to existing assessment tools, documents and website. The fees will be charged in accordance with Council's Goods and Services Pricing Schedule.

Any requests to review and approve a new SQID will be at Council's discretion based on Council's capacity to engage an independent consultant to undertake this task. This will be determined on a case-by-case basis.

13 Calculation of the Stream Erosion Index (SEI)

13.1 When to use the SEI

The SEI is required to be satisfied for all lands within the growth centre areas and employment zones where permanent on-lot water quality treatment is required. This includes:

- industrial
- business
- R3 and R4 residential zonings
- employment lands
- some SP2 special uses such as community or aquatic centres, substations, pumping stations or similar - it is required in these areas even where regional detention basins are provided within the catchment.

The SEI will not be required at the time of land subdivision only. It will be required at time of development of the individual lots.

13.2 Deemed-to-comply solutions for the North West Growth Area

SEI is deemed-to-comply and will not be required where permanent on-site stormwater detention is provided using Blacktown City Council's OSD spreadsheet or S3QM. This is in accordance with Version 4 of the UPRCT that provides a low flow orifice sized at the 50% AEP (approximately 1.5 year ARI) and with a total minimum site storage volume of 300 m^3/Ha for the 50% AEP event. Temporary detention basins cannot be used in our city to achieve the target SEI.

Where permanent on-site stormwater detention is not provided as described above. Council will accept the following approaches to a deemed-to-comply solution for the SEI:

- **Industrial, business, SP2 special uses**: a stormwater treatment train, which includes a rainwater tank that supplies 80% of non-potable demand, together with a bioretention basin sized in line with Section 11.8 that meets the water quality targets as the only tertiary treatment device, is an acceptable solution for these developments. Non-potable demand includes all landscape watering (including the bioretention basin, swales, landscape beds and turf), together with all internal uses (toilet flushing), plus any site-specific uses such as truck washing or other industrial processes
- **R3 or R4 residential development**: a stormwater treatment train, which includes a rainwater tank that supplies 80% of non-potable demand, together with a bioretention basin sized in accordance with Section 11.8 that meets the water quality targets as the only tertiary treatment device, is an acceptable solution for these developments. Non-potable demand includes all landscape watering (including the bioretention basin, swales, landscape beds and turf).

13.3 How to estimate the SEI

The SEI is defined as:

SEI = sum of the post development volume of mean annual stormwater flows greater than the 'stream-forming flow' divided by the sum of the pre-development (for the catchment under natural conditions) volume of mean annual stormwater flows greater than the 'stream-forming flow'.

The higher the SEI, the more damage can occur to the natural creek system.

For areas in the growth centre areas including the Marsden Park industrial precinct and Western Sydney Employment Areas (where on-lot water treatment is required), then the SEI is to be determined and measures undertaken to ensure that the SEI less than or equal to 3.5.

Blacktown City Council uses the method developed in the NSW *MUSIC Modelling Guidelines 2015* that is adapted from D. Blackham and G. Wettenhall (2010).

Due to sensitive environmental issues for areas draining to Little Creek, the SEI must be less than or equal to 1. This is unless alternative low flow systems are in place that direct low environmental flows to alternate drainage catchments.

WSUD strategies are typically modelled using the MUSIC. MUSIC can be used to estimate the SEI for a developments stormwater management strategy to determine compliance with the SEI objective.

Blacktown City Council requires that the post development duration of stream forming flows will be no greater than 3.5 times the pre-developed duration of stream forming flows with a stretch target of 1.

The 4 steps for estimating SEI:

Step 1: estimate the critical flow for the receiving waterway above which mobilisation of bed material or shear erosion of bank material commences

Step 2: develop and run a calibrated MUSIC model of the area of interest for pre-development conditions to estimate the mean annual runoff volume above the critical flow

Step 3: develop and run a MUSIC model for the post developed scenario to estimate the mean annual runoff volume above the critical flow

Step 4: use the outputs from Methods 1 or 2 in Section 13.6 to calculate the SEI for the proposed scenario.

13.4 Estimating the critical flow for the receiving waterway

The critical flow for a waterway is defined as the flow threshold below which no erosion is expected to occur within the waterway. This has been estimated (EarthTech, 2005) as a percentage of the pre-development 50% AEP (2 year ARI) peak flow at the location in question. For our City, this percentage is 25% based on the dispersive characteristics of the typical local clay soils. The peak flow from the 50% AEP storm event corresponding for pre-developed conditions is to be calculated using the probabilistic rational method as described in *Australian Rainfall and Runoff 1987 (ARR)*:

Step 1: using the area of the site (in square kilometres), calculate the time of concentration using the probabilistic rational method from equation 1.4 of *ARR 1987* Volume 1, Book 4.

$t_c = 45.6 \times A^{0.38}$ A(km^2 =ha/100), t_c (minutes)

Step 2: select I_2 (mm/hr) from the rainfall intensity chart in the *Blacktown City Council's Engineering guide for development 2005* based on the 50% AEP (2 year ARI) and the calculated t_c in minutes

Step 3: determine the 50% AEP (2 year ARI) runoff coefficient C_2 using equation 1.5 of *AR&R* Volume 1, Book 4.

$C_2 = C_{10} \times FF_2 = 0.6 \times 0.74 = 0.444$

where C_{10} is the 10 % AEP (10 year ARI) runoff coefficient from Figure 5.1 from *ARR 1987* Volume 2 = 60%, and FF_2 = the 50% AEP (2 year) frequency factor from Table 1.1 of *ARR 1987* Volume 1, Book 4 = 0.74

Step 4: using the rational method $Q_2 = 0.278 \times C_2 \times I_2 \times A$, substitute results from 2 and 3 above.

Q_2 (m^3/s) = 0.278 x 0.444 x I x A = 0.1234 x I (mm/hr) x A (km^2)

Step 5: $Q_{critical}$ = Q_2 x 25% (m^3/s)

13.5 Developing SEI MUSIC models for pre and post-development

2 MUSIC models are to be prepared.

The pre-development model will consider the site fully pervious using the Urban node for Revegetated Land.

The post development MUSIC model is the same model required to meet the water quality systems targets but modified slightly to determine the SEI depending upon the method chosen.

13.6 Estimate the mean annual flow for pre and post-development and calculating the SEI

13.6.1 Method 1. Generic node with high flow by-pass

At the downstream end of each of the pre-development and post-development models, create a Generic node with the high flow by-pass set to $Q_{critical}$ in m^3/s. Direct the primary drainage link to the Receiving node and the high flow by-pass as a red secondary drainage link to a new Junction node which then flows to the Receiving Node. After running the model:

1. left click the Junction node
2. click on ‘Mean Annual Loads’
3. copy the flow outflow value in ML/yr.

The SEI is calculated as the output mean annual flow at the Junction Node for the post-developed model divided by the corresponding value for the pre-development model. The SEI has to be less than 3.5 with a stretch target of 1.

SEI = Post developed output flow $_{Junction}$ / Pre-developed output flow $_{Junction}$

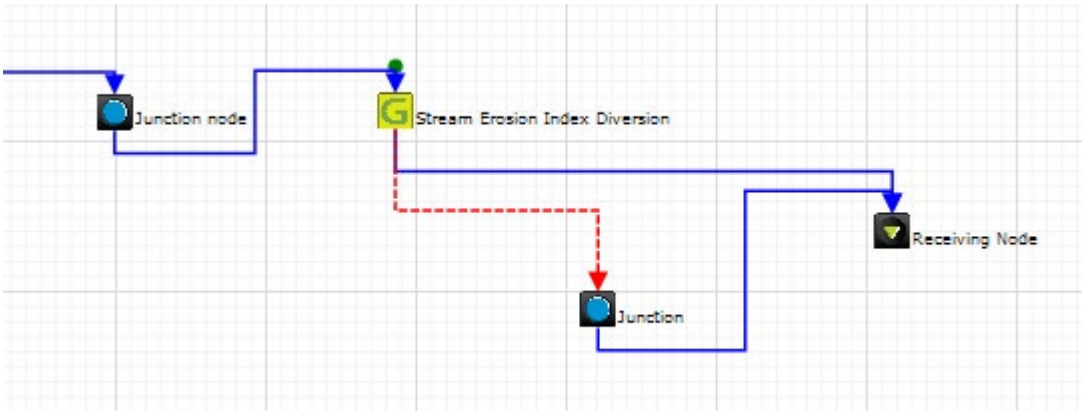


Figure 28. Example of MUSIC with flow bypass

13.6.2 Method 2. Modification of the generic node flow transfer function

The data required for estimating SEI can be directly extracted from MUSIC by interrogating a Generic node that is added to the treatment train immediately upstream of the receiving node. The Generic node in MUSIC provides a flow transfer function which can be simply defined to easily calculate the annual volume of flow above the critical flow. The Generic node should be set up to convert all inflows at, or below, the critical flow to 0 outflows. Flows above the critical flow will be passed through the node at the magnitude by which flow exceeds the critical flow, as described below:

- $Q_{out} = 0$ if $Q_{in} < Q_{critical}$
- $Q_{out} = Qin - Q_{critical}$ if $Q_{out} > Q_{critical}$

The generic flow transfer function is to be defined by 3 points:

- 0 (input): 0 (output)
- $Q_{critical}$ (input): 0 (output)
- Q (input): Q - $Q_{critical}$ (output)

The maximum inflow Q plotted should be 5 to 10 x $Q_{critical}$ to provide a good scale.

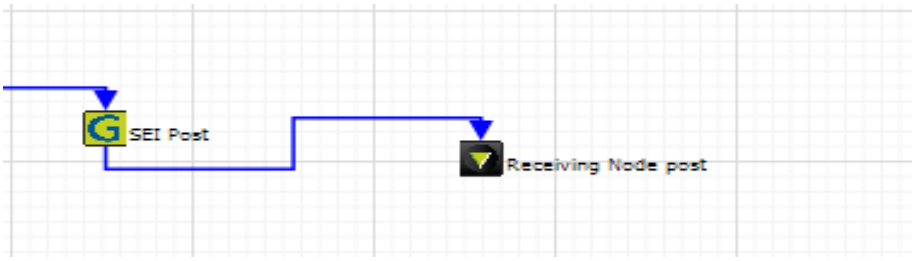


Figure 29. Example of MUSIC SEI setup with modified generic node

Check the flow transfer generic node at the downstream end of the MUSIC models for pre and post-development conditions by:

1. left click the Generic node
2. click on ‘Mean Annual Loads’
3. copy the flow output value in ML/yr.

The SEI is calculated as the output mean annual flow at the Generic node for the post-developed model divided by the corresponding value for the pre-development model. The SEI has to be less than 3.5 with a stretch target of 1.

SEI = Post developed output flow $_{Generic}$ / Pre-developed output flow $_{Generic}$

Because of the modified flow transfer function in Method 2, you will not obtain a correct pollutant removal rate for the development if you click on the Receiving node.

14 WSUD signage

Under Blacktown City Council's *Development Control Plan* Part J, there is a requirement to submit a WSUD sign design for approval and evidence it has been installed.

If you have 1 or more WSUD devices or systems in your development, a condition of consent will be placed in your development determination document that indicates a WSUD sign is required. This signage applies to all scales of development.

Prior to 30 June 2023, the condition required the sign to be approved prior to release of the construction certificate. From 1 July 2023, this must occur prior to the release of the occupation/subdivision certificate. To meet this condition, the applicant must have received the following approvals from the WSUD Compliance team wsud@blacktown.nsw.gov.au:

1. approval of the WSUD sign design using our template prior to printing and fabrication
2. confirmation of received evidence the sign has been appropriately installed.

To assist in timely approval of the WSUD sign, we have created templates and a guide on to how to fill it out. These details are on Council's website Water sensitive urban design (WSUD) signage - Blacktown City

Larger sites with multiple WSUD devices or systems may require multiple signs. A sign is not required for development sites requiring temporary on-site water quality treatment.

The safety regulatory signage for these signs is provided in Blacktown City Council's WSUD standard drawing 6101, sheet 2.

The WSUD sign will assist Blacktown City Council's WSUD Compliance Officer with identifying the elements of the WSUD system in relation to the site layout. The sign will also help the occupants and visitors to the site understand the components and function of these devices that protect our waterways. Signage is useful for maintenance purposes and in the event of a chemical spill or a flood event.

15 Flood modelling of local catchments

15.1 Flood impact and risk assessment

A Flood impact and risk assessment (FIRA) must be prepared for sites subject to riverine, local runoff, overland flooding, or sites situated in a defined low point with the potential for overland flows. The FIRA must:

- outline all the data, processes used and assumptions made in developing the flood model based on the specific requirements of this handbook
- demonstrate that neighbouring properties are not adversely impacted by the new development for flood events up to, and including the 1% AEP flood event. Maximum allowable isolated rise is 0.02 metres
- demonstrate that the hazard within the development site (outside of the Hawkesbury-Nepean backwater) is within H1 or H2 as identified in Figure 6.7.9 Combined Flood Hazard Curves from *Australian Rainfall and Runoff 2019 (ARR*). In some cases, see also Section 15.7.2 for further limitations for H1 and H2 hazards. If the hazard originally exceeded H1 or H2, the proposed development must ensure the hazard is not further exacerbated, even if the depth or velocity is increased through the site
- identify areas of the site with hazard H3 as identified in Figure 6.7.9 Combined Flood Hazard Curves from *Australian Rainfall and Runoff 2019* (ARR). Areas of H3 within the site may be considered on their merit, provided they are not part of the evacuation route
- identify areas of the site with hazard H4 or greater from Figure 6.7.9 Combined Flood Hazard Curves from *Australian Rainfall and Runoff 2019* (ARR) and how these areas are to be isolated/restricted from the development
- set appropriate floor levels to protect the proposed development from flood impacts, as detailed in Section 15.3 below
- reference all ground and flood levels to Australian Height Datum (AHD)
- assess non-critical development sites and buildings isolated by the 1% AEP flood for flood/flow depth and hazard in a Probable Maximum Flood (PMF). Such assessment is to be used only for assisting in developing an external flood evacuation or shelter-in-place management plan in accordance with NSW Government: Shelter-in-place guideline for flash flooding
- applications may be required to indicate that permanent, fail-safe and low-maintenance measures are incorporated in the development. This is to ensure the timely, orderly and safe evacuation of people from the area should a flood occur. In addition, it may also be necessary to demonstrate that the displacement of these people during times of flood will not significantly add to the overall community cost and community disruption caused by the flood
- assess essential services, sensitive or critical developments and sites for flood/flow depth and hazard in a PMF
- be signed off by a suitably qualified practicing civil or hydraulic engineer with CPEng accreditation.

Note:

- the digital full package (electronic) flood models are to be provided to Blacktown City Council for assessment with the flood report
- failure to supply the required information above may result in refusal of the application.

15.2 Categories of flood affectation

Each category of flood has a different approach, such as:

- riverine flooding (medium to high risk) with open channels and no pipes - this includes some areas identified as SEPP flooding
- riverine flooding (medium to high risk) with piped trunk drainage
- local overland flooding-major drainage - this includes some areas identified as SEPP flooding
- local runoff (minor flows)
- flood planning levels above the 1% AEP flood
- currently unmapped overland flooding. The process of Blacktown City Council identifying properties that may be impacted by surface flows is ongoing and not all such properties are notated. Individual sites need to be assessed by the applicant as due diligence. The applicant needs to review site topography that directly concentrates flows to the site including trapped drainage low points.

15.3 Design standards

15.3.1 Design flood/flood planning level

The 5% AEP flood event is the design flood standard for non-habitable rural structures less than 100 square metres and residential carports (with some conditions).

The 1% AEP flood planning level is the design flood standard for most general development.

The flood planning level is also the design flood standard for the following developments subject to consideration of the PMF for elements of the design:

- critical developments that provide support or essential services to the community such as hospitals, telecommunication towers, large power supply stations, police, ambulance and fire stations. Protection of critical elements to the PMF may be required
- sensitive developments that require a higher standard of flood protection due to the age of and potential risk to the occupants, such as nursing homes, aged hostels, preschools, primary schools or child care centres. In most of these, sufficient area above the PMF may need to be demonstrated for all occupants to shelter-in-place. Emergency back-up generators (if provided) and flood related infrastructure are to be installed above the PMF
- provision of a second storey or attic, or raising the floor level for non-critical developments where the PMF is more than 0.5 metres above the habitable floor level (but less than the record floor level). Also, where there is no continually rising evacuation route (escape path) from the door and the recommendation is to shelter-in-place in accordance with NSW Government: Shelter-in-place guideline for flash flooding
- where shelter-in-place is specified as a flood management strategy then a structural engineer, registered on NER, is to certify that the structure is safe to the PMF level.

In some sites subject to overland flows, a true PMF level may not have been determined through the available modelling. For small scale developments, consideration may be given to a simplified approach using an extreme flood considered as 3.5 times a 1% AEP flood flow.

15.3.2 Minimum floor levels (residential-habitable, business/industrial)

- general (local drainage, no flood affectation) -- minimum 0.225 metres above finished ground level
- local runoff -- 0.3 metres above the 1% AEP flood level
- local overland flooding-major drainage -- 0.3 metres above the 1% AEP flood level
- riverine flooding (business/industrial) -- 0.3 metres above the 1% AEP flood level
- riverine flooding (residential – outside growth centres) -- 0.5 metres above the 1% AEP flood level
- riverine flooding (residential – growth centres) -- 0.5 metres above the 1% AEP flood level with climate change consideration
- riverine flooding (residential behind levees) -- 0.5 metres above the 1% AEP flood level
- flood planning level -- 0.5 metres above the 1% AEP flood level
- flood refuge where required (shelter-in-place) -- the PMF level (no freeboard).

15.3.3 Minimum carport/carpark level

- above-ground detention storage of 0.2 metres (with no flooding)
- subject to V x D < 0.3 with velocity <= 2 m/s and depth <= 0.2 m below the 1% AEP flood level.

15.3.4 Minimum driveway level

- above-ground detention storage of 0.2 meters (with no flooding)
- subject to V x D < 0.3 with velocity <= 2 m/s and depth <= 0.3 m below the 1% AEP flood level.

15.3.5 Minimum garage floor level

- garages - 0.1 metres above the 1% AEP flood level.

15.3.6 Minimum basement carpark/below-ground garage crest level

- general (no flood affection) -- 0.3 metres above the higher street gutter invert level
- trunk drainage gutter flows -- 0.3 metres above the 1% AEP street gutter flow level
- local runoff -- 0.3 metres above the 1% AEP flood level
- local overland flooding-major drainage -- 0.3 metres above the 1% AEP flood level
- riverine flooding -- 0.5 metres above the 1% AEP flood level
- flood planning levels -- 0.5 metres above the 1% AEP riverine flood level.

15.3.7 Farm sheds (farm equipment and material storage only)

- farm sheds less than 150 square metres -- 0.1 metres above the 5% AEP flood level
- farm sheds greater than 150 square metres, but less than 300 square metres -- 0.1 metres above the 2% AEP flood level
- farm sheds greater than 300 square metres -- 0.1 metres above the 1% AEP flood level.

15.3.8 Pools

- there is no minimum level requirement for the pools themselves, however mud and rubbish may be deposited into the pool from a flooding event
- where practical, the power points for pumps and lights are to be set at the 1% AEP flood level plus freeboard
- an automatic safety cut-off switch in the meter box is required for the connection of all power to the pool area.

Note:

- above-ground pools are prohibited within the 1% AEP Hawkesbury-Nepean backwater flood extents
- an application for an in-ground pool alone (no other development) where the coping around the pool is a maximum of 150 millimetres above existing ground level do not require flood modelling.

15.3.9 Additions to existing buildings in flood areas

- minor additions

  All additions are considered on merit. Blacktown City Council will consider matching existing floor levels (even where these are below the flood level) for additions of a small single room or enlarging of an existing room. For example, creating a larger kitchen where otherwise it would create a step in the room. Otherwise, the addition should meet the flood planning controls

- larger additions

  The finished floor level of such additions should be to the flood planning level, but Blacktown City Council will consider each application on its merits having regard for the proposed use of the additions, the existing levels and flood behaviour

- additions on land below 17.3 metres AHD

  Section 10.4.3.1 of Blacktown City Council *Development Control Plan (DCP) 2015 Part A*, requires that where additions are proposed to an existing building which is located below the Hawkesbury-Nepean backwater flood level, Blacktown City Council will generally permit additions to a maximum of 10% of the existing floor area. The finished floor level of such additions should be 500 millimetres above the designated flood level, but Blacktown City Council will consider each application on its merits, having regard for the proposed use of the additions, the existing levels, and flood behaviour.

15.3.10 Building on land located 2.5 m or more below the 1% AEP flood level

Section 10.4.3.2 of Blacktown City Council *Development Control Plan (DCP) 2015 Part A,* the finished floor level requires that the maximum height between natural ground level (that is, ground level prior to any filling of the land) is 3 metres. Based on a 0.5 metres freeboard, this means that habitable development is prohibited on any land located 2.5 metres below the 1% AEP flood level. In the Hawkesbury-Nepean backwater flooding area, no habitable development is permitted on an area of land below 14.8 metres AHD.

15.3.11 Subdivision of land within riverine flood risk precinct

Section 10.4.3 of *Blacktown City Council Development Control Plan (DCP) 2015 Part A*, requires that where an application is made for the subdivision of land creating additional lots, Blacktown City Council will only grant consent if it is satisfied that future development on that land could be undertaken in line with the provisions of the DCP in relation to development on flood prone land. Where subdivision is approved in industrial and business zones, the land must be filled to 300 millimetres above the designated flood level. For subdivision in residential zones, Blacktown City Council will require land to be filled to 500 millimetres above the designated flood level. The street gutter invert is to be above the 1% AEP flood level at all points.

15.3.12 Subdivision of land within overland flow flood risk precinct

A subdivision of existing land, subject to overland flows or local runoff, will only be permitted after development of a flood model using a realistic footprint of future dwellings and including allowance for filling for future driveways and ramps to garages. The building and driveway footprints are considered as a full blockage in the model. A covenant will be required to ensure future building works conform to the approved modelling. Safe evacuation is to be provided with a hazard level H2 or less in a 1% AEP storm.

15.3.13 Building on C4 land

As a minimum, a pad area of sufficient size for a large house is to be provided within the C4 land filled to 0.5 metres above the 1% AEP flood level with considerations of climate change for each C4 lot. Where other larger development is proposed, the pad size needs to increase to reflect this.

15.4 Information available from Blacktown City Council

15.4.1 Maps online

To identify any known flood affectation for a site undertake the following steps:

Step 1: type 'Blacktown City Council Maps Online' into your search engine

Step 2: review the disclaimer and if acceptable click on 'I agree'

Step 3: type in the address of the property in the top left-hand corner and click on the appropriate location from the drop-down menu

Step 4: under 'Layers', click on 'Flooding Precincts'. The colours detailed below represent:

- dark blue – riverine high risk flood precinct
- medium blue – riverine medium risk flood precinct
- light blue – riverine low risk flood precinct
- dark orange – overland flow high risk flood precinct
- medium orange – overland flow medium risk flood precinct
- light orange – overland flow low risk flood precinct
- Light blue hatching – SEPP State Government flood maps – 1% AEP
- White – no known flood issue

You should always verify our flood precincts with independent technical advice because not all overland flooding investigations have been completed in our city.

15.4.2 Flood related information

To purchase any of the following items, contact floodadvice@blacktown.nsw.gov.au

Blacktown City Council will provide an initial response and indicate any additional details that may be required and how payment can be arranged.

Current prices are outlined in Blacktown City Council Goods and Services Pricing Schedule.

15.4.3 Flood advice letter

A Flood advice letter provides preliminary and indicative flood-related information to support early planning and feasibility investigations for a site.

A Flood advice letter may include, where available:

- flood mapping and adopted flood levels
- requirements for finished floor levels
- scaled, localised contour mapping
- indicative locations of Council's stormwater drainage infrastructure on and around the site
- identification of known flood constraints
- general advice on flood risk and potential development considerations.

This advice is intended to assist owners and developers in understanding flood constraints that may affect a future development but does **not** constitute certification of a potential development.

15.4.4 Flood control lot certificate

We also offer a Flood control lot certificate, which is a formal certification used where development is proposed under private certification.

A flood control lot certificate only applies to land that is identified as a Flood control lot under Council's adopted flood planning framework. If the land is not a Flood control lot, a Flood control lot certificate cannot be issued.

We only issue these at our discretion, based on the flood affectation of the lot and the proposed development. Where applicable, the certificate confirms that the lot, or the part of the lot subject to development, is not any of the following:

- a flood storage area
- a floodway area
- a defined flow path
- a high hazard area
- a high risk area

This certificate provides formal assurance for compliance and is typically required to support development applications and construction certificates issued by private certifiers.

15.4.5 Aerial laser survey

Aerial laser survey (ALS), sometimes referred to as LiDAR, provides thousands of ground level points for an area that enables the derivation of cross-sections or digital terrain data for input into a variety of flood models.

Blacktown City Council can supply tiles in an American Standard Code for Information Interchanger (ASCII) format that covers large areas of our city for a fee.

15.4.6 Blacktown City Council flood model

Blacktown City Council has a range of existing hydrology and hydraulic digital flood models available in certain areas in a variety of formats including RAFTS, DRAINS, HEC-RAS, TUFLOW and others depending upon location.

Before using Council's model for your site-specific flood modelling, please ensure that the suitable parameters applied align with the modelling guideline and technique in section 15.5 and 15.6.

15.4.7 Blacktown City Council flood model - results only

We can supply flood modelling results from existing digital Blacktown City Council flood models.

15.4.8 Blacktown City Council flow information

We may be able to supply a peak flow or hydrograph. This information may be from an existing digital Blacktown City Council flood model, or from a flood study report.

15.4.9 Contour plans and general pipe layout

We can supply contour plans and general pipe layouts in the catchment that drains to the proposed development site.

Blacktown City Council can identify certain pipe sizes from our geographic information system (GIS). However, they are indicative only and have to be verified on site by a registered surveyor at your expense. No responsibility will be taken for the accuracy of the data provided. There are no pipe inverts or other pipe information available.

Contour with 0.5 metres contour data can be used to assess flow directions and potential impacts from upstream flows not previously identified by Blacktown City Council. For example, drainage pits with trapped low points.

15.5 General design and flood modelling guidelines

15.5.1 Site survey

A survey plan signed (certified) by a registered surveyor is to accompany the development application and flood report. It must:

- include details of the levels over the whole property including front footpath, kerb and road levels, plus drainage pit information if present
- include all building improvements including existing floor levels
- show the origin and level of the benchmark used
- provide a local benchmark on top of the kerb in front of the site to AHD
- extend the survey upstream and downstream of the site as well as to the side to the full extent of the flow path - determine any control structures for input into the flood model and ensure model stability or ALS data is required outside the site as detailed in Section 15.4.4.

15.5.2 Manning's n roughness value

Consider the following generic Manning's n roughness values for modelling:

- hard paving impervious including road and driveways n = 0.025
- pervious landscaped areas n = 0.05.

Note:

- these values include an additional allowance for temporary or non-structural obstructions to the floodplain including cars, bins, barbeques and raised garden beds
- where advised by Blacktown City Council, an alternate Manning's n may be determined by the consultant when developing a flood model to match a specific flood level.

15.5.3 Buildings, structures and extensions as obstructions

All building works must be modelled as fully blocked in the floodplain even where the development is proposed on piers. This includes any filling for driveways to garages. No allowance will be made for any active flow under a building and driveway.

15.5.4 Fences as obstructions

For the palisade and louver fencing percentage, obstructions for each fence in the floodplain are noted below. The openings are to be at least to the 1% AEP flow depth with freeboard. These percent obstructions apply to the general floodplain and not to specific structures in the floodway such as culverts:

- pallisade (pool fence type) fencing – 15%
- horizontal louvers (single vane) – 25%
- horizontal louvers (double vane or arrow head) – 30%
- colorbond and timber paling fences – 50% (note that these fence types will typically collapse during a major flood event)
- brick or blockwork fences – 100%.

Note:

- consider a merits-based approach to assess the worst impact. For example, apply the blockage to the fence downstream of the development, not upstream
- where there is an assessment undertaken based on the procedure in *ARR 2019* use the higher blockage factor from *ARR 2019*, or that nominated above.

15.5.5 Pit blockages

Where a pipe hydraulic model is used in the 1% AEP flood event allow for blockages of:

- on-grade pit with grate and lintel - 20%
- on-grade pit with grate only - 35%
- sag pit with grate and lintel - 50%
- sag pit with grate only - 75%.

The pit blockages do not apply where the pit charts have already been adjusted for the blockages.

15.5.6 Pipe blockages

The pipe blockages nominated below are only to be applied at an individual pipe or culvert that will result in the maximum overflow to the site under investigation. Typically, this is the pipe travelling through or immediately upstream of the site. It is not to be applied to the whole pipe network in the catchment.

For pipes or culverts under roadways, or elsewhere that has open channel sections upstream where large debris can enter, the blockage factor to apply is:

- culverts < = 300 mm high; pipe < = 375 mm 100% blocked
- culverts > 300 mm high; pipe > 375 mm 50% blocked.

Note: The blocked percentage above allows for the possible future installation of angled bars to prevent people being washed in.

For pipes or culverts under roadways that are only filled through grated pits and lintels (that is, do not have an open channel section upstream where large debris can enter), the blockage factor to apply is:

- pipe < = 375 mm 100% blocked
- 375 mm < pipe < = 825 mm 50% blocked
- 825 mm < pipe < = 1,200 mm 33% blocked
- 1,200 mm < pipe < = 1,650 mm 25% blocked
- pipe > 1,650 mm 10% blocked.

Note:

- the above reductions also apply to culverts of a similar area
- for some digital (electronic) models such as DRAINS it is not possible to directly enter a percentage pipe blockage. The pipe sizes will need to be manually adjusted in the model.

15.5.7 Floodplain storage and filling in the floodplain

Blacktown City Council generally requires that there be no net loss of floodplain storage below the 1% AEP flood level for both Riverine and overland flow locations. A flood storage plan with sufficient level/contours is to be provided in addition to the normal fill plan. This should clearly show that the flood storage below the 1% AEP flood level has been maintained within the site. See exemptions to providing floodplain storage in below section.

Excavation of material from one part of a site below the 1% AEP flood level and placement elsewhere within the same site below the 1% AEP flood level to ensure no net loss of floodplain storage is allowed. However, such arrangements will require a flood study to verify that there is no adverse hydraulic impact. Similar arrangements could also be undertaken with Blacktown City Council approval where such material is excavated from a nearby lot and filled within another lot. The sites must be close enough to ensure they are both part of the same floodplain.

The limitation on any excavation is that it must be self-draining to ensure the area excavated for flood storage is available for the next flood event. It cannot be a dam. Any filling required above the 1% AEP flood level can be imported from off-site.

Where structures are built on a lot below the 1% AEP flood or flow level (this includes part or all of the site) then:

- design the structures to have a floor with a void under (suspended slab or bearers and joists) to maintain floodplain storage

- apply the area of void only to the extent of inundation
- minimise the extent of underfloor inundation by over-excavating under 1 part of a structure to allow fill under other parts of the structure such as garage or elsewhere, providing the volume below the 1% AEP level is maintained and the lowered area is self-draining. Self-draining can mean an accessible collection pit with an oversize pipe connection to a significant water source such as a Blacktown City Council drainage system
- a minimum of 25% of the outside wall area below the flood, or critical flow level along the line of inundation is to be open down to the ground level
- secure the underfloor openings with vertical metal bars, horizontal metal louvers, or coarse metal mesh with minimum 50 millimetre square openings
- in the hydraulic modelling, these openings are considered as fully blocked (for example, there is no active flow under the building)
- nominate finished surface levels within the underfloor area to show that it is self-draining, at 1% minimum
- provide details and levels with the submission to Blacktown City Council.

Exemptions to providing floodplain storage include the following:

- where 2D hydraulic modelling results in no adverse flood impact after modelling the area under the structure and any raised driveway as a full obstruction. 2D models include TUFLOW and HEC-RAS 2D. Buildings have the option to be constructed on fill with drop edge beams or piers as desired. This only applies to areas outside the 1% AEP Hawkesbury-Nepean floodplain
- under driveways at entry to garages for single dwellings, where the:
    - maximum driveway height above existing ground level is 500 millimetres
    - vertical retaining walls or drop edge beams are used at the sides
    - driveway grade is as steep as practical (10% minimum) to minimise fill
- developments where the average 1% AEP floodplain storage depth under a structure over the area of inundation is less than 0.2 metres.

15.5.8 Fencing in the floodplain – urban zones

Set the open area of the flow-through fencing or gates across the flow path to a minimum of:

- 0.3 metres above the 1% AEP flow level for overland flows
- 0.5 metres above the 1% AEP flow level for riverine flooding.

Within urban areas provide a 0.05 to 0.1 metres gap at the base of the flow-through fence.

All boundary fencing, other than along the front boundary and front building setback, must comprise horizontal metal (typically Colorbond or aluminium) louvers with a maximum gap of 100 millimetres as they provide privacy.

Horizontal metal louvers can be either:

- single vane louvers, or
- double vane louvers (arrow head type) - double vane louvers are preferred as they provide a privacy component in both directions.

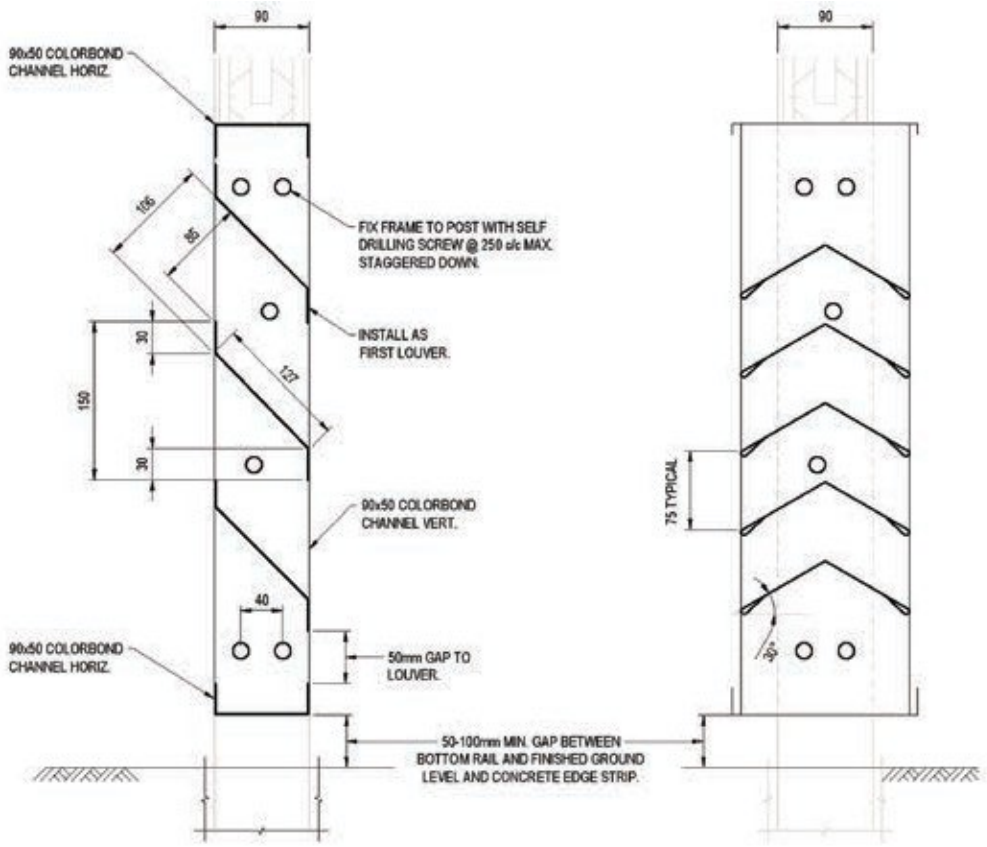


Figure 30. Single vane louvers and double vane louvers

Along the front boundary, within the front building setback, or anywhere else within the site above the base gap of the fence or gate, the flow-through fence openings may be secured with:

- vertical metal bars with approximate 100 millimetres openings (pool fence or palisade style), or
- horizontal metal louvers as described above.

Within residential areas, for fencing parallel to the flow direction, provide a 0.05 to 0.1 metres gap at the base of the fence to allow flood flows to equalise both sides of the fence.

15.5.9 Fencing in the floodplain – E2, E3 and E4 zones

Fencing across the flow path can use either of the fence types described above. However, no special restrictions apply to boundary fencing.

Traditional rural fencing such as post and rail, or posts (or star pickets) and horizontal wires are also accepted.

15.6 Flood modelling techniques

15.6.1 Hydrology

The flood study hydrology will be based on the local catchment draining to the site. For smaller catchments less than 2 hectares, the Rational Method can be used. For larger catchments, an agglomerated or simplified DRAINS or RAFTS model can be easily assessed by Blacktown City Council. Maximum travel time is 14 minutes. In calculating the design surface flows, deduct any allowable pipe flow (if applicable) after considering pipe blockage as per Section 15.5.6.

For larger studies along the major waterways, Blacktown City Council can provide more detailed hydrology as per Section 15.4.6.

15.6.2 Pre to post modelling and tolerances

Flood levels: A maximum isolated rise of 0.02 metres is permitted in a flood section or cell external to the site. Multiple rises to this level will not be permitted.

Velocity: Generally, a 5% increase in velocity is permitted in a flood section or cell external to the site. 10% may be considered in certain circumstances.

Hazard (V * D): No increase in hazard is permitted in a flood section or cell external to the site. An internal increase in hazard within the site is permitted subject to the use being appropriate to the H1, H2 or H3 hazard based on Figure 6.7.9 Combined Flood Hazard Curves from *ARR 2019.*

The maximum 0.02 metres isolated rise applies to the proposed development and all future additions to the same development. When assessing subsequent development, the pre development is to consider the original extent of development not the latest extent to avoid flood creep.

15.6.3 HEC-RAS 1D

This model is freely available and relatively easy to use. However, Council generally requires 2D Hec-Ras model in most of the cases. It generally uses a steady state peak flow, although unsteady state is also available. It is best suited for linear flows with consistent cross-sections. As the model fills the lowest part of the cross-section first, this causes distortions where these low points vary significantly across the sections particularly from one side of a building to another (which is not physically possible). This sometimes can be represented in 1D by using ‘levees’ in the model or multiple HEC-RAS models to represent the split flows. However, this is very difficult to manage effectively and for Blacktown City Council to assess.

The HEC-RAS 1D section locations are to be clearly shown on the plans, maintaining where possible the same location for the pre and post model sections. The zero position for each section is to be positioned along a recoverable line such as a side boundary. Show all buildings as fully blocked obstructions, including those on neighboring properties, and provide ineffective flow areas for sections close to the obstructions at 45% to the flow.

The sections should always extend far enough upstream and downstream of the development to stabilise the model.

15.6.4 TUFLOW and HEC-RAS 2D

These models are ideal for the modelling of urban catchments where complex interactions of building structures may result in water moving in unexpected directions. These models either rely on boundary conditions with upstream flows represented as hydrographs, or rain on the grid. The pipe flow can be represented with a 1D link, but for most smaller systems the pipe flow is generally ignored. Flow boundaries are to be digitised approximately perpendicular to the flow direction.

Note:

- currently Blacktown City Council cannot use TUFLOW models using HPC, and models presented to Blacktown City Council should be in classic format
- Blacktown City Council does not accept XP-Storm models
- it has been determined that severe water volume imbalances can occur where rain on the grid is used and this will significantly underestimate flows. Blacktown City Council consequently requires the use of flow boundaries throughout the model. Rain on the grid should not be used, or at most used for the local area immediately surrounding the development.

Grid size: The grid or cell size needs to be sufficiently small to model the hydraulics reflecting the site conditions. For most small urban models representing the restricted flow between buildings, the grid size should be a maximum of 0.5 metres. For sites with very narrow gaps, an even smaller grid size should be used.

To model open channels in 2D, each channel needs: at least 3 to 4 cells across where the channel flow is parallel to the grid, but 5 or 6 where the channel is angled to the grid. Where this cannot be achieved, the channel is to be modelled as a 1D component within the 2D model.

Similarly, where pipes or narrow open channels are modelled these are to be modelled as a 1D element within the 2D model.

All buildings are to be modelled as full obstructions, and it is unacceptable to simply increase the Manning's n roughness value.

The model report is to include a volume validation. Continuity of volume is to be maintained and details provided in the model report log.

Provide a differences map for flood levels including 0 to 0.02 metres.

Provide a provisional hazard and difference map for H1, H2, H3 and H4 or greater based on Figure 6.7.9 Combined Flood Hazard Curves from *ARR 2019.*

15.7 Flood emergency evacuation and management plan

15.7.1 General principles

Blacktown City Council has 3 principles for management of the residual impacts of flooding (that is, the remaining flooding impacts once floor levels are raised to the flood planning level). These are, in order of importance:

1. protection of people from injury
2. protection of building structures from damage
3. protection of building contents.

15.7.2 Protection of people from injury

Blacktown City Council's principal aim is to protect people from injury especially the very young and very old.

Blacktown City Council's general standards for hazard for areas outside the Hawkesbury-Nepean backwater flooding area are based on Figure 6.7.9 Combined Flood Hazard Curves from *ARR 2019.* These include:

- **vehicle parking (H1)** - V * D <= 0.3 where V <= 2 m/s and D <= 0.2 m
- **driveways (H1)** - V * D <= 0.3 where V <= 2 m/s and D <= 0.3 m
- **pedestrian access (H2)** - V * D <= 0.6 where V <= 2 m/s and D <= 0.5 m

Vulnerable people in the community require a higher standard than that noted above. Blacktown City Council will not support new childcare centres, kindergartens, primary schools, or aged care facilities (for example nursing homes) within any:

- **high hazard riverine flood areas** defined as dark blue on Blacktown City Council's flood maps
- **medium hazard riverine flood areas** where the depth (D) > 0.3 m and/or velocity (V) > 2 m/s and/or V * D > 0.3
- **local runoff/overland flow areas** where the depth (D) > 0.3 m and/or velocity (V) > 2 m/s and/or V * D > 0.3
- **hazard rating H2 or greater** based on Figure 6.7.9 Combined Flood Hazard Curves from *ARR 2019.*

Even where the development satisfies the criteria above, it still needs to satisfy the requirement for a continually rising evacuation route.

Other developments will be permitted within the hazard areas identified above. However, Blacktown City Council generally requires a continually rising evacuation route up to the PMF level. This means people can exit from the building by walking or driving, and as they exit, the depth of floodwaters becomes shallower until they can reach safety. In some situations, this will follow the natural slope of the land where flood waters are deeper at the rear and shallower at the front. In other situations where the land falls to the rear, engineered solutions such as elevated walkways set at or above the 1% AEP flood level may be possible. If this approach is taken, it must not cause an adverse impact on flooding nor cause adverse planning issues such as overlooking of neighbours.

Where a continually rising evacuation route is not possible from the 1% AEP flood affected site, Blacktown City Council will generally:

- refuse any new residential development or subdivision
- refuse more intensified business or industrial redevelopment (including schools, places of public worship or recreational clubs)
- not refuse the above developments where a flood refuge can be provided above the PMF, that is, shelter-in-place. The shelter- in -place must be designed in comply with NSW Government: Shelter-in-place guideline for flash flooding. This will not be possible within most areas of backwater flooding in the Hawkesbury-Nepean floodplain where the PMF level is considerably higher than the 1% AEP flood level. However, in other areas, the provision of shelter-in-place by raising of floor levels to heights significantly above neighbouring properties, may be refused on planning or privacy concerns.

Where Blacktown City Council has previously approved a single dwelling on a site and there is no continually rising evacuation route, this will not prevent approval of a replacement single dwelling at the site. This is subject to the requirements of Blacktown City Council *Development Control Plan 2015 Part A* being satisfied. Appropriate conditions of consent will apply.

15.7.3 Protection of building structures from damage

Prior to the issue of a construction certificate, a structural design certificate from a chartered structural engineer on the NER must be provided. The certificate must state that the structure has been designed to withstand all loads and forces acting upon it up to the design level including:

- scour
- impact of debris
- hydrodynamic pressure
- hydrostatic forces
- buoyancy forces.

The design level is generally the flood planning level. However, where shelter-in-place is proposed, the structural design certificate is to consider a flood at the PMF (or extreme flood) plus freeboard in addition to the standard flood level.

To ensure the building remains as serviceable as possible during or after a flood, the following items should be set at or above the flood planning level:

- meter box
- hot water heater
- air conditioning units
- solenoid switching device for rainwater tanks
- all electrical power points
- rainwater tanks (except where underground)
- large gas cylinders (where this is impractical, secure them to overcome buoyancy).

All building materials below the flood planning level, must have flood compatible building components that are either suitable or mild-effects as per table 4.3.1.3 of *Reducing Vulnerability of Buildings to Flood Damage, Hawkesbury- Nepean Floodplain Management Steering Committee, June 2006*.

15.7.4 Protection of building contents

The protection of moveable contents inside the buildings are addressed through the Flood emergency plan. Where time permits, occupants are encouraged to raise items that may be damaged by floodwaters to higher levels before evacuating and to remove important family items, such as legal papers and medicines, when evacuating.

Insurance for building contents that includes flood cover is also an option.

15.7.5 Flood risk management plan

A Flood risk management plan must be prepared by an experienced chartered hydraulic engineer registered on NER, to address emergency flood management of the site. This should include, as appropriate, warning signs, notices of procedures and depth gauges (if required).

The plan should recommend whether to shelter-in-place or evacuate during the flood event. Blacktown City Council does not support shelter-in place in critical and sensitive developments.

The plan should incorporate maintenance-free measures into the development to ensure the timely, orderly and safe evacuation of people from the area if required.

The plan should detail the evacuation route.

Any requirements of the plan are to be implemented prior to occupation.

A colour laminated copy of the Flood management risk plan is to be permanently affixed within the building to the inside of a kitchen or laundry cupboard door, or to any owner's corporation community noticeboard.

15.7.6 Sample flood emergency plan

NSW State Emergency Service (SES) has online tools to assist in development of emergency plans for home or business. You can visit SES website for more information.

In critical and sensitive development, a comprehensive flood emergency plan needs to be prepared by an experienced hydraulic flood engineer to consider specific local affectations that may require higher standards of protection. Local considerations may include steep driveways that may require earlier vehicular evacuation due to slippery conditions, or evacuation routes that may be cut early. Where the structure is unsafe to remain during a PMF, then mandatory evacuation is required. The general considerations of the flood emergency plan are shown in below. A sample flood emergency plan from SES is provided in section 15.7.7.

Background

Blacktown City Council has advised that this property is in a flood prone area and is subject to flooding in a 1% AEP (sometimes referred to as the 1% AEP) storm event. The property is also subject to the Probable Maximum Flood (PMF). Relevant levels to Australian Height Datum (AHD) are:

Front boundary level = m AHD

Rear boundary level = m AHD

1% AEP flood level at front boundary = m AHD

1% AEP flood level at rear boundary = m AHD

Habitable floor level at front = m AHD

Habitable floor level at rear = m AHD

Habitable upper floor level = m AHD

PMF flood level at front/rear = m AHD

Where information for a PMF is not available, consider details for an extreme flood event set to a minimum of 3.5 * 1% AEP. Sample critical points are included here. However, the aim is to include any relevant level or location that will assist in informing the resident how risky the flood is and if they should evacuate. The recommendation to shelter-in-place should only be given in this Plan where the lower or upper floor is above the PMF and the structural engineer has certified that the building is stable in a PMF + freeboard.

Flood emergency plan

XXX Address XXX

The property is within the upper reaches of XXX Creek and flows are moving from YYY Street, through the property and down towards ZZZ Street.

This property is in an area of flash flooding and short intense storms lasting only a few minutes to a few hours can cause major flooding. There may be a risk if you stay in the building during a major flood. When the storm finishes, the flood levels will usually reduce quickly.

Before a flood occurs

1. Prepare an emergency kit containing the following items:
    - emergency phone numbers
    - legal papers including insurance details and passports
    - USB or similar containing photos, videos of family and friends
    - a waterproof bag for valuables and mementos.
2. Develop your own family flood plan and be prepared and know what each person has to do if flooding should occur while the kids are coming home from school, or when you are returning from work. Know what route you are going to use to evacuate.
3. Permanently store personal items and mementos (such as photos) and dangerous items (such as chemicals and poisons) as high as possible.
4. Ensure there are no items blocking the gaps under the building or fencing. This is to ensure floodwater can flow through the site.
5. For further information on being flood prepared visit www.ses.nsw.gov.au

During heavy rain and prolonged storms

6. Listen to local radio stations such as ABC 702 AM for weather updates and advice. Please be aware that flash flooding can occur quickly and there may not be any specific flood warning issued for your local area.
7. Secure objects that can float and cause damage, for example garbage bins.
8. Monitor any flow of water around the home and be prepared to evacuate.

If water starts to run down the driveway or adjacent to the house

9. Collect your previously prepared emergency kit and add:
    - mobile phones and chargers
    - medications
    - special requirements for babies or the elderly
    - raincoats/umbrella.
10. Put on your strong shoes or gumboots.
11. Turn off and disconnect any large electrical items such as a dishwasher or washing machine that cannot be raised to a higher level.
12. Continually monitor the depth of water flowing down the driveway or adjacent to the house.
13. Evacuate immediately if the Police or the SES personnel instruct you to.

By the time the water level in the driveway or adjacent to the house is 200 millimetres deep, you must decide whether to evacuate the site, or remain within the building.

If you choose to evacuate

14. Collect your **emergency kit** with the additional items noted at number 9 (above), then evacuate the site by vehicle and go to higher ground.

15. Take care when evacuating as many streets and roads will be cut-off by floodwaters. It is dangerous travelling through floodwaters on foot, or even in a vehicle, as: the water may be polluted, obstructions can be hidden under the floodwaters, or you or your vehicle could be swept away in fast flowing water.

If you choose to remain within the building

16. Do not attempt to evacuate once the floodwaters are 500 millimetres deep or more unless instructed to do so by the Police or by SES personnel.
17. In the case of a medical emergency, ring 000 as normal and explain about the flooding.
18. Raise any items within the home that may be damaged by water (for example, photo albums).
19. If floodwaters approach the floor level, place damp towels across the bottom and lower sides of external doors to slow down the entry of water through the doors.
20. If floodwaters enter the building, move to the highest location in the house.

Other Items

21. A coloured laminated copy of this **Flood emergency response plan** must be permanently attached (glued) on an inside cupboard door in the kitchen or laundry.
22. Have an electrician check any electrical items before use if affected by floodwaters.

Flood emergency response plan should be reviewed every 5 years after that.

15.7.7 Sample flood emergency plan – State Emergency Service

The SES has developed the “My Emergency Plan” tool, a personalised resource that provides clear, step-by- step instructions outlining exactly what you and your household should do to stay safe if a flood, storm or tsunami occurs.

To prepare your “My Emergency Plan”, please go to SES website and search “Prepare for a flood”. For further flood information, please go to “Flooding in Blacktown City” on our website.

A laminated colour copy of the My Emergency Plan should be permanently affixed to the inside of a kitchen or laundry cupboard door.

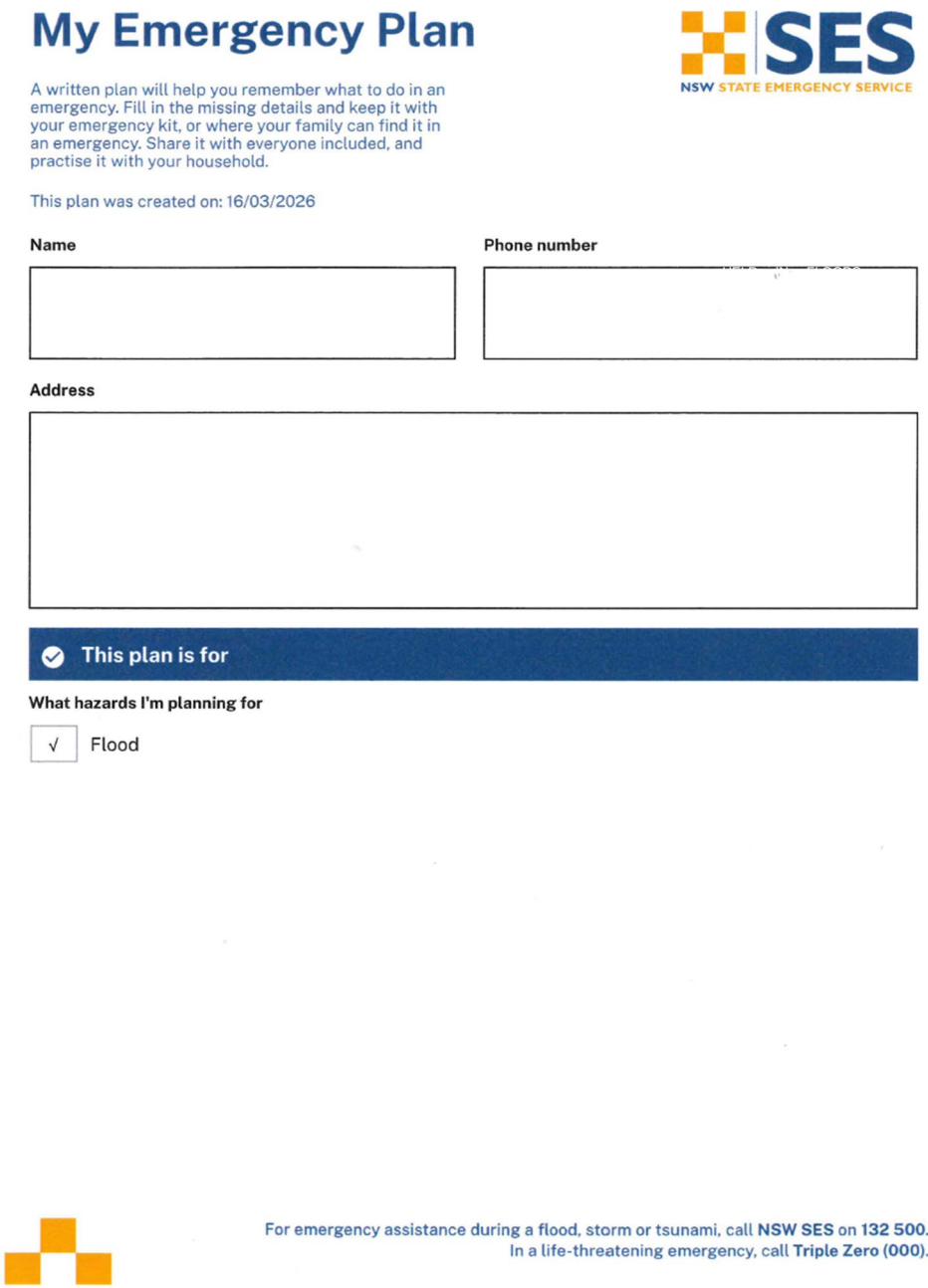


Figure 31. My Emergency Plan – State Emergency Service

16 WSUD checklists

WSUD checklists have been developed to outline all essential steps and requirements in the design, construction and certification of WSUD systems. It helps the engineer/designers to evaluate the WSUD design, ensuring it meets Blacktown City Council's requirements throughout each stage of the development.

WSUD checklists show detailed information of the design of WSUD systems that must be submitted with the development application to Blacktown City Council. The checklist must also be submitted throughout each stage of the development to Blacktown City Council or a certifying authority.

Provision of the information contained within these checklists and sign off by a chartered professional engineer (CPEng) (Civil/Environmental College) who has membership to Engineers Australia will enable the development application and subsequent stages to be signed off in a timely manner.

Note: Not all WSUD checklists will be required for your development application. WSUD checklists only need to be completed for those components of a WSUD system that is being proposed.

WSUD checklists that need to be completed after the development has been approved will be identified in the conditions of consent.

The following is a full list of the WSUD checklists:

WSUD checklist category	Development stage			
	Submit development application	Before the issue of a construction certificate	While building work is being carried out	Before the issue of an occupation /subdivision
Rainwater/stormwater tank	100	200		400
Wetland	101	201	301	401
Pond	102	202	302	402
Swale	103	203		403
Sediment basin	104	204	304	404
Bioretention system (standalone)	105	205	305	405
Bioretention system (Stage 1 sediment basin)	106	206		
Bioretention system (Stage 2 – temporary)			307	
Bioretention System (Stage 3 – final planting after catchment development)			308	
Bioretention and detention basin (combined)	109			
Cartridge Filter - StormFilter	120	220		420

	Development stage			
WSUD checklist category	Submit development application	Before the issue of a construction certificate	While building work is being carried out	Before the issue of a construction /subdivision
Cartridge Filter – AtlanFilter	121	221		421
Cartridge filter jellyfish	122	222		422
Cartridge Filter HumeCeptor	123	223		423
Atlan FlowGuard	124	224		424
HumeFilter	125	225		425
GPT	130	230		430
AtlanBasin	131	231		431
High flow media filtration - Filterra	132	232		432
Pit Inserts StormSack	140	240		440
Pit Insert OceanGuard	141	241		441
Pit Insert - Enviropod	142	242		442
On-site stormwater detention (Standalone)	150	250	350	450
Flood report	160	260		460
Groundwater report	161	261	361	461
Stormwater management report	162			

17 Glossary and acronyms

1% AEP flood - the flood that is likely to occur, or be exceeded, on average, once in every 100 years. It has a probability of 1% (or a 1 in 100 chance) of occurring in any given year.

AEP – Annual exceedance probability

Activity - any development on land and may include, but is not limited to, any 1 of the following:

- the erection of a building
- the carrying out of work in, on, over or under land
- the use of land or of a building or work
- the subdivision of land whether involving earthworks or not
- any soil disturbing activity in or on a public place or on lands owned by Blacktown City Council which may or may not be the subject of an approval
- any act, matter or thing for which provision may be made under Section 26 of the *Environmental Planning and Assessment Act 1979* and which is prescribed for the purposes of this definition but does not include any act, matter or thing referred to in Section 26 for which development consent is required or which is prohibited under an environmental planning instrument.

AHD – Australian height datum.

Amenity - the quality of being attractive, something that is intended to make a place more pleasant or comfortable.

Aquifer - a body of saturated rock or soil containing a system of interconnected voids from which significant (economic) quantities of groundwater may be extracted.

ARI – Annual recurrence interval.

AtlanBasin - an advanced stormwater treatment system designed to efficiently remove pollutants using natural processes. It comes as a prefabricated, compact, and self-contained unit, utilising horizontal flow bioretention technology and multistage pre-treatment.

AtlanFilter - a type of filter cartridge developed by Atlan Stormwater (formerly SPEL Stormwater) which is housed in an underground tank. It captures nutrients, fine sediments and oils from stormwater before it enters the drainage system.

Biofilter - an alternative name for a bioretention system or raingarden. It is a type of WSUD device used to remove nutrients and sediment through passive physical and biological treatment techniques.

Bioretention - a stormwater treatment technology whereby water is directed to a vegetated depression in the landscape for treatment. Water temporarily ponds on the system surface before seeping through a permeable growing media. Treatment occurs in this media, facilitated by the media itself, the vegetation growing within it and microbial life present. Treated water is typically collected at the base of the system and discharged to the stormwater network. Bioretention comes in many different shapes, sizes and visual configurations. Refer to ‘bioretention system’, ‘bioretention basin’, ‘bioretention street tree’. These systems are sometime referred to as ‘biofilters’, and ‘raingarden’.

Bioretention basin - a medium to large scale (that is, typically greater than 50 square metres) application of bioretention technology ideally located with public open space. Bioretention basins are typically vegetated with either native grasses and sedges and/or trees.

Bioretention raingarden - an alternative name for biofilter or raingarden and improves the quality of stormwater running off the site through settlement, filtration, sediment capture, and biological processes. Stormwater will pond on the surface after the storm for 1-4 hours before infiltrating through the filter material. Healthy dense surface vegetation improves stormwater treatment.

Bioretention system - a collective term for all types of bioretention (for example, 'bioretention basins', 'bioretention street tree', 'biofilters' and 'streetscape raingardens').

Bioretention street tree - a small bioretention system with filter media and underdrainage fitted into a tree pit. They are very similar to a bioretention raingarden in terms of design, operation, maintenance and monitoring requirements.

Building - any part of a building. It also includes any structure or part of a structure (including any temporary structure or part of a temporary structure). However, it does not include a moveable dwelling or associated structure or part of a manufactured home, moveable dwelling or associated structure.

Business development - generally defined in the *Blacktown Local Environment Plan 2015*. This includes not-for-profit and government organisations undertaking developments such as schools, aged or disabled care facilities, nursing homes, hospitals, affordable housing, childcare centres, meeting halls, places of public worship, entertainment and sporting facilities or similar.

Catchment - an area of land which drains all run-off water to the same lowest point.

CDS GPT - a continuous deflective separation (CDS) is a gross pollutant trap developed by Rocla. It is a below-ground system designed to catch oils, coarse sediment, litter, debris and other larger pollutants conveyed within the stormwater drainage network.

CFU – Colony Forming Units.

Civil Works Specification - the document dated February 2005, developed by Blacktown City Council and entitled Civil Works Specification as amended or superseded.

Confined space - an enclosed or partially enclosed space that:

- is not designed or intended primarily to be occupied by a person
- is, or is designed or intended to be, at normal atmospheric pressure while any person is in the space
- is or is likely to be a risk to health and safety from:
    - an atmosphere that does not have a safe oxygen level
    - contaminants, including airborne gases, vapours and dusts, that may cause injury from fire or explosion
    - harmful concentrations of any airborne contaminants engulfment.

Construction security - a bank guarantee provided by the applicant for the purpose of securing fulfillment of its obligations under the construction phase.

Contributions plan - a plan prepared, exhibited and adopted by Blacktown City Council in accordance with Section 7.11 of the *Environmental Planning and Assessment Act 1979*, authorising the conditioning of contributions.

Cooling towers - heat removal devices used to transfer and process waste heat to the atmosphere. This is done by evaporating water to remove process heat and cool the working fluid to near the wet-bulb air temperature. Alternatively, it can be done by relying solely on air to cool the working fluid to near the dry-bulb air temperature. Common applications include cooling the circulating water used in oil refineries, chemical plants, power plants and building cooling.

CSTR – Continuously stirred tank reactors.

CUPDR – Committee on the Uniformity of Plumbing and Drainage Regulations.

Deemed-to-comply certificate - a certificate of compliance produced by Blacktown City Council's deemed to comply tool.

Deemed-to-comply tool - the online tool developed by Blacktown City Council that can demonstrate compliance with the water conservation, water quality and/or on-site stormwater detention controls.

Detention - the temporary storage of stormwater generated within an allotment. This restricts the discharge from the site to a predetermined rate to reduce flooding both in the local drainage system immediately downstream of the site and along the creeks and watercourses further downstream.

Detention basin - an excavated area installed on, or adjacent to, waterways and their tributaries to protect against flooding (and in some cases, downstream erosion) by storing water for a limited period of time.

Detention time - the time it takes for water to flow from the inlet to the outlet.

Developers Toolkit for WSUD - a set of resources contained on the Blacktown City Council website that gives guidance to developers and the community on Council's requirements for stormwater management and water sensitive urban design.

Development footprint - the area of new work that is the subject of the development application. It refers only to new external work. It includes new paved areas and new buildings and new for old building replacements.

Device - a thing made for a particular purpose. It is often used to describe individual water quality and stormwater treatment systems.

Dewatering - the removal of water from solid material or soil by a solid-liquid separation process. This is often done during the site development phase of a major construction project due to a high-water table and may involve the use of pumps.

Drainage - the natural or artificial movement of surface and sub-surface water from a given area.

Dual reticulation - the supply of water from 2 different sources using separate pipes. For example, when an alternative water source such as sewage or stormwater is used for non-potable purposes such as irrigation and toilet flushing, or Sydney Water's supply is used for potable purposes such as drinking and cooking.

Dwelling - a room or suite of rooms designed to be used as a separate domicile.

EC – Electrical conductivity

Ecologically sustainable development - the same meaning as the definition in the *Local Government Act 1993.*

EDD - Extended Detention Depth.

Engineering guide for development - the document dated February 2005, developed by Blacktown City Council and entitled *Engineering guide for development* as amended or superseded.

Erosion - the process by which the detachment, entrainment, suspension and transport of soil occurs by wind, water or gravitational effects. Erosion leads to sedimentation.

Erosion and sediment control plan - a plan as described in Managing Urban Stormwater: Soils and construction – volume 1.

Flow duration - the time period over which flows occur.

Flow path - the route that water draining from an area will take.

Flow rate - the volume of fluid, in this instance water, that passes through a given point per unit of time.

Gross pollutant- contaminants equal to or greater than 5 millimetres in diameter that when introduced into an environment, cause instability, disorder, harm or discomfort to the physical systems or living organisms. This may include for example trash, litter and vegetation.

Gross pollutant and hydrocarbon trap - a device used to trap gross pollutants and hydrocarbons transported through the stormwater system.

Groundwater - water contained within the voids and spaces in rocks or soils.

Groundwater management system - the processes or practices used to control groundwater.

HED – High early discharge.

High flood risk precinct - the land subject to a high hydraulic hazard (in accordance with the provisional criteria outlined in the NSW Government Flood Risk Management Manual 2023) in a 1% AEP flood event and/or subject to potential evacuation difficulties during a flood.

HLR – Hydraulic loading rate.

Hydrocarbon - an organic chemical compound that is composed exclusively of hydrogen and carbon atoms. They are naturally occurring and form the basis of crude oil, natural gas, coal and other important energy sources.

Impervious areas - areas which have no or very limited ability infiltrate water through the surface through to the subsurface. Impervious areas occur where the soil surface is sealed, eliminating rainwater infiltration and natural groundwater recharge. They consist mainly of artificial structures such as pavements, rooftops, sidewalks, roads and parking lots covered by materials such as asphalt, concrete, brick and stone. Soils compacted by urban development are also highly impervious.

Impervious surfaces - the hard surfaces (for example roofs and pavement) within a catchment.

Integrated water management - an approach to the management of all aspects of water including rainwater, stormwater, groundwater, water supply and use, reuse and treatment.

Local overland flooding - occurs on the floodplains of original watercourses (which may be piped, channelised or diverted) or sloping areas where overland flows develop along alternative paths. This occurs when the system capacity has been exceeded and/or water depths generally in excess of 0.3 metres for the critical 1% AEP storm.

Local runoff - the direct surface runoff, surcharges and overflows from smaller stormwater pits/pipes and low points in kerbs. Local runoff invariably involves shallow depths of less than 0.3 metres in the critical 1% AEP storm.

Low flood risk precinct - all land within the floodplain that is within the extent of the probable maximum flood (PMF) but not identified as either a high flood risk or medium flood risk precinct. Therefore, the low flood risk precinct is all the land between the 100 year and the PMF flood extents.

Managing Urban Stormwater: Soils and Construction (The Blue Book) - the document dated 2004 and developed by Landcom. This is commonly referred to as The Blue Book as amended or superseded.

Medium flood risk precinct - the land below the 1% AEP flood level subject to a low hydraulic hazard (in line with the provisional criteria outlined in the *NSW Government Flood Risk Management Manual 2023*).

Model for urban stormwater improvement conceptualisation (MUSIC) - a modelling tool for both simple and highly complex urban stormwater systems using WSUD.

NER – National engineering register.

Non-potable water - water that is not fit or suitable for drinking and consumption purposes but may be used for purposes such as laundry, toilet flushing and air conditioning cooling towers. Non-potable water can be sourced from rainwater, stormwater and recycled sewage. The source determines appropriate uses, with rainwater having more fit-for-purpose uses than the other 2 sources, subject to treatment methods.

NPWR – Non-potable water reuse.

NTU – Nephelometric Turbidity Units.

On-site stormwater detention (OSD) measures - practices that control the peak, post development flows of stormwater discharged from a site to below pre-developed flows of stormwater being discharged from a site.

Perched aquifer - an aquifer in which infiltrating water remains separated from an underlying main body of groundwater, with an unsaturated zone existing between them.

Potable water (or mains water) - water reserved for or suitable for drinking purposes/consumption. In this region potable water comes from Warragamba Dam and is supplied by a Sydney Water reticulation system.

Probable maximum flood (PMF) - the largest flood that could conceivably occur at a particular location. The PMF defines the extent of flood prone land, that is, the floodplain. The risk of it happening in any 1 year is extremely rare and is considered to be between a 1 in 10,000 to 1 in 10,000,000 chance.

PSD - Permissible site discharge.

Public open space - land zoned as RE1 Public Recreation in the *Blacktown Local Environment Plan 2015.*

PWL – Permanent water level.

Rainwater tank - a reservoir or container that is used to collect and store (harvest) rain that runs off impervious surfaces such as roofs, via gutters and downpipes. This water can be re-used for toilet flushing, car washing, general wash down and irrigation. Rainwater tanks also play a part in local flood control by temporarily storing run-off from roof surfaces.

Receiving watercourses - a watercourse into which stormwater runoff flows. The method of delivery may be from subsoil drainage, local overland flow paths, pipes or other watercourses.

RWT – Rainwater tank.

S3QM – Small scale stormwater quality model.

Sediment - material of varying size (both mineral and organic) that is being or has been moved from its site of origin by the action or wind, water or gravity and comes to rest.

Sedimentation - the deposition of sediment, usually in locations such as a channel, along a fence line, in an area of low slope, depression, watercourse or sediment trap.

Soil - a natural material consisting of layers, amalgamates or individual particles or mineral and/or organic constituents, of variable thickness. It differs from its parent material in morphological, physical, chemical and mineralogical properties and biological characteristics.

SQID – Stormwater Quality Improvement Device.

SQIDEP – Stormwater Quality Improvement Device Evaluation Protocol.

Stormwater - the runoff from all hard surfaces in the urban environment, which is carried away through a system of pipes that is separate to the sewerage system.

Stormwater harvesting and reuse scheme - a process of collection, treatment, storage and use of stormwater.

Stormwater management - the processes or practices used to control stormwater.

Stormwater tank - a reservoir or container that is used to collect and store (harvest) stormwater. The stormwater can run off a variety of surfaces including landscape areas, driveways, pedestrian access areas (including roof podiums), as well as overflows from rainwater tanks. The designer must ensure the stormwater is treated to be appropriate for its intended use.

Stormwater treatment measure - the hard and soft engineering practices that treat and improve the quality and quantity of stormwater.

Stream forming flow - the following percentage of the 2-year ARI (50% AEP) flow rate estimated for the catchment under natural conditions:

- 10% for cohesionless (for example sandy) bed and banks
- 25% for moderately cohesive bed and banks
- 50% for cohesive (for example stiff clay) bed and banks.

Note: 25% is the default percentage to be used in our city due to dispersive characteristics of the typical local clay soils. However, 50% may be used if proven to the satisfaction of Blacktown City Council that the use of the 50% threshold is appropriate. However, to achieve this, the applicant will need to demonstrate that the soils are capable of withstanding the erosive forces generated through the increased flow.

Subsoil drainage - drainage of the layer of soil under the surface of the ground.

TDS – Total dissolved solids.

TFR – Treatable flow rate.

Total nitrogen (TN) - the sum of the nitrogen present in all nitrogen-containing components in the water column. This includes large and small phytoplankton and zooplankton, suspended microphytobenthos, dissolved inorganic nitrogen (nitrate and ammonia), dissolved organic nitrogen, labile detritus (both at the Redfield ratio and the Atkinson ratio) and refractory detritus. Total nitrogen concentration is determined by a balance between inputs (diffuse catchment loads, point source loads) and loss terms (export from the site to a watercourse and within the sediments).

Total phosphorous (TP) - the sum of the phosphorus present in all phosphorus-containing components in the water column. This includes large and small phytoplankton and zooplankton, suspended microphytobenthos, dissolved inorganic phosphorus (both absorbed and desorbed), dissolved organic phosphorus, labile detritus (both at the Redfield ratio and the Atkinson ratio) and refractory detritus. Total phosphorus concentration is determined by a balance between inputs (diffuse catchment loads and point source loads) and loss terms (export from the site to a watercourse and within the sediments).

Total suspended solids (TSS) - a measure of the mass of solid material (organic and inorganic) suspended in the water column. Suspended solids can include a range of inorganic and organic particles suspended in the water column which can be defined as the filterable residue retained on a 2.0-micron pore size filter dried at 1050C.

Voluntary planning agreement (VPA) - a planning agreement under Section 93F of the *Environmental Planning and Assessment Act 1979*.

Wastewater - water transferred through the sewer system, treated in sewerage treatment plants (STPs) and then discharged into our creeks.

Watercourse - the same meaning as 'river' under the *Water Management Act 2000* as amended or superseded.

Water conservation measure - practices that contribute to a reduction in water usage.

Water sensitive urban design (WSUD) - an approach that integrates water cycle management into urban planning and design. It is used to help mitigate and reduce the impacts of development on our local waterways.

Water Sensitive Urban Design (WSUD) checklists - the series of checklists containing information that needs to be submitted at the development stage and throughout each stage of the development. These checklists are contained in the WSUD developer handbook.

Water sensitive urban design (WSUD) developer handbook - this document developed by Blacktown City Council as amended or superseded.

Water Sensitive Urban Design (WSUD) sign - a sign that must be displayed on a property to identify the individual devices that form part of the WSUD system in relation to the site layout.

Water Sensitive Urban Design (WSUD) system - refers to devices and systems constructed and/or installed to manage stormwater runoff, including but not limited to on-site stormwater detention systems, rainwater tanks, bioretention systems, swales, gross pollutant traps, and all other stormwater treatment measures approved by Council.

Water table - the top level of water stored underground. It is the surface of groundwater in the soil.

Waterways - includes rivers, creeks and streams.

Waterway stability - the ability of a watercourse to withstand erosive forces.

18 References

Argue, J., (2013). WSUD: *Basic Procedures for Source Control of Stormwater Procedures*.

Australian Guidelines for Water Recycling: Managing Health and Environmental Risk (Phase 2) Stormwater Harvesting and Reuse, 2008.

Ball J, Babister M, Nathan R, Weeks W, Weinmann E, Retallick M, Testoni I, (Editors), 2019, *Australian Rainfall and Runoff: A Guide to Flood Estimation*, Commonwealth of Australia.

Blackham, D., and Wettenhal, G. (August 2010). *NSW MUSIC Modelling Guide*.

Blacktown City Council, 2005. *Engineering guide for development*.

Blacktown City Council, 2024. *WSUD standard drawings*.

Department of Environment and Conservation, NSW. Dec 2006. *Managing Urban Stormwater: Harvesting and Reuse.*

Environment Protection & Heritage Council, National Health & Medical Research Council, Natural Resource Management Ministerial Council, May 2018. *Australian Guidelines for Water Recycling: Augmentation of Drinking Water Supplies*.

eWater. *MUSIC User Guide*.

Fletcher, T., et.al., Cooperative Research Centre for Catchment Hydrology, 2004. *Stormwater Flow and Quality, and the Effectiveness of non-proprietary Stormwater Treatment Measures – A Review and Gap Analysis.* Technical Report, report 04/8.

Hawkesbury-Nepean Floodplain Management Steering Committee, June 2006. *Reducing Vulnerability of Buildings to Flood Damage.*

HumeGard, March 2017. *GPT Technical Manual Issue 4*.

Institution of Engineers, Australia (1987) *Australian Rainfall and Runoff (ARR): A Guide to Flood Estimation*, Vol. 1, Editor-in- chief D.H. Pilgrim, Revised Edition 1987 (Reprinted edition 1998), Barton, ACT.

Landcom, 2004. *Managing Urban Stormwater: Soils and Construction* (also known as The Blue Book).

Melbourne Water, 2017. *Wetland Design Manual. Supporting document: Wetland form and function.*

NSW Government. 2023. *Flood Risk Management Manual*.

Sydney Coastal Councils Group, September 2016. *Groundwater Management Handbook*.

OceanProtect. March 2019. *Jellyfish Filter Technical Design Guide, Rev. 1*.

Upper Parramatta River Catchment Trust, 2005. On-*site Stormwater Detention Handbook Version 4.*

Blacktown
City Council

62 Flushcombe Road
Blacktown NSW 2148
PO Box 63
Blacktown NSW 2148
Ph: 02 9839 6000
Fax: 02 9831 1961
Email: council@blacktown.nsw.gov.au
Website: www.blacktown.nsw.gov.au
Facebook: Blacktown City Council
Twitter: @blacktowncc
Flickr: Blacktown City Council
YouTube: Blacktown City Council
Instagram: Blacktown City Council